

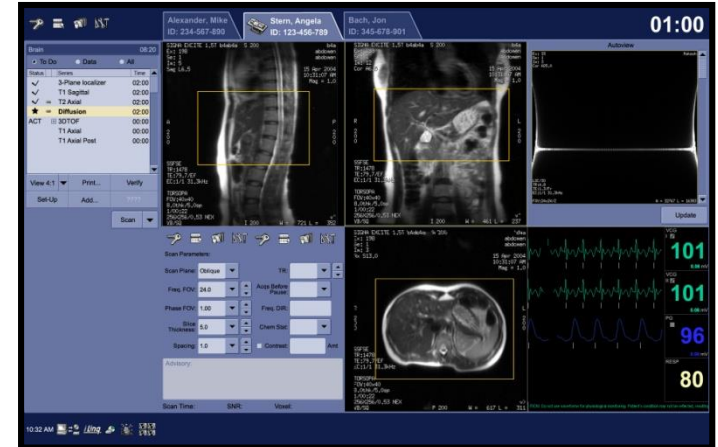


GE Optima/Discovery MR450 & MR750

Objectives

- Identify the impact of DISCOVERY and Optima MR on the Customer, Service Engineer, GE business
- Recognize the differences between DISCOVERY MR and other systems.
- Recognize the differences between Optima MR and other systems.
- The student will understand the high level layout and functionality of the system.

- 32 channels with full clinical value
- 128-channel architecture
- New Modular HD coils suite
- New gradient subsystem; 50/200 gradient coil and driver
- New optical RF chain
- Silent patient table with touch-and-go landmarking
- Point-and-shoot user interface
- Significantly faster exam time



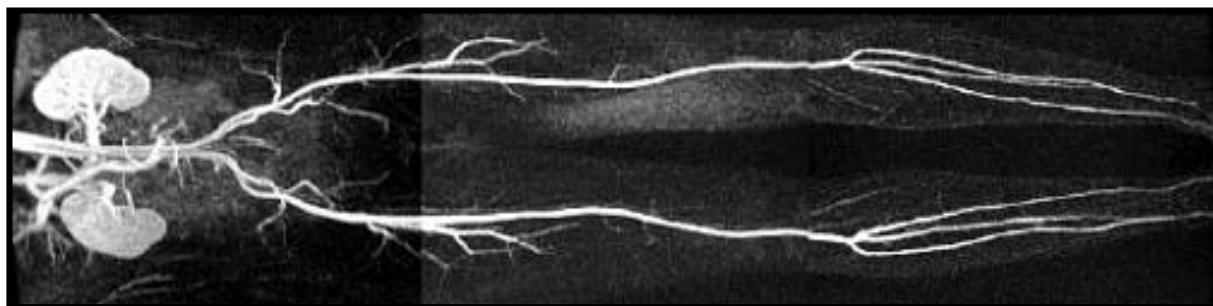
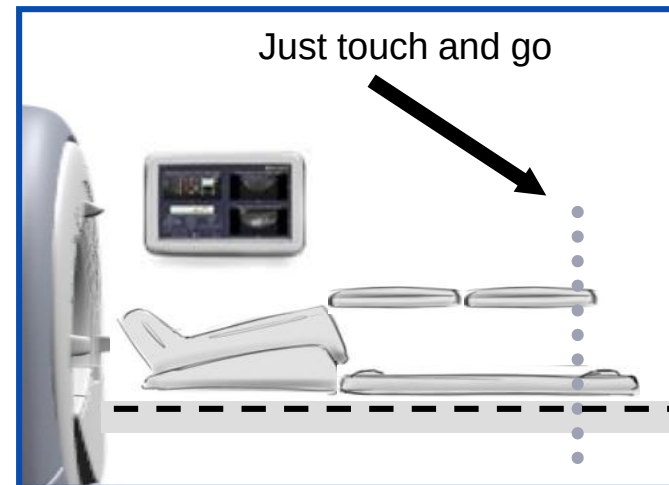
Point-and-shoot user interface



Complete Abdominal exam
in 6 breath-holds;
currently takes 20 minutes

Volumetric Vascular Scan

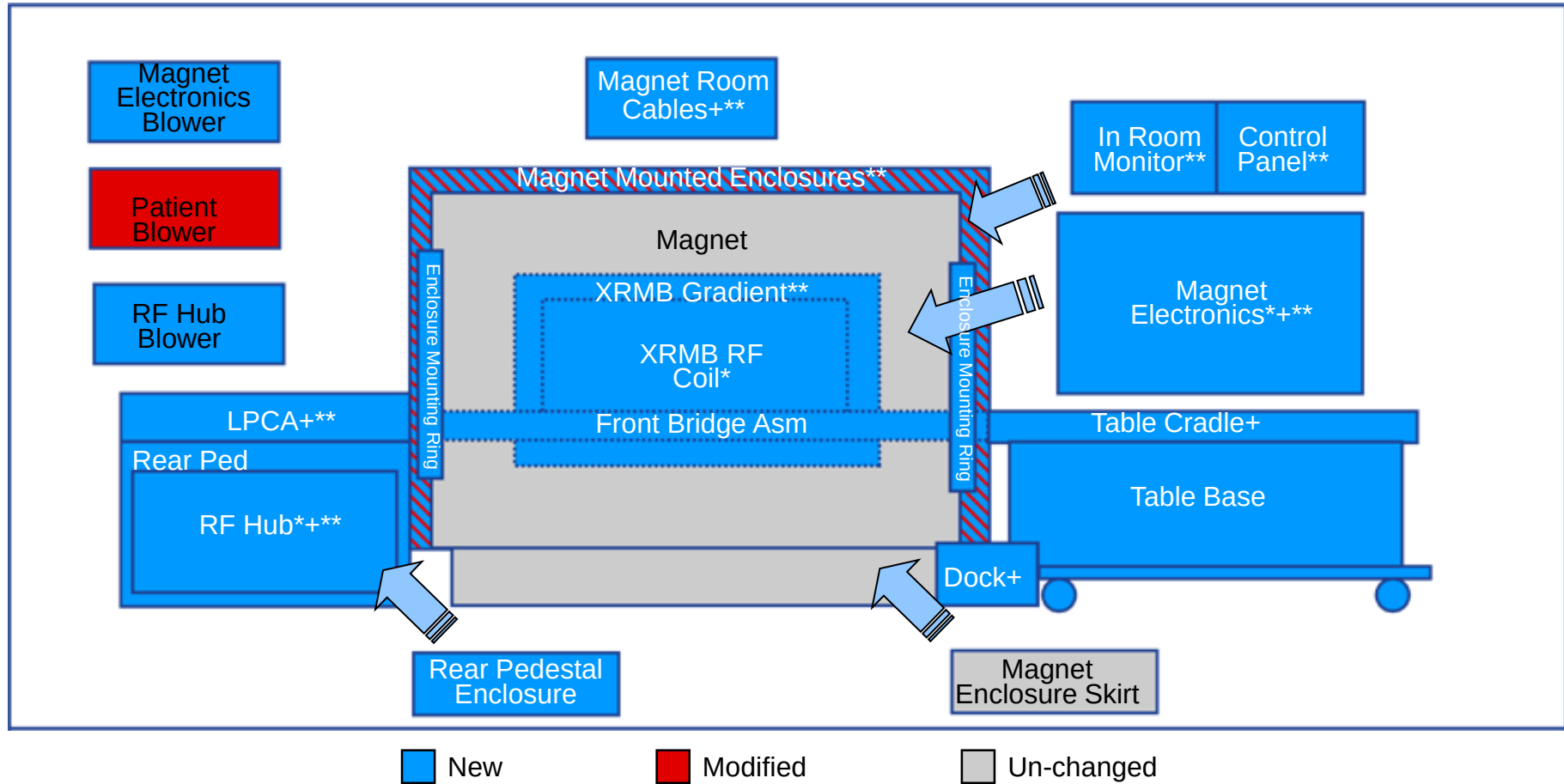
- time reduced from 45 minutes to 6 minutes



Volumetric vascular (run-off) scan

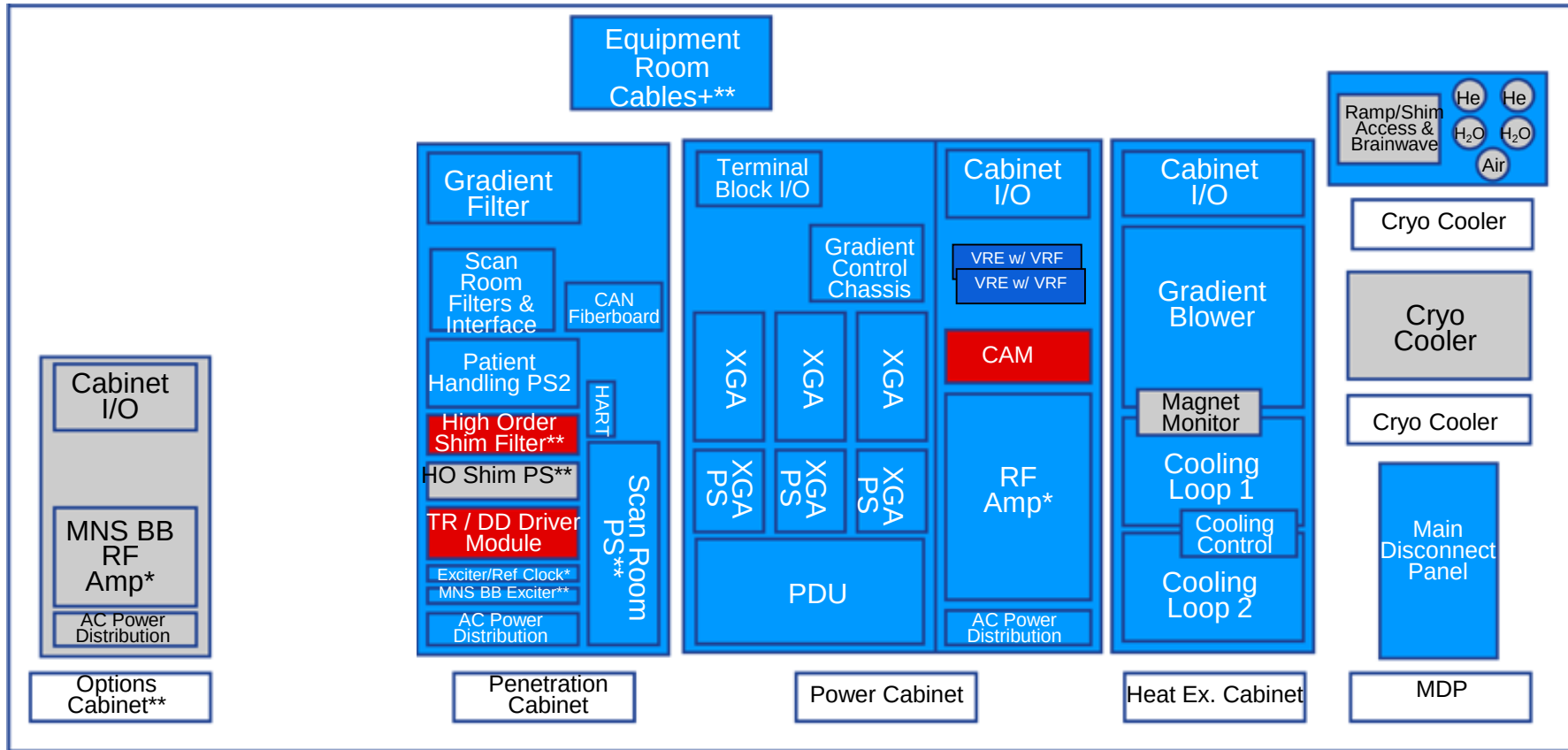
System Re-architecture Approach

- Scanner room – represented is Discovery. (Optima is similar – not identical).



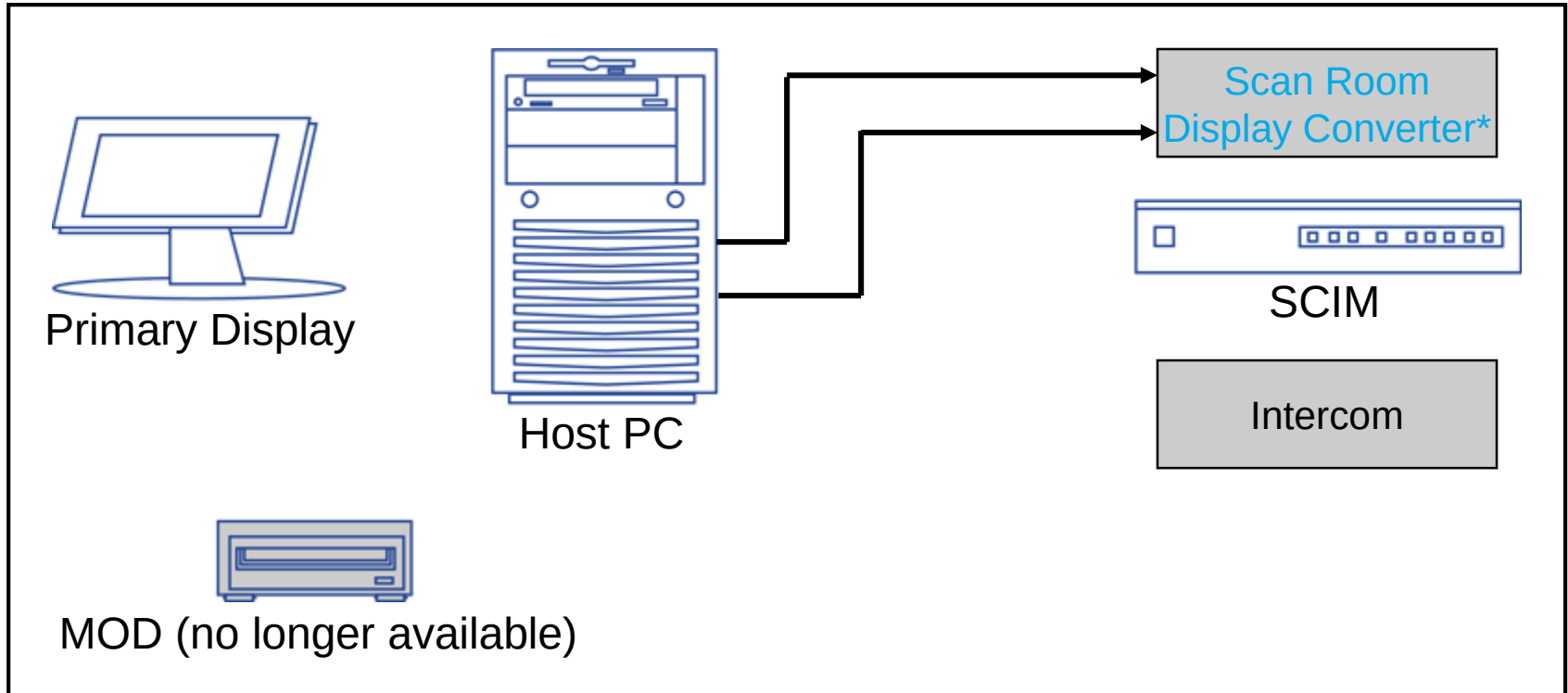
System Re-architecture Approach

- Equipment room – represented is Discovery and Optima



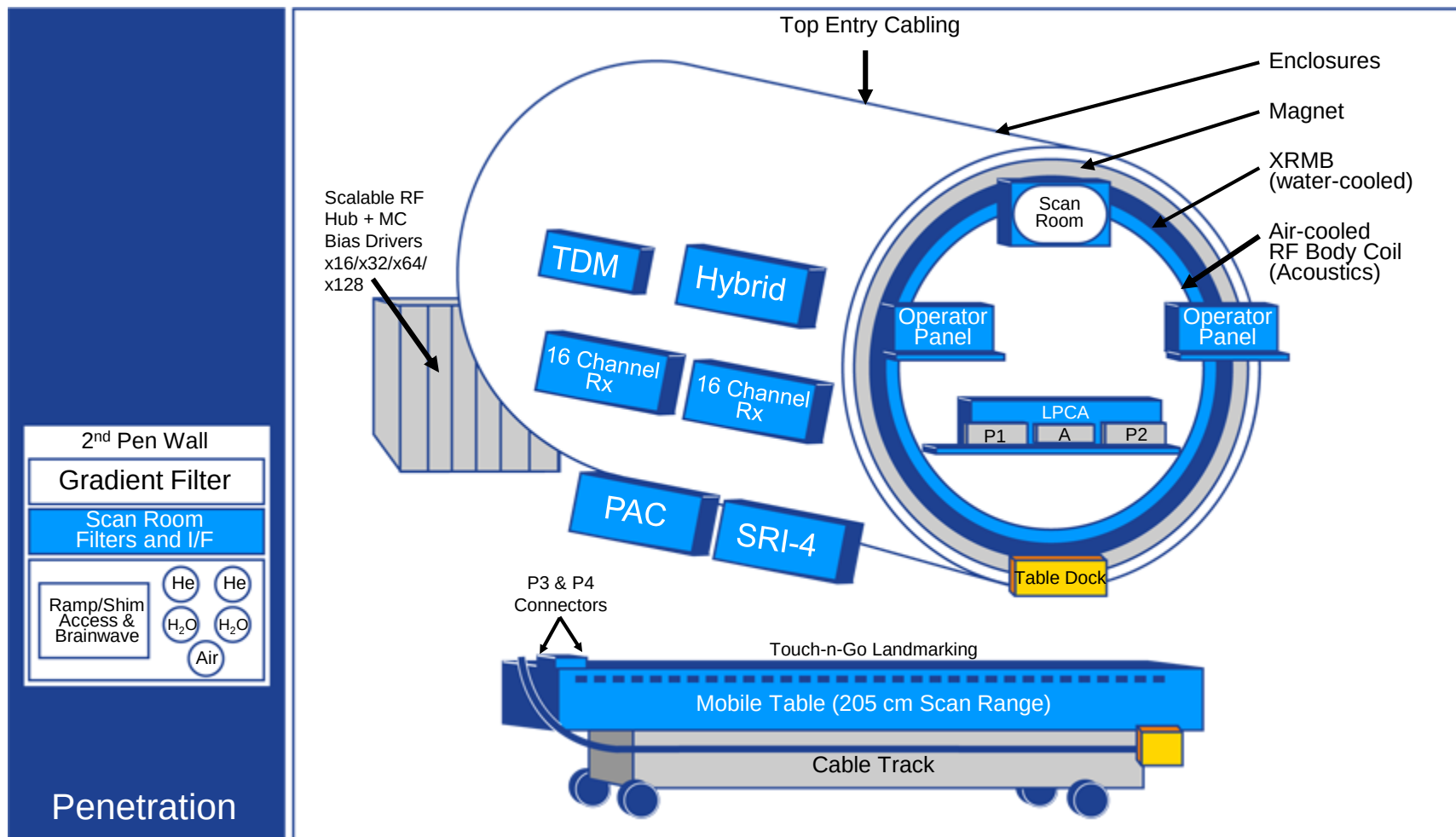
System Re-architecture Approach

- Operator room – Discovery and Optima

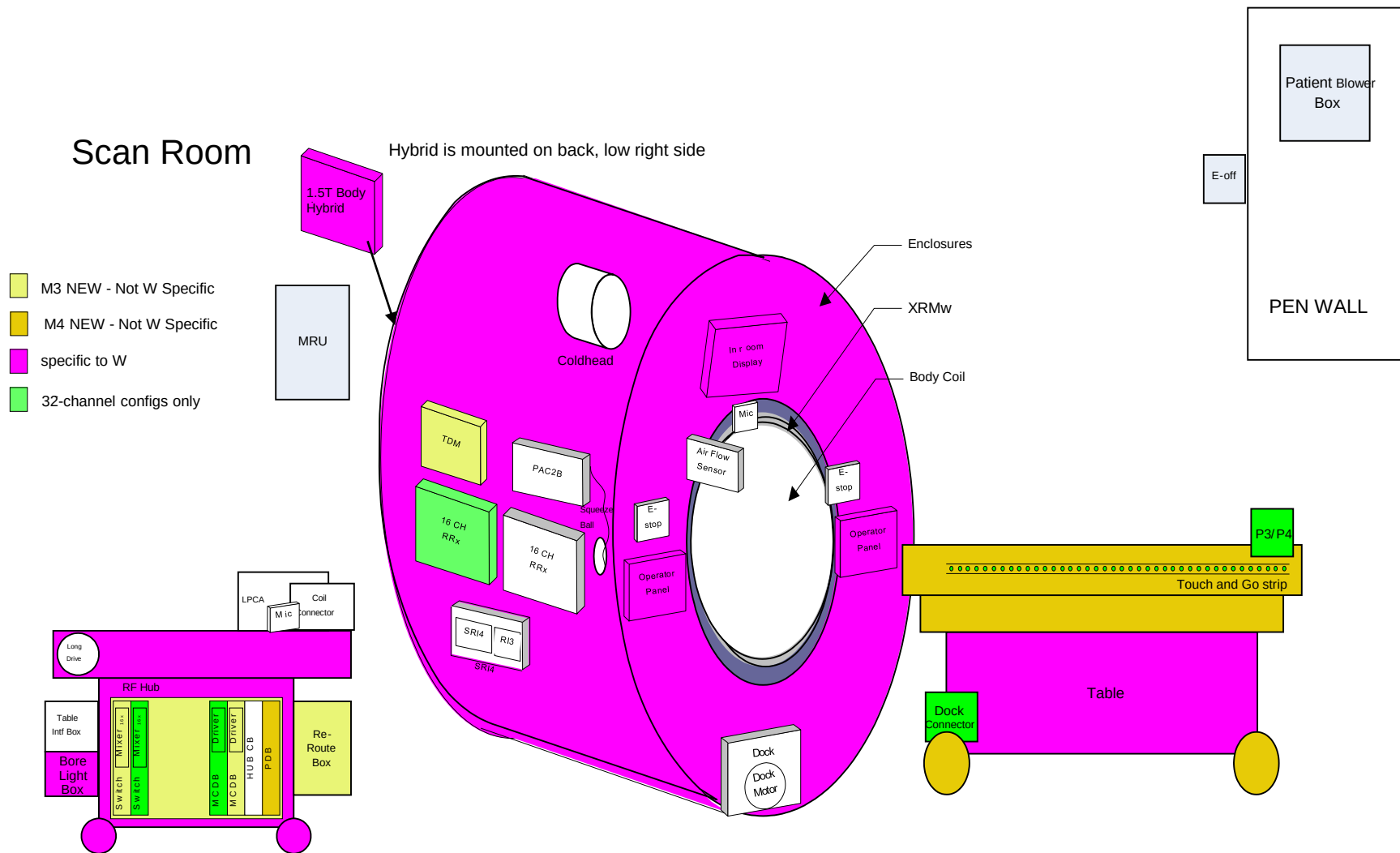


Discovery Scan Room Layout

- Scan room – Discovery MR 750 or 450. Optima (450w) is different



Optima Scan Room Layout

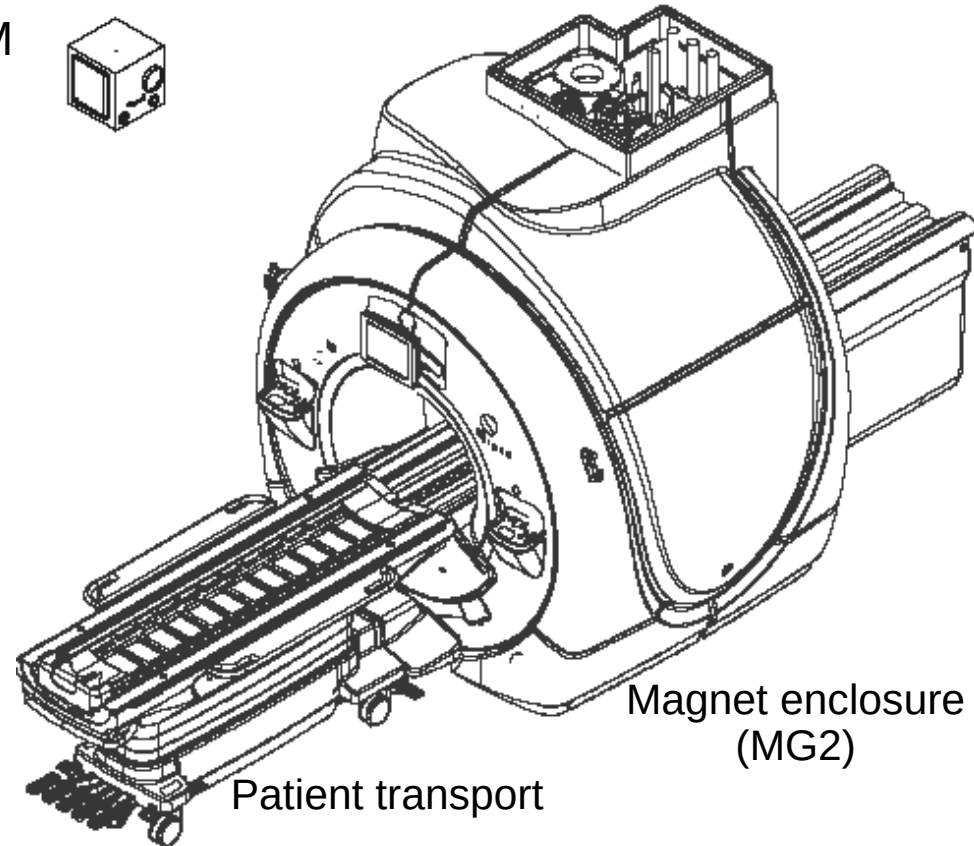


Basic components

- **The basic HDv system consists of the following major equipment:**
 - LCC magnet with magnet enclosure
 - Magnet accessories
 - Patient transport table and cradle
 - Identical Gradient Coil (XRM)
 - and unique RF body coils and head coil



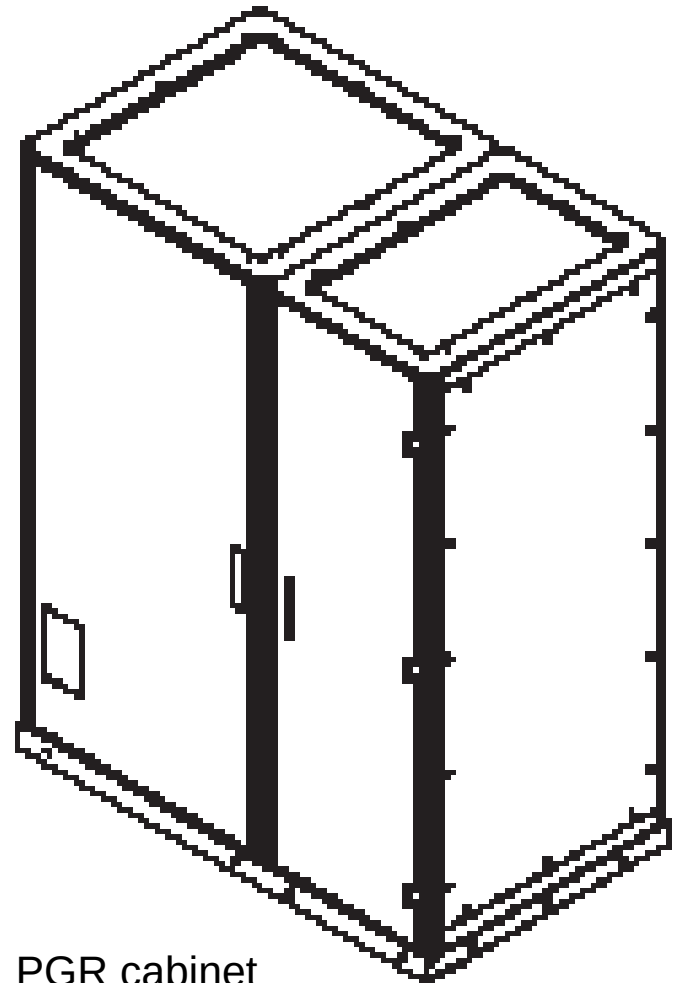
Storage cart for the
SPT phantom set



Magnet enclosure
(MG2)

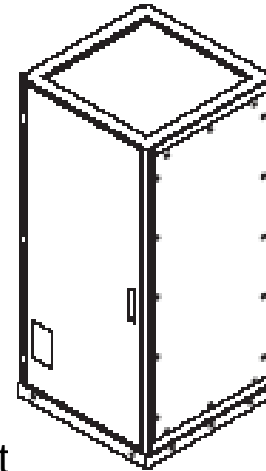
Patient transport

- System electronics cabinets include:
 - **Power Electronics (PGR) cabinet** containing:
 - XGD gradient drivers
 - Power Distribution Unit module with unregulated transformer: 380/400/415/480 Volt, 50/60 Hz with power filter
 - CAM chassis with boards for either 16 or 32 channel
 - Components for the Volume Reconstruction Engine (VRE)
 - Narrow band RF amplifier (different 1.5 and 3T)

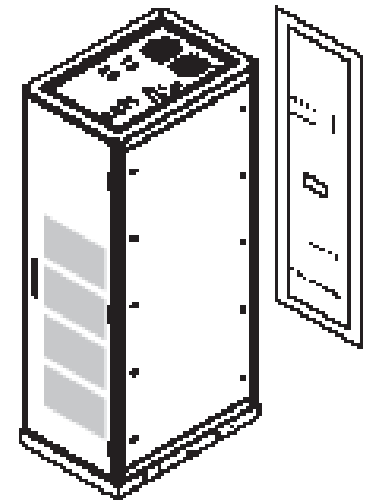


PGR cabinet

- System electronics cabinets include (continued):
 - **HEC cabinet** containing the cooling loops and gradient blower
 - **PEN cabinet** containing:
 - PHPS
 - Reference clock / Exciter (different for 3T /1.5T)
 - Scan room power supply
 - CAN fiber I/F
 - T/R DD driver module
 - HO shim filter (optional)
 - **Main Disconnect Panel** (GE supplied)
 - **Magnet Monitor (MON)**

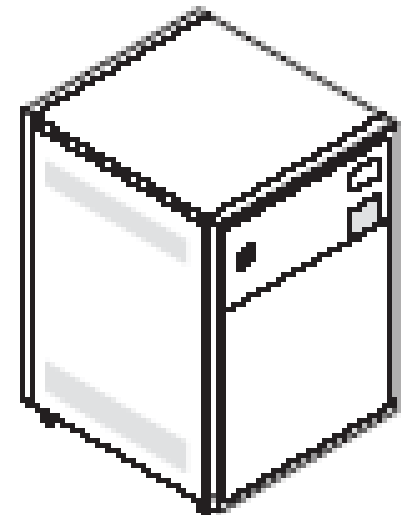
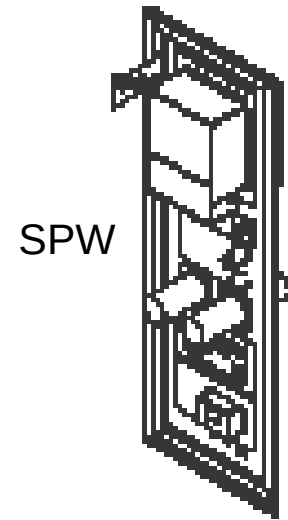


HEC cabinet



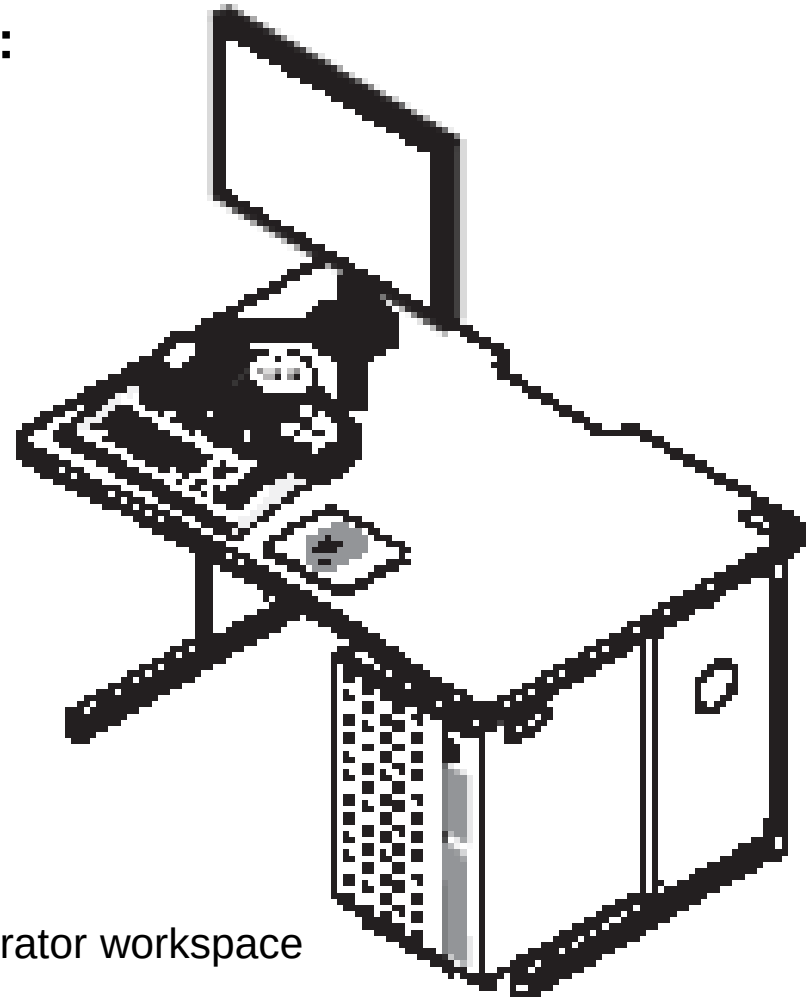
PEN cabinet

- System electronics cabinets include (continued):
 - **SPW (Second Penetration Wall)** containing:
 - Gradient filter
 - Scan room filters and I/F
 - Ramp/shim access
 - **Shield/cryo cooler compressor cabinet**



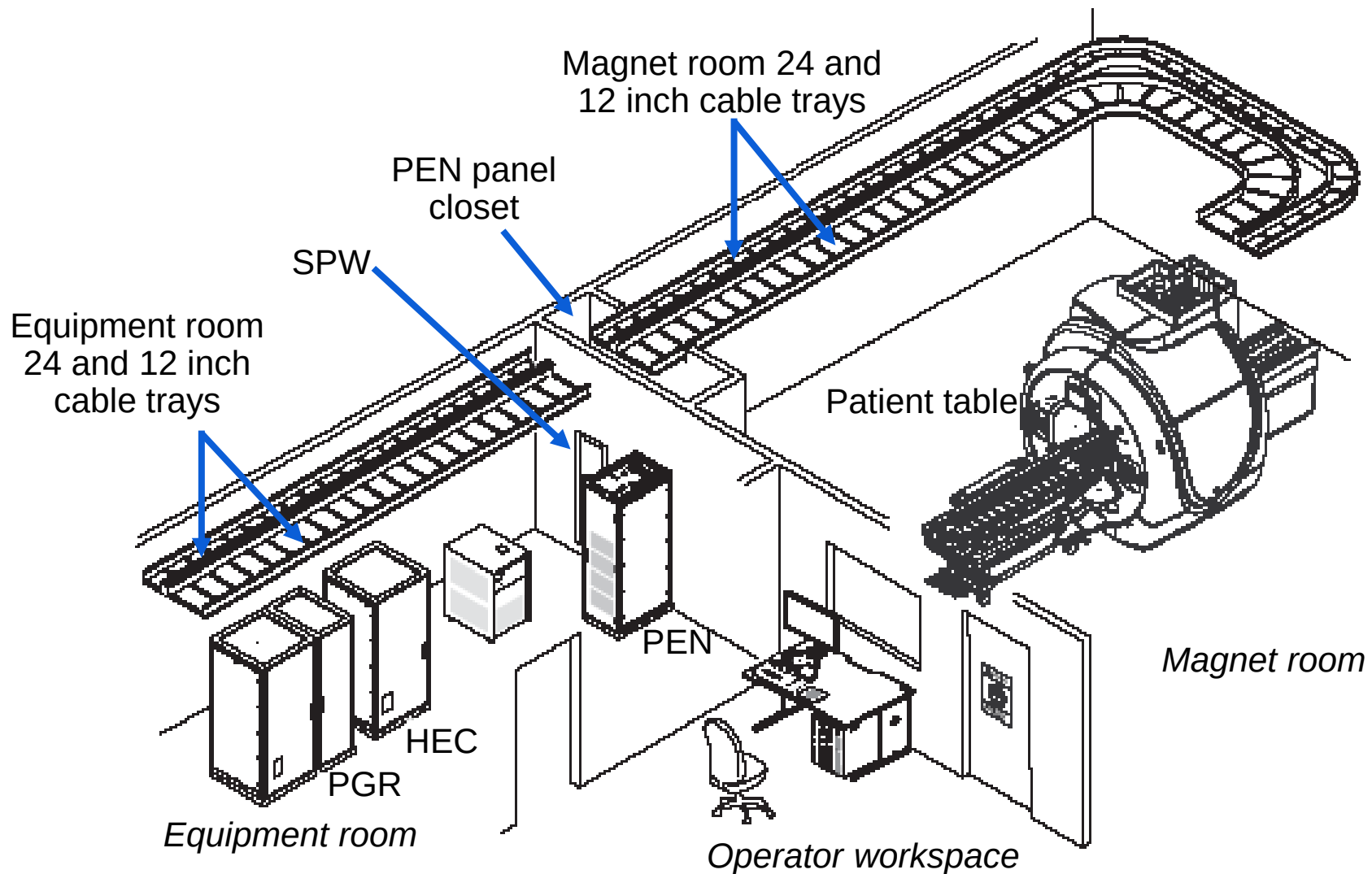
Operator Workspace

- **The operator workspace includes:**
 - LCD widescreen monitor
 - Keyboard
 - SCIM
 - Mouse and mouse pad



Operator workspace

New system layout



- **Software platform changes**

- Support for DVMR hardware

- XRM-B/XGD, New Gradient Heating model, New Body Coil and RF Amp models
 - 128-channel capability (not yet supported)
 - In-room monitor
 - Automatic coil element selection
 - Touch-and-go landmarking, more

- Complete UI and workflow overhaul

- Workflow manager/multiple sessions
 - Enhanced protocol management
 - Integrated post-processing apps/tasks
 - Graphic Rx enhancements

- **Quality/Reliability Features**

- Process Failure Recovery
- Test automation, Agitar adoption
- Full Obj-C removal & Code clean-up

- **Applications**

- 2D GEM Support for Volume acq.
- TAO – Dual-Echo DIXON
- IDEAL –MSK
- Propeller Enhancements(NPW, Non-axial, Body)
- 3T PURE
- EZStep
- MR Echo on 3T

- **Applications Visualization Features:**
 - R2* measurement
 - Functool Fusion
 - Inline visualization and auto-launch
 - Cartilage segmentation
 - MR pasting
 - ReportCARD 4.0
 - Brainwave 4.0
 - CadStream 3.0

- **Service Features:**
 - XGD failure detection
 - RF receive failure detection
 - RF transmit failure detection
 - Patient handling failure detection
 - FE Cockpit-3
 - Prognostics (MR)
 - All system tools support
 - RF transmit system FRU isolation
 - Environmental shutdown
 - IBIS support

Configuration 1

- 32 channels
- 2 ICNs
- IRM
- 32-channel table
- High order shim

- All G3 to Signa MR750 Upgrades (PSI = MDU009)
- Forward Production (PSI = MDF013)

Configuration 2

- 16 channels
- 1 ICN
- IRM
- 16-channel table (no P connectors)
- High order shim

- Forward Production (PSI = MDF011)

Configuration 3

- 16 channels
- 1 ICN
- 16-channel table (no P connectors)

- Forward Production (PSI = MDF012)

Configuration C

- 32 channels
- 2 ICNs
- IRM
- 32-channel table
- High order shim

M7450A	32ch no intellitouch	1	32ch no intellitouch
M7450B	32ch intellitouch, no	3	32ch intellitouch, no HOS
M7450C	32 ch intellitouch HO	6	HOS
M7450D	U/G	4	U/G

- All LCC to MR450 Upgrades (PSI = M7450D)
- Forward Production (PSI = M7450C)

Configuration A

- 32 channels
- 2 ICN
- No IRM
- No electric dock (no P connectors)
- High order shim

- Forward Production (PSI = M7450A)

Configuration B

- 32 channels
- 2 ICN
- IRM
- Electric dock (P connectors)
- No HOS

- Forward Production (PSI = M7450B)

Configuration A

- 16 channels
- 1 ICN 5127452-6 w/ VRF2 & w/o Infini
- No IRM
- No P Ports in table (no Elec. Dock)

Configuration B

- 16 channels
- 1 ICN 5127452-6 w/ VRF2 & w/o Infini
- IRM
- No P Ports in table (no Elec. Dock)

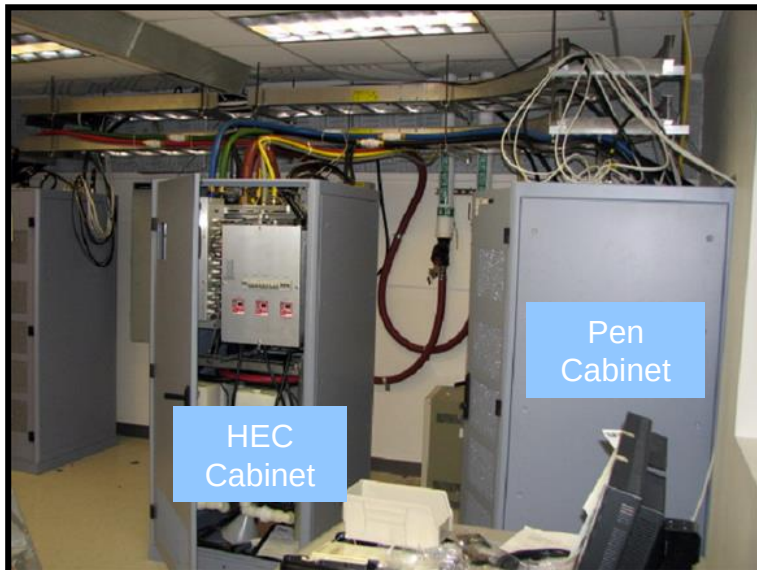
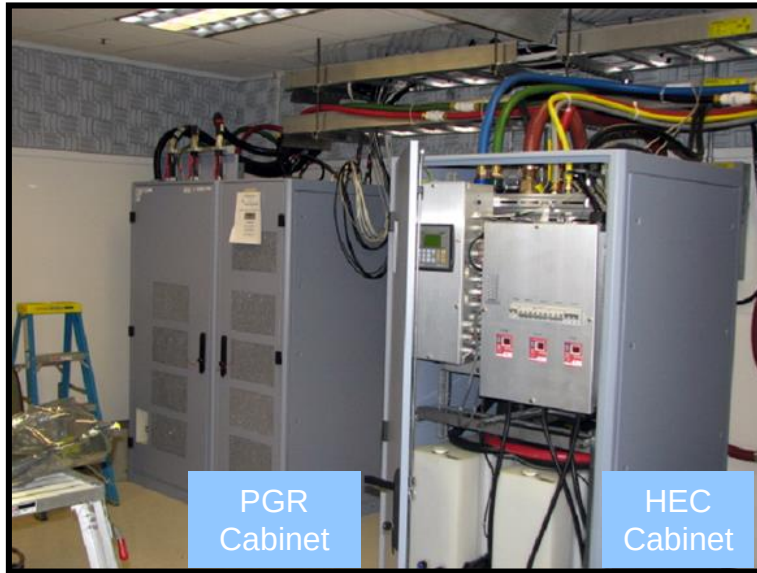
Configuration C

- 32 channels
- 2 ICNs 5127452-2 w/ VRF & w Infini
- IRM
- No P Ports in table (no Elec. Dock)

Configuration D

- 32 channels
- 2 ICNs 5127452-2 w/ VRF & w Infini
- IRM
- P Ports in table (Elec. Dock)

Signa MR750 Cabinets



Power Gradient RF (PGR) Cabinet

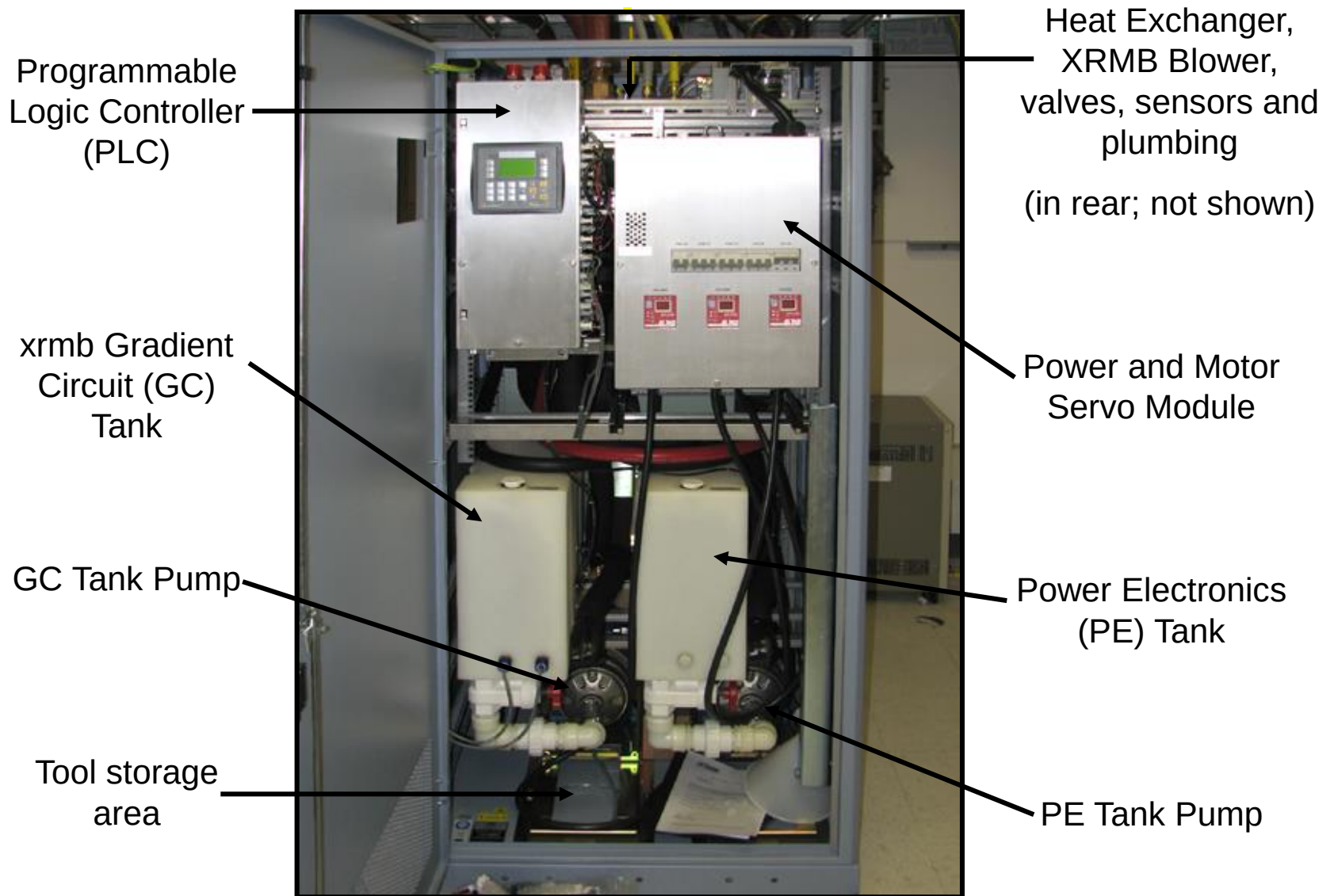


PGR Cabinet Contents



Heat Exchanger Cabinet (HEC)





Penetration (Pen) Cabinet



Sited cabinet – Front view

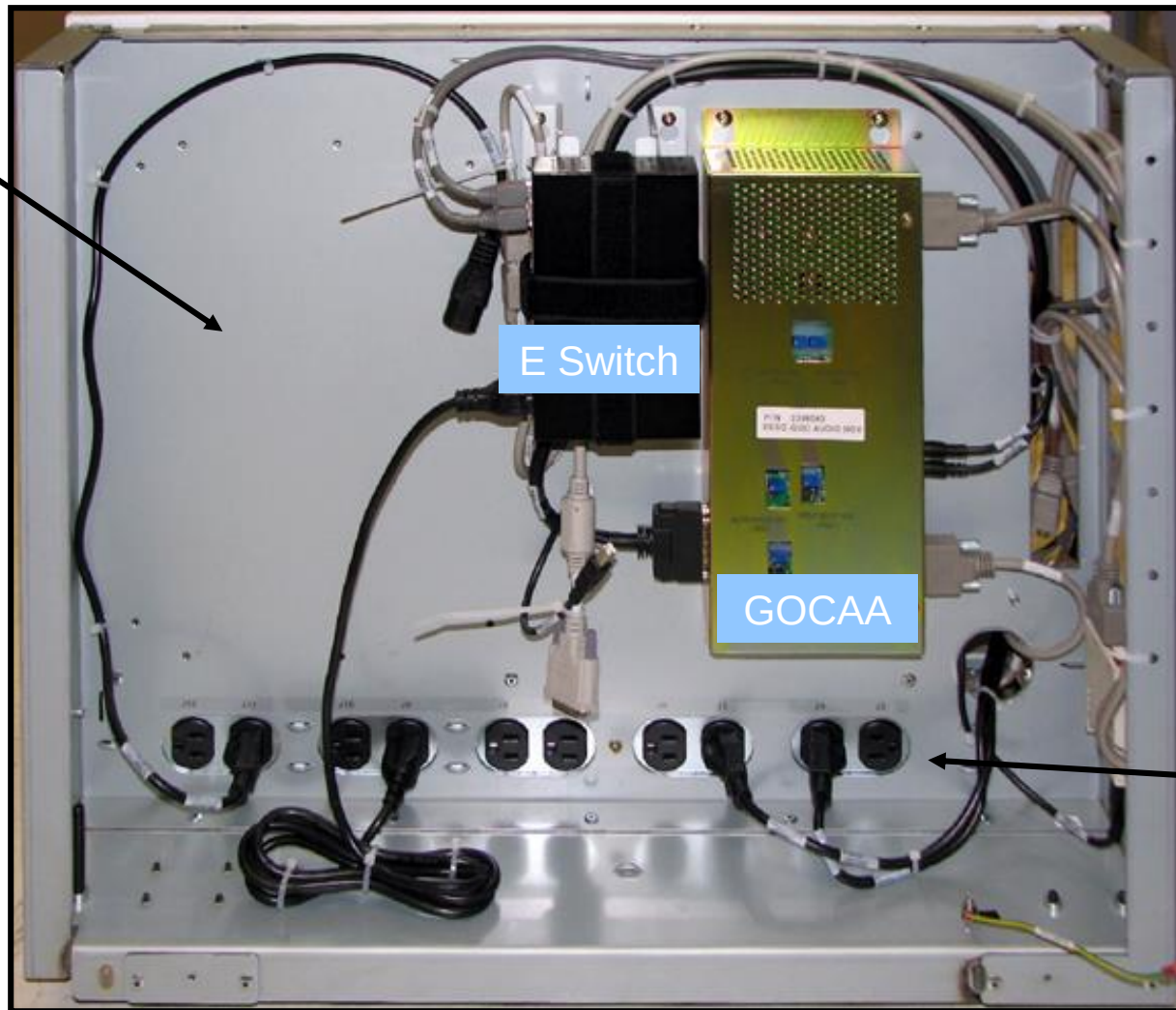


Un-sited cabinet – Rear view

Global Operator Console (GOC)



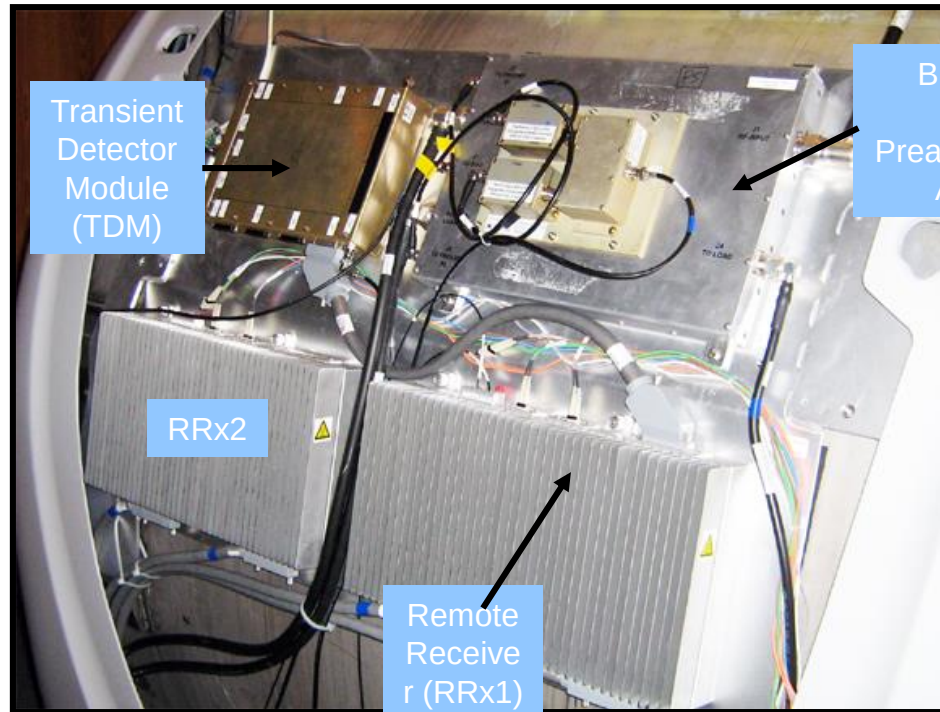
Location for
optional
IRDM



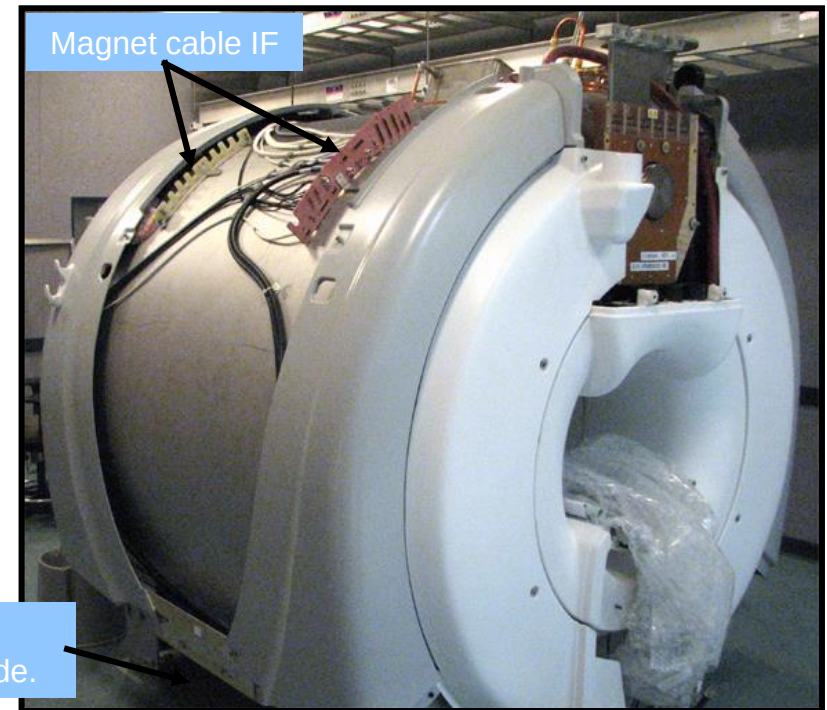
View with right side cover removed.

Magnet

- Magnet arrives from Magnet Plant with all electronics pre-installed.

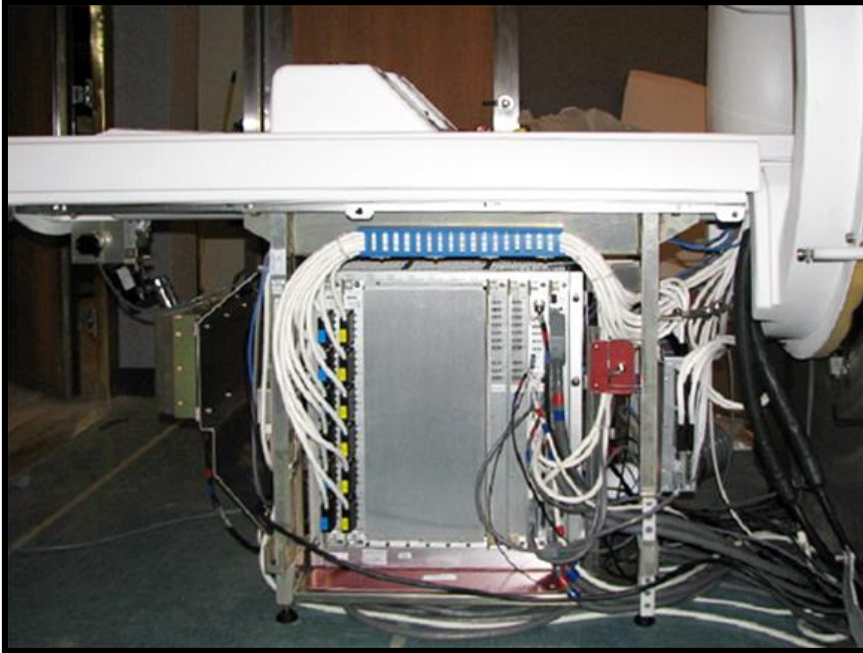


Left side from front

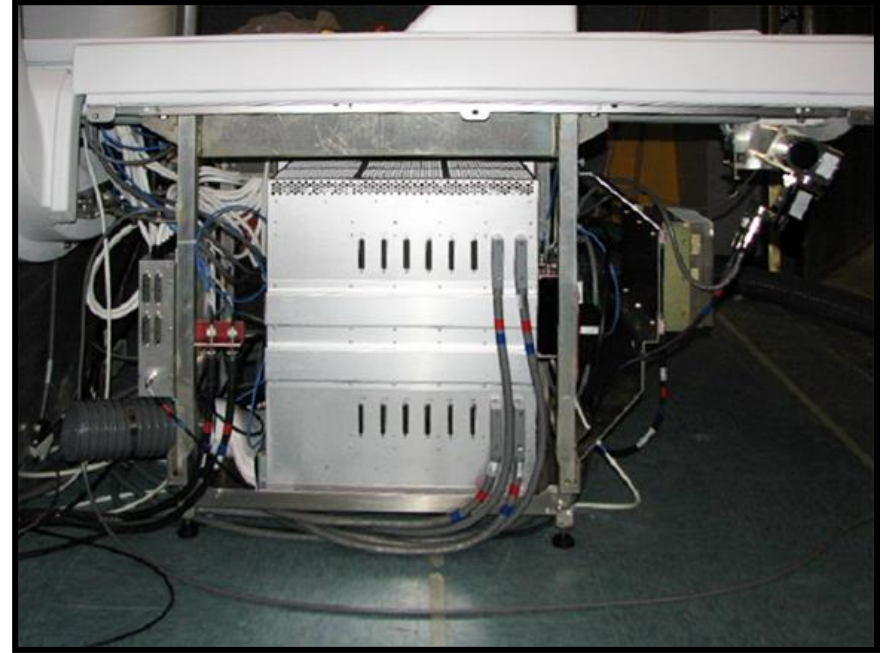


Left side from rear

Rear Pedestal

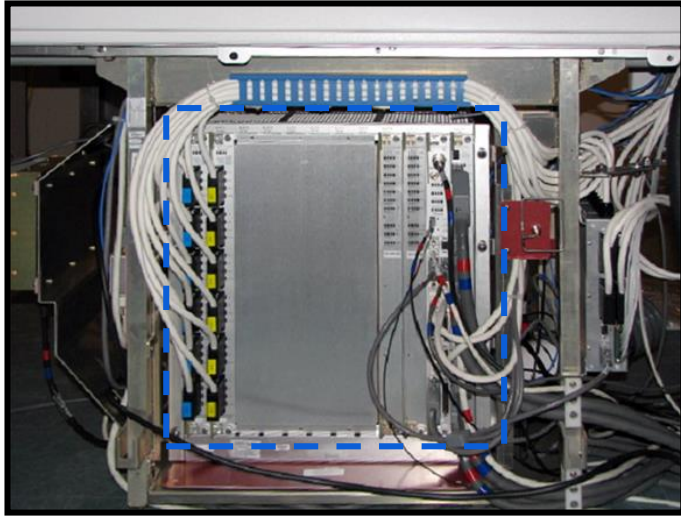


Left side from front

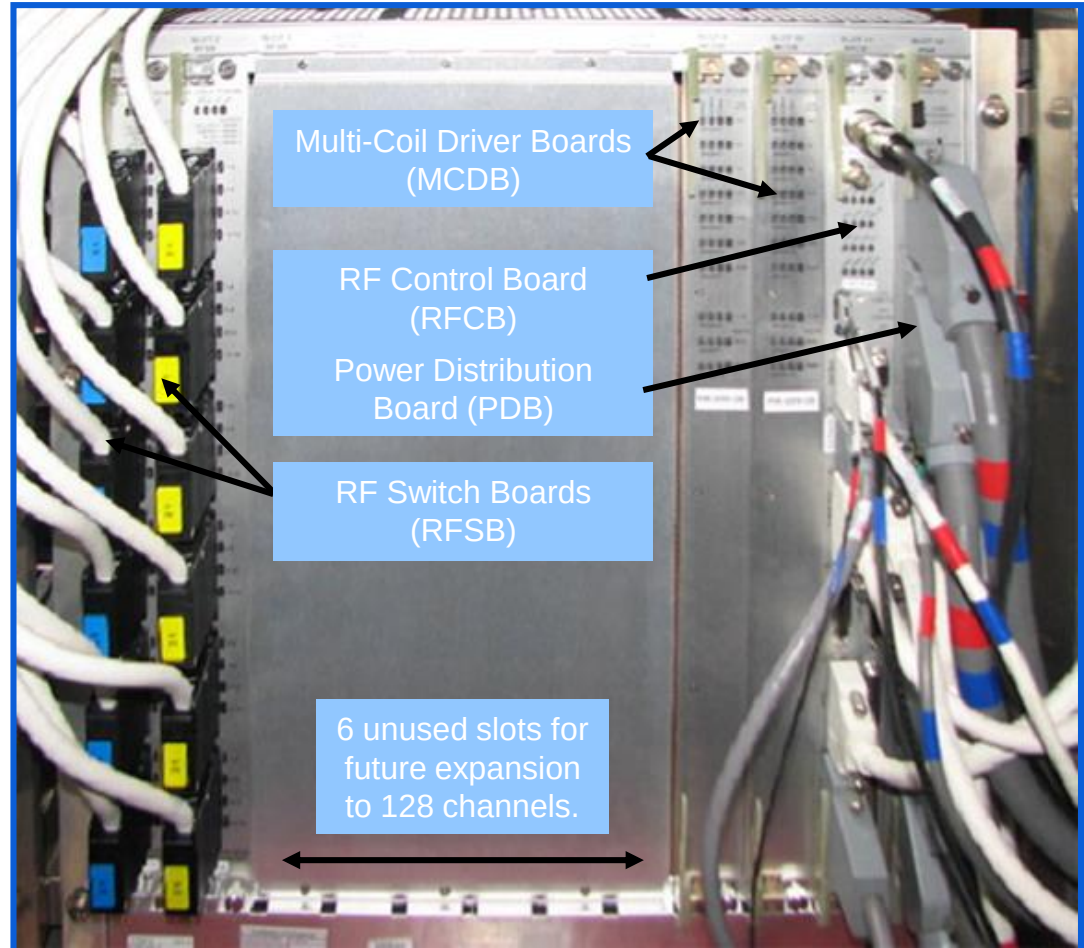


Right side from front

Rear Pedestal

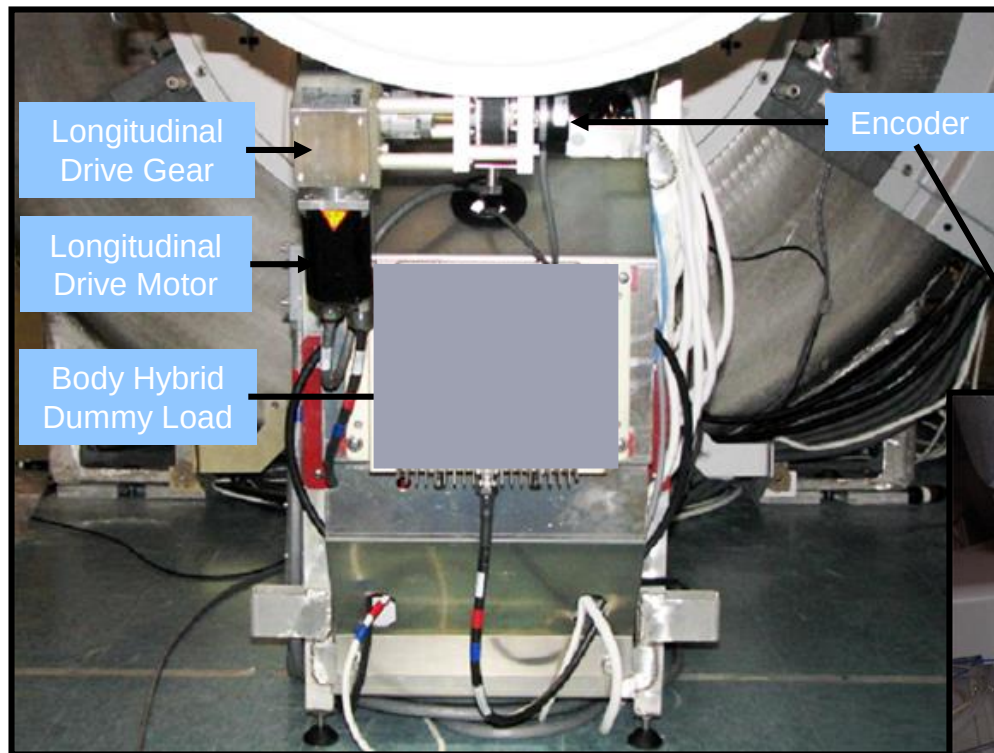


- **16 channel RF Hub contains one of each:**
 - RFSB
 - MCDB
 - RFCB
 - PDB

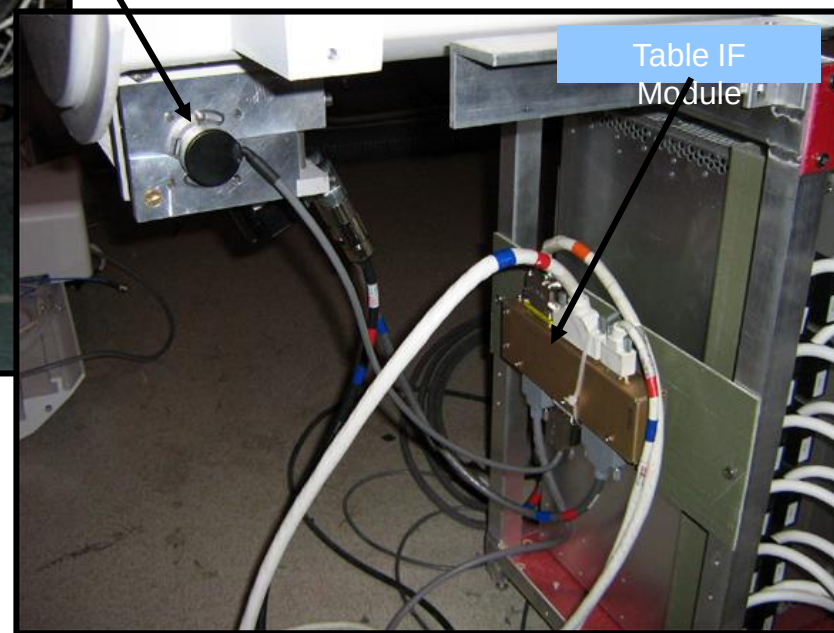


32 channel RF Hub

Rear Pedestal



End view



Left side view- dummy load removed

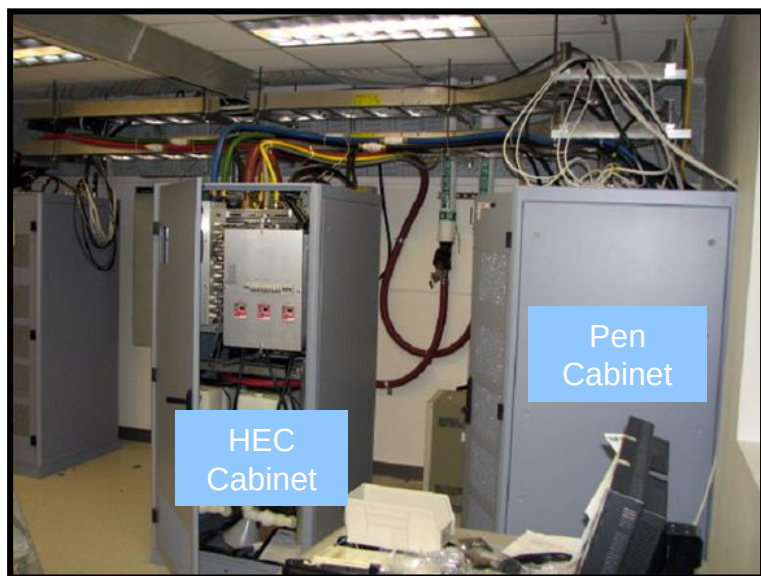
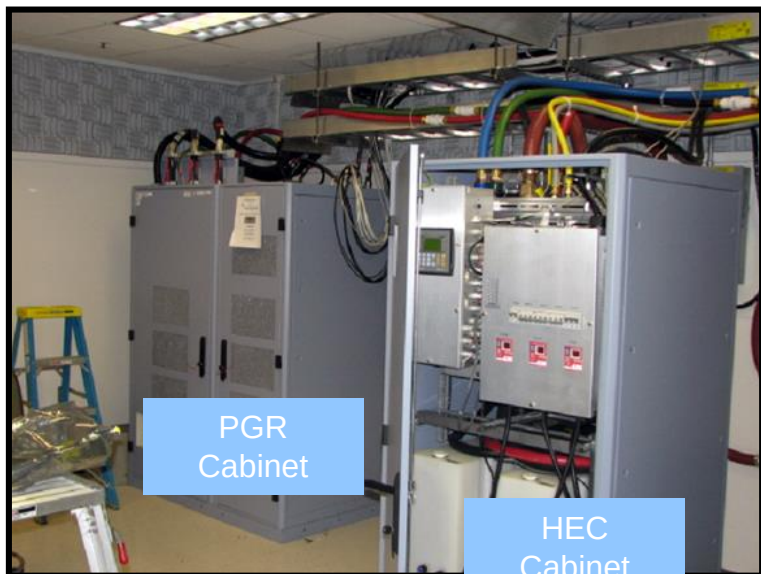




GE Optima/Discovery MR450 & MR750

- Identify the three main cabinets of the Discovery MR system (PGR, HEC, Pen)
- Identify the three optional cabinets (8kW AMT MNS, Brainwave Cabinet, Focused Ultrasound)
- Preview the Cooling Loop
- Locate and identify components in the PGR
- Locate and identify components in the HEC
- Locate and identify components in the Pen cabinet
- Locate and identify components in the GOC console

Discovery MR750/450 Cabinets

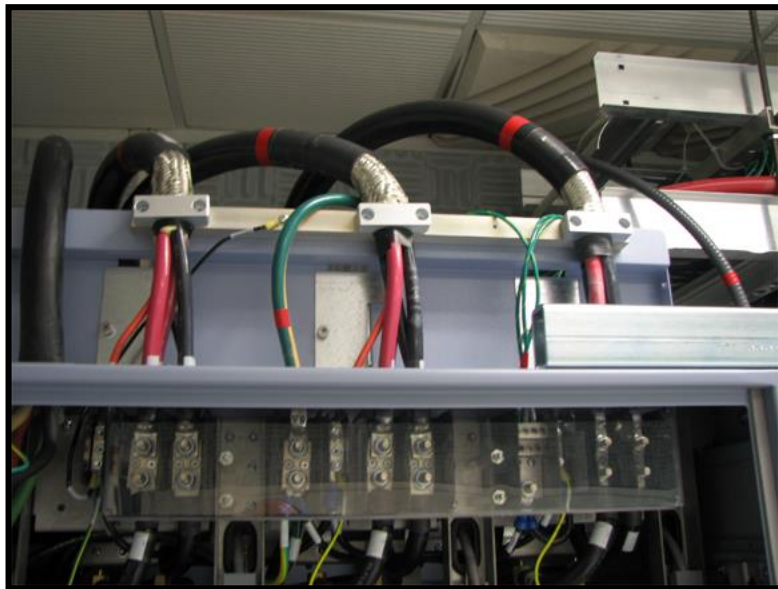


- All cabinets, with exception of Pen Cabinet, are front access ONLY.
- All cabling penetrates through top of cabinets.
- Cabling in Equipment and Scan Rooms is routed overhead in dual cable troughs.
- All Discovery cabinets are sited on sub-floor; not sited on raised flooring.
- All cabinets are stationary and immobile; not mounted on casters.

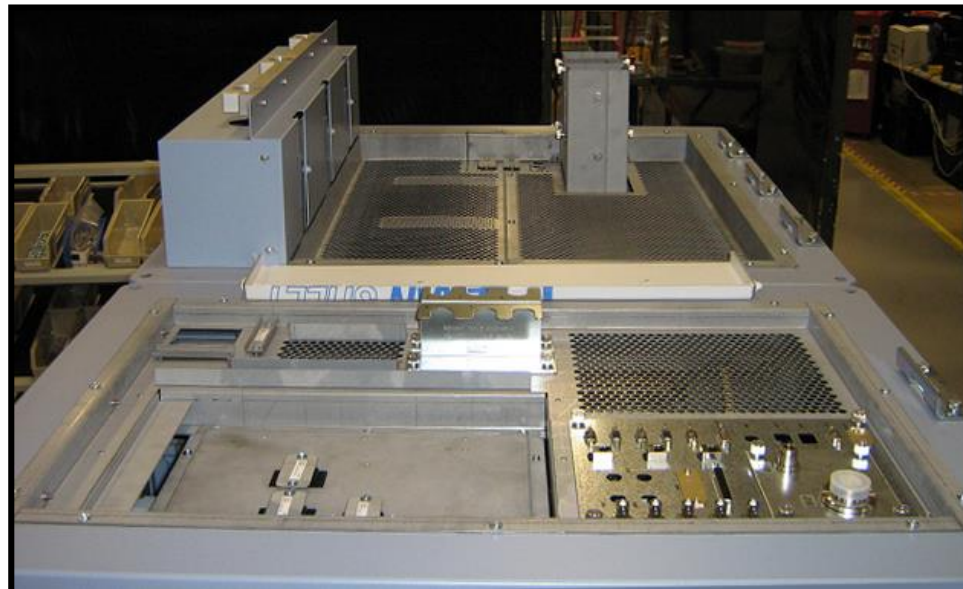
- Standard system contains 3 cabinets:
 - Power Gradient RF (PGR) Cabinet
 - Heat Exchanger Cabinet (HEC)
 - Penetration Cabinet (Pen Cabinet)

- Optional cabinets are:
 - 8kW AMT MNS Cabinet (same as used w/G3)
 - Brainwave Cabinet (same as currently used)
 - Focused UltraSound (FUS – same as currently used)

- All cables penetrate cabinet from top.



This is a Left front view showing gradient cables



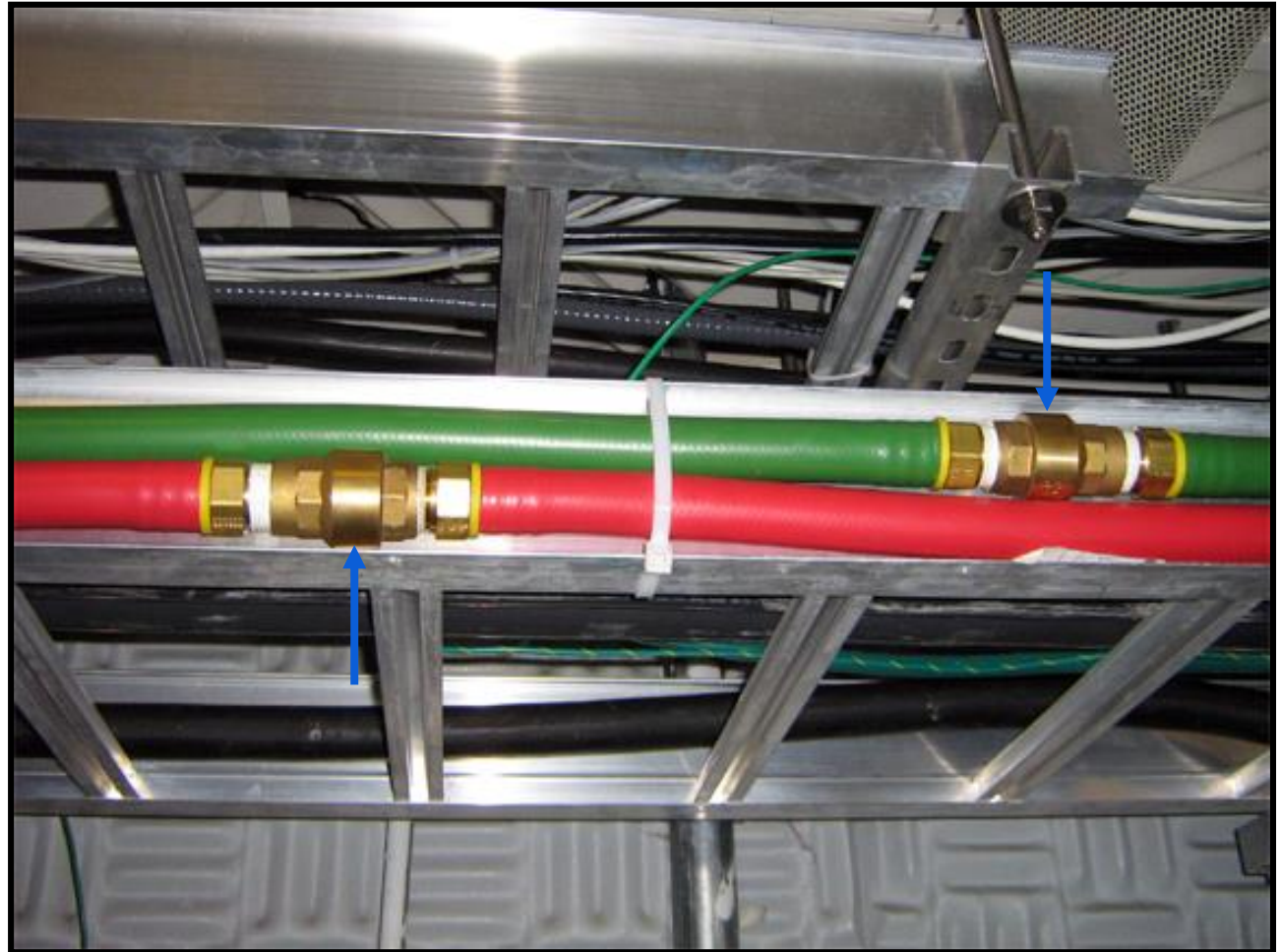
The Right top view shows the Interface panels for RF, data, and options connections

- 2 troughs allow the cables & hoses to be segregated



PHILIPS Overhead Cable Troughs – Equip. Room

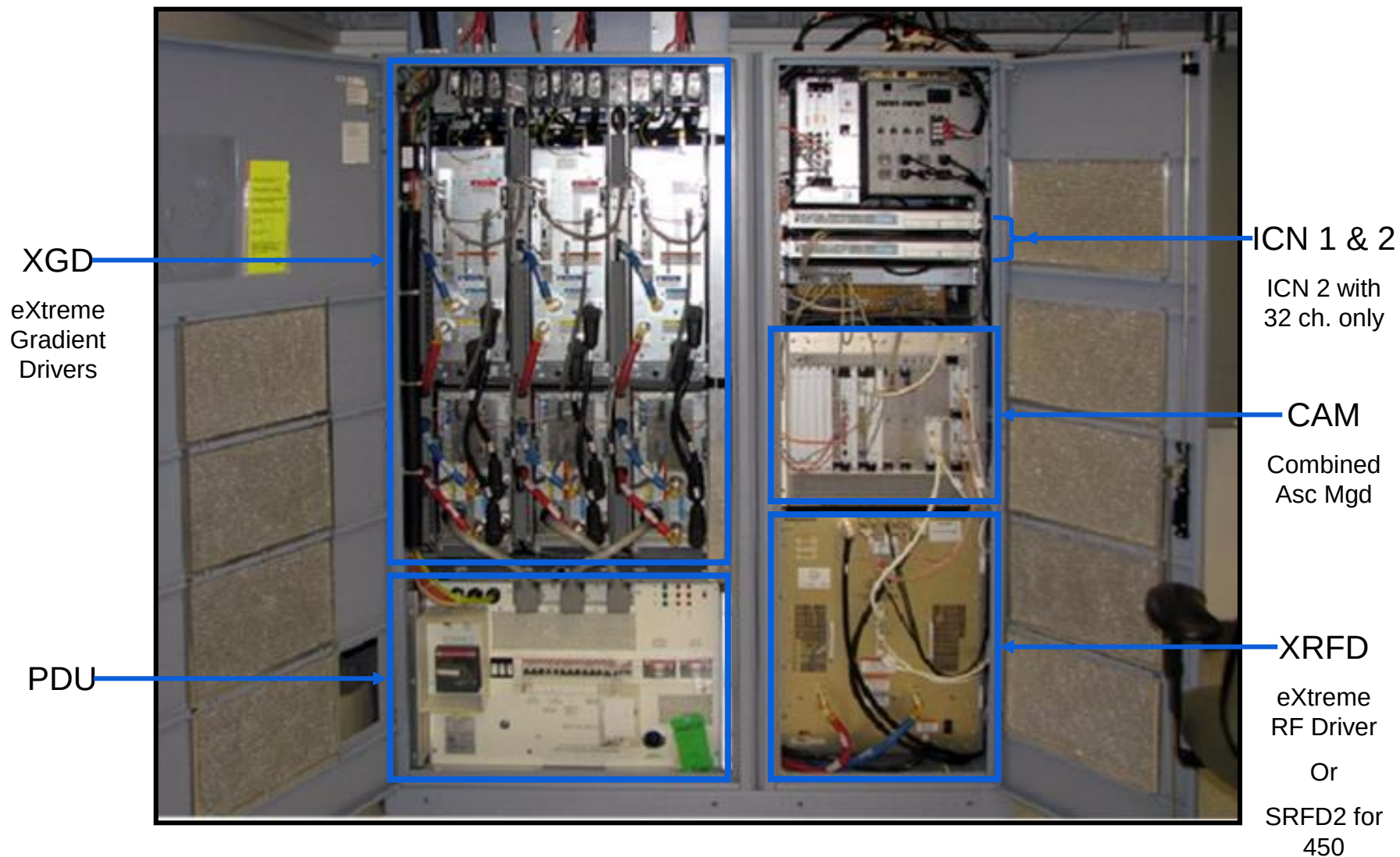
- Brass flow check valves installed in PGR Cabinet coolant line circuit and XRMB coolant line circuit to limit backflow during service to HEC



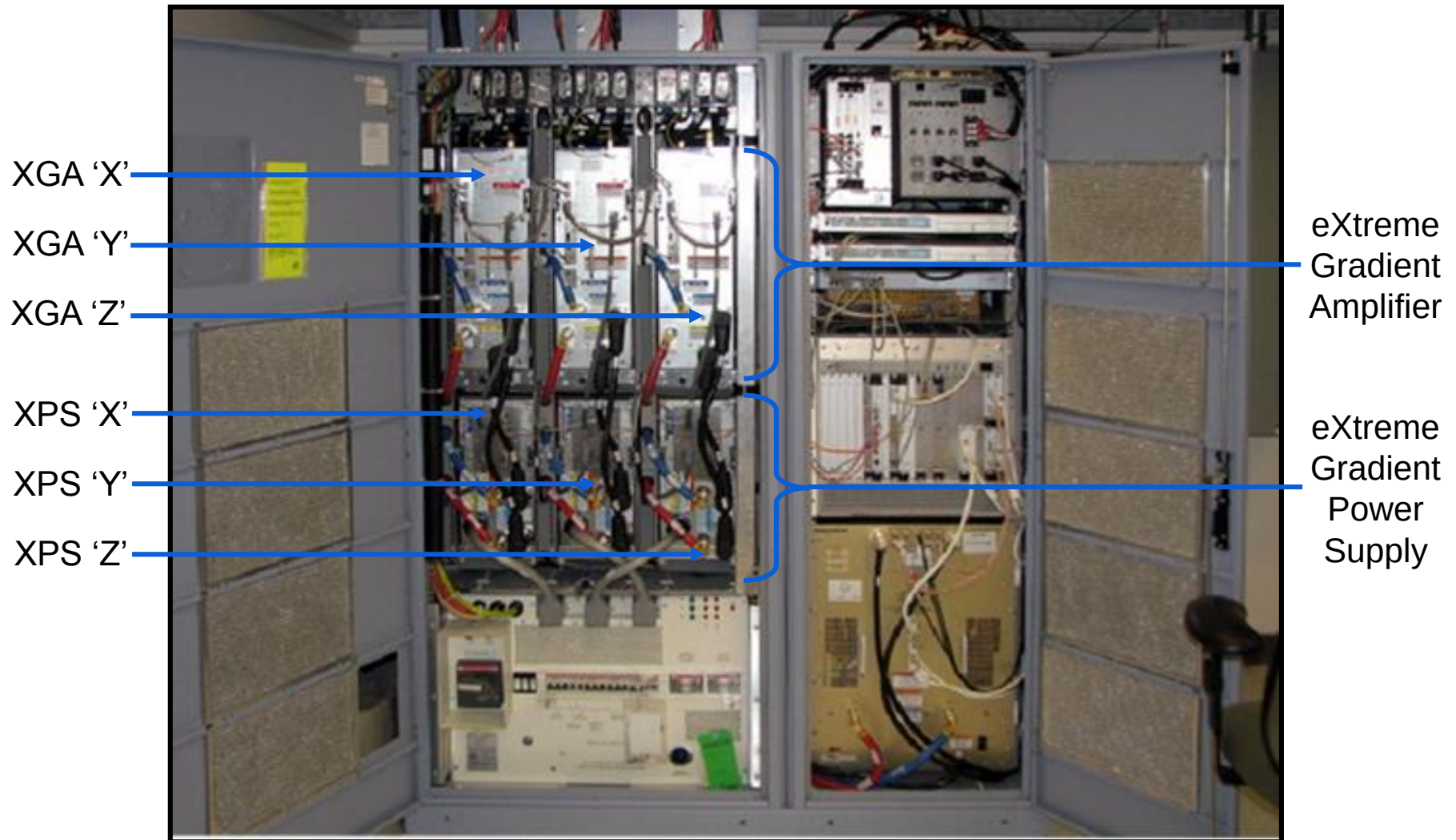


The cabinet group shown is an example of the PGR. The new design allows for a much more streamlined installation. In older models the cables were routed through raised flooring.

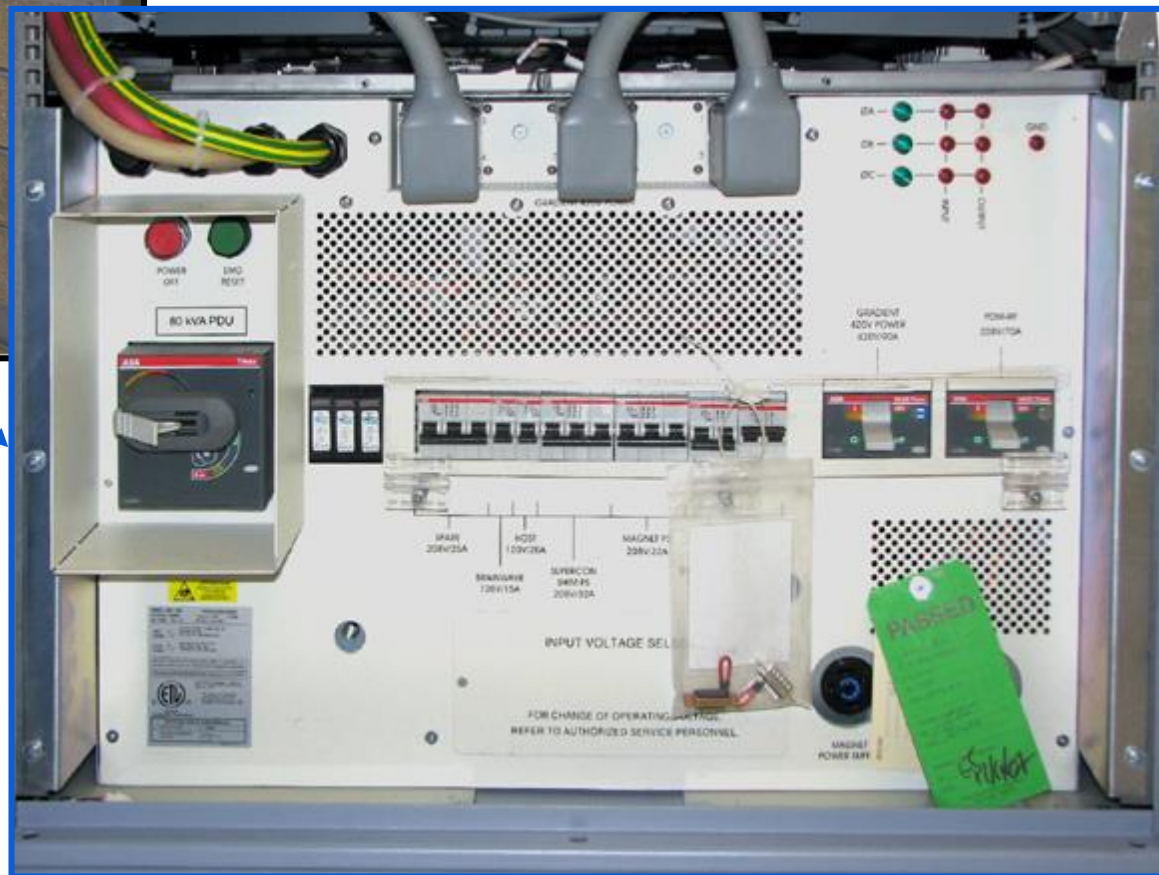
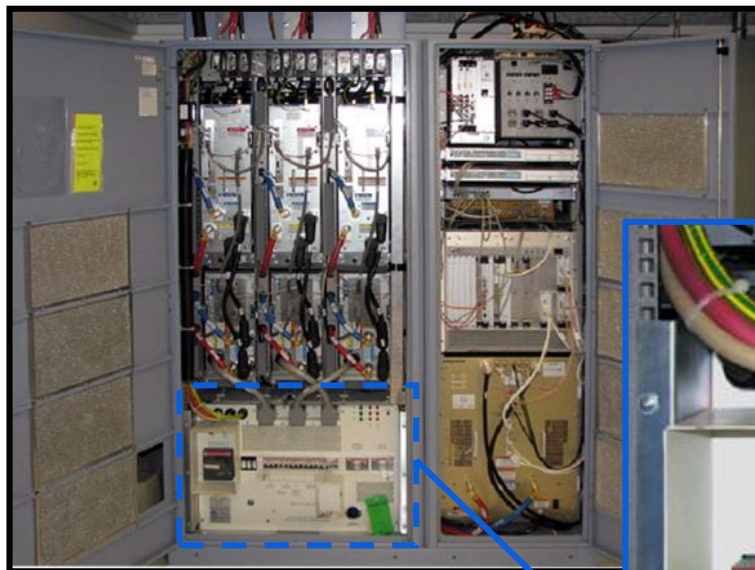
PGR Cabinet Contents



PGR Cabinet Contents - XGD



PGR Cabinet Contents - PDU

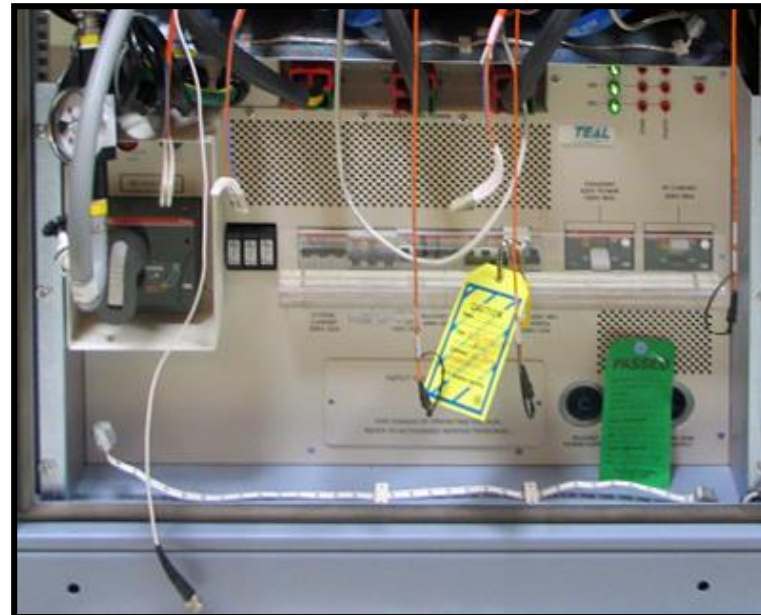


PDU

Power Distribution Unit (PDU)

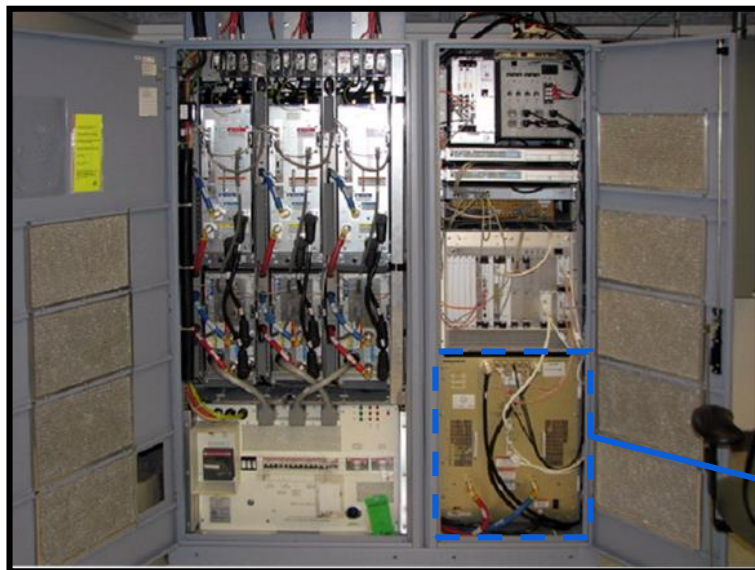
- The PDU consists of a transformer with a multi-tapped primary and two secondary's, a filter and a load center with circuit breakers to distribute power to the various peripherals in the MR system. It also provides the facility for single point grounding of the MR system. Service and FRU's are accessed from the front.
- The MDP and the PDU are now a matched set which translates to a PDU that can handle fewer voltage options. (See Table Below)

PDU Voltage Options
480
415
400
380



PGR Cabinet Contents – XRFD*

*for initial roll-out, the 450 used a modified SRFD2

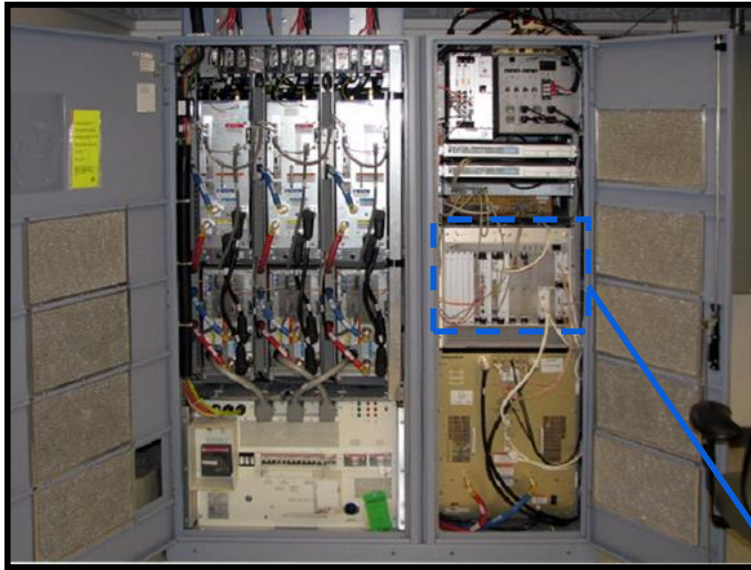


XRF Power Supply



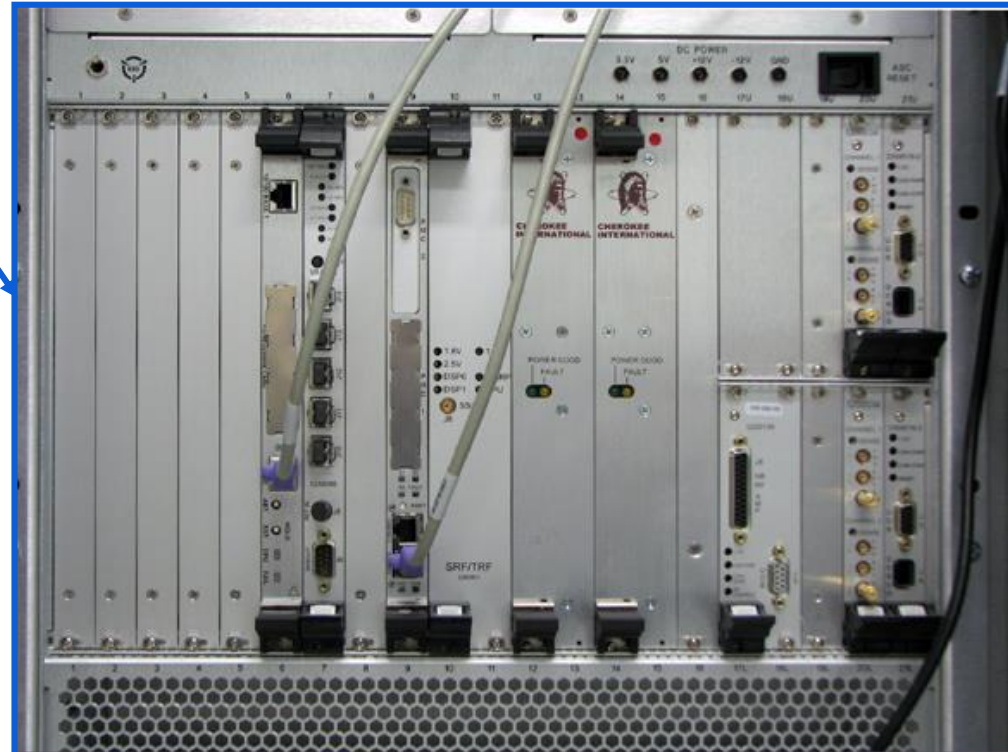
eXtreme RF (XRF) Driver

PHILIPS PGR Cabinet Contents – CAM Chassis



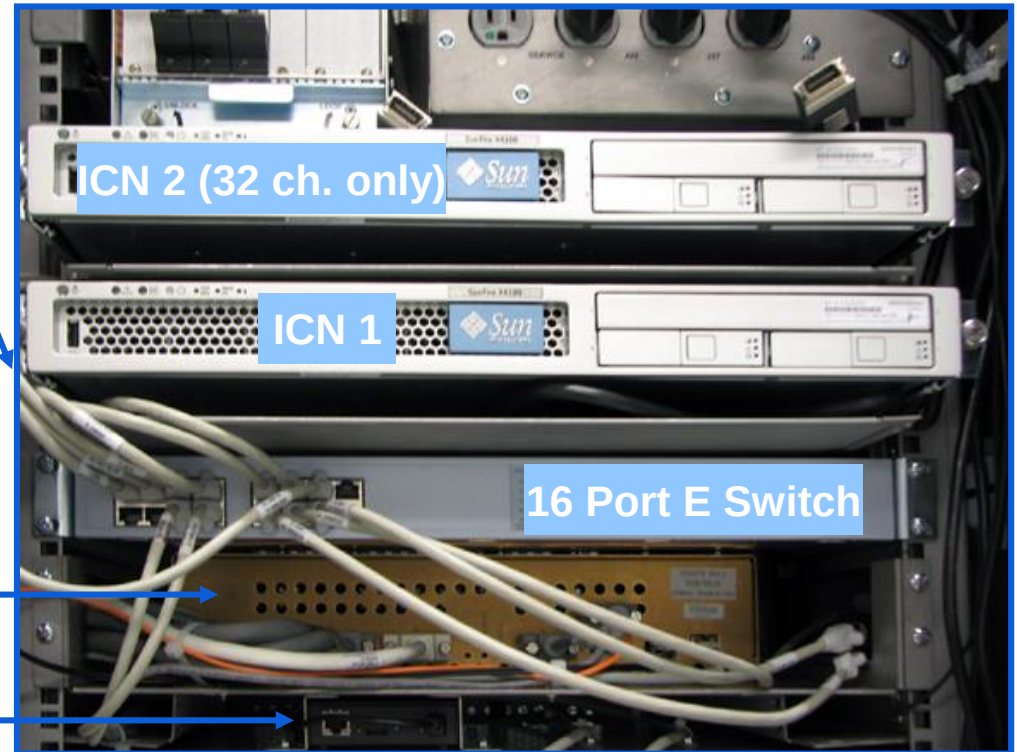
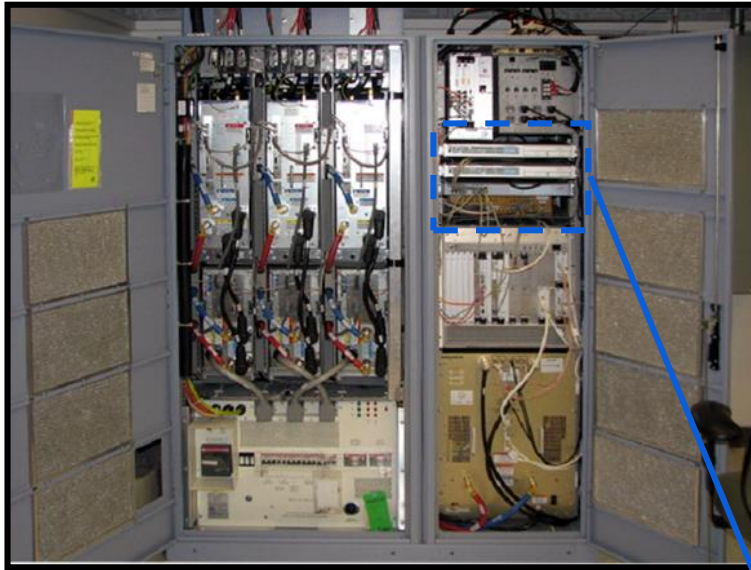
- CAM contains:
 - IRF3 Board
 - Bridge Board minus LED's, Older rev had LED's

- The CAM does not contain:
 - APS Board
 - Infiniband Board
- Both of which are used on the HDxt



**Combined Asc Mgd (CAM)
Chassis**

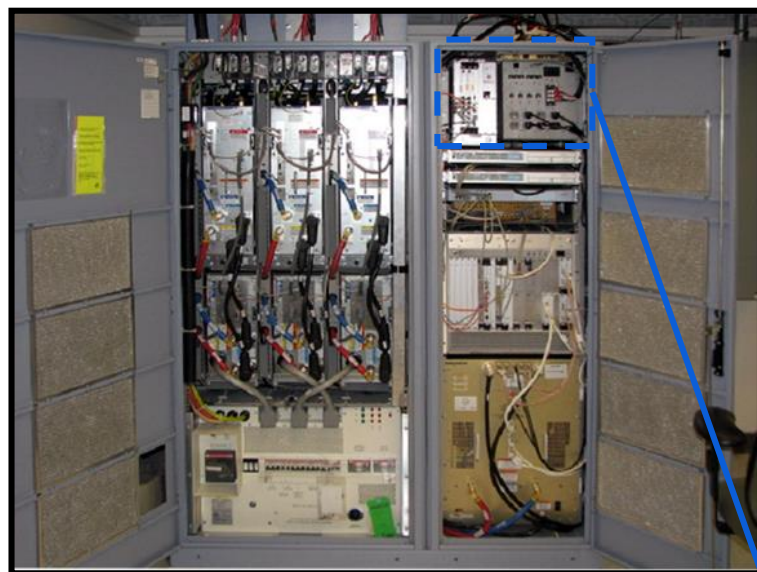
PHILIPS PGR Cabinet Contents – Recon, Comm.



Auxiliary Module

Terminal Server

PHILIPS PGR Cabinet Contents – Cronus, PDM

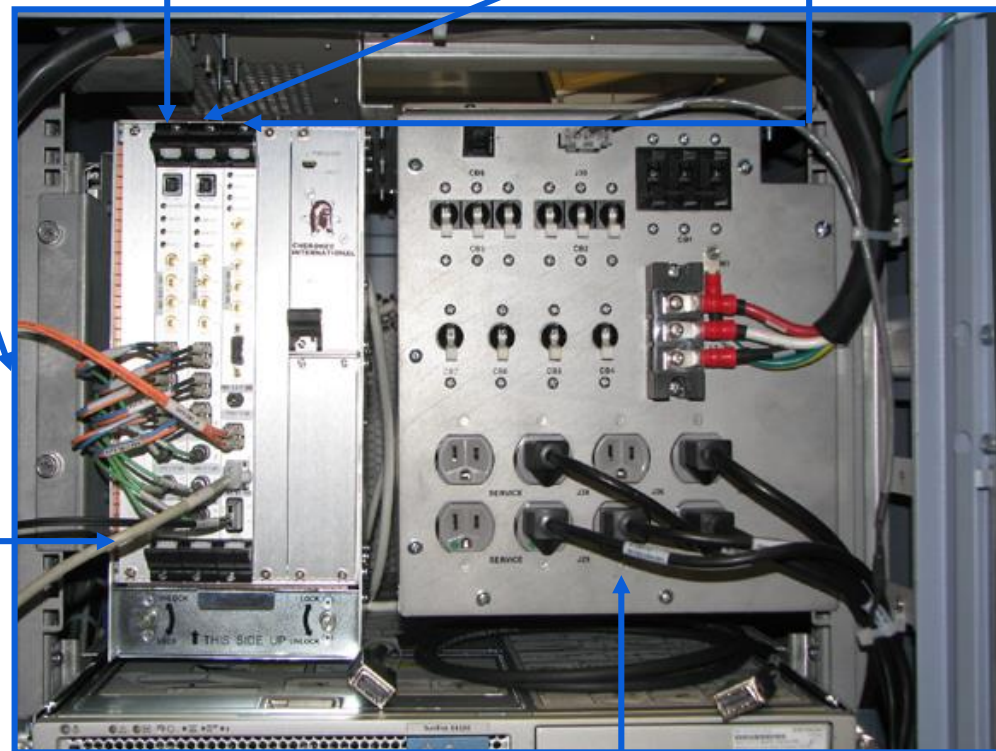


Gradient Master Controller (GMC) or “Cronus” Chassis

eXtreme Power Supply (XPS) Control Board

eXtreme Gradient Amplifier (XGA) Control Board

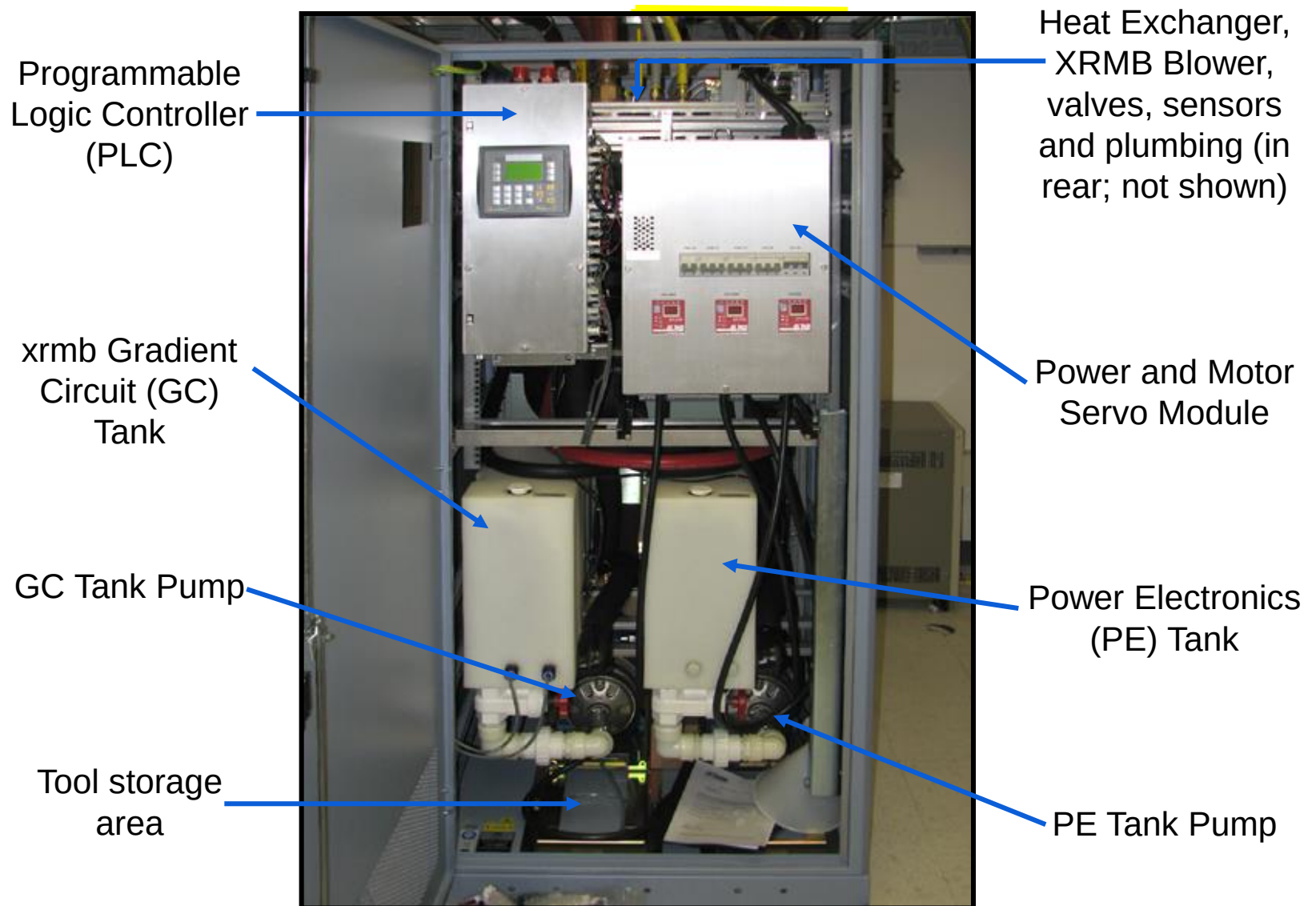
GMC Board

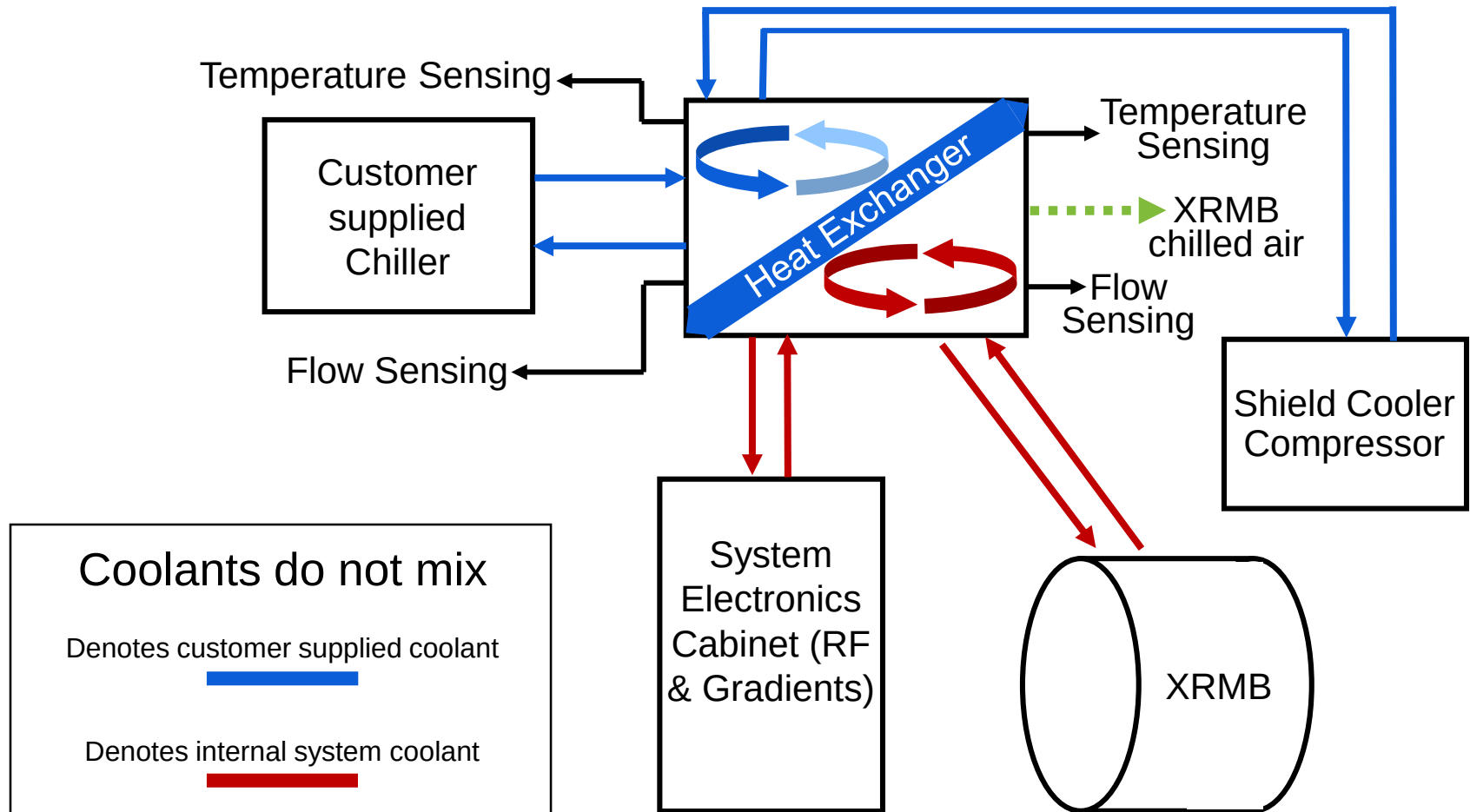


Power Distribution Module (PDM)

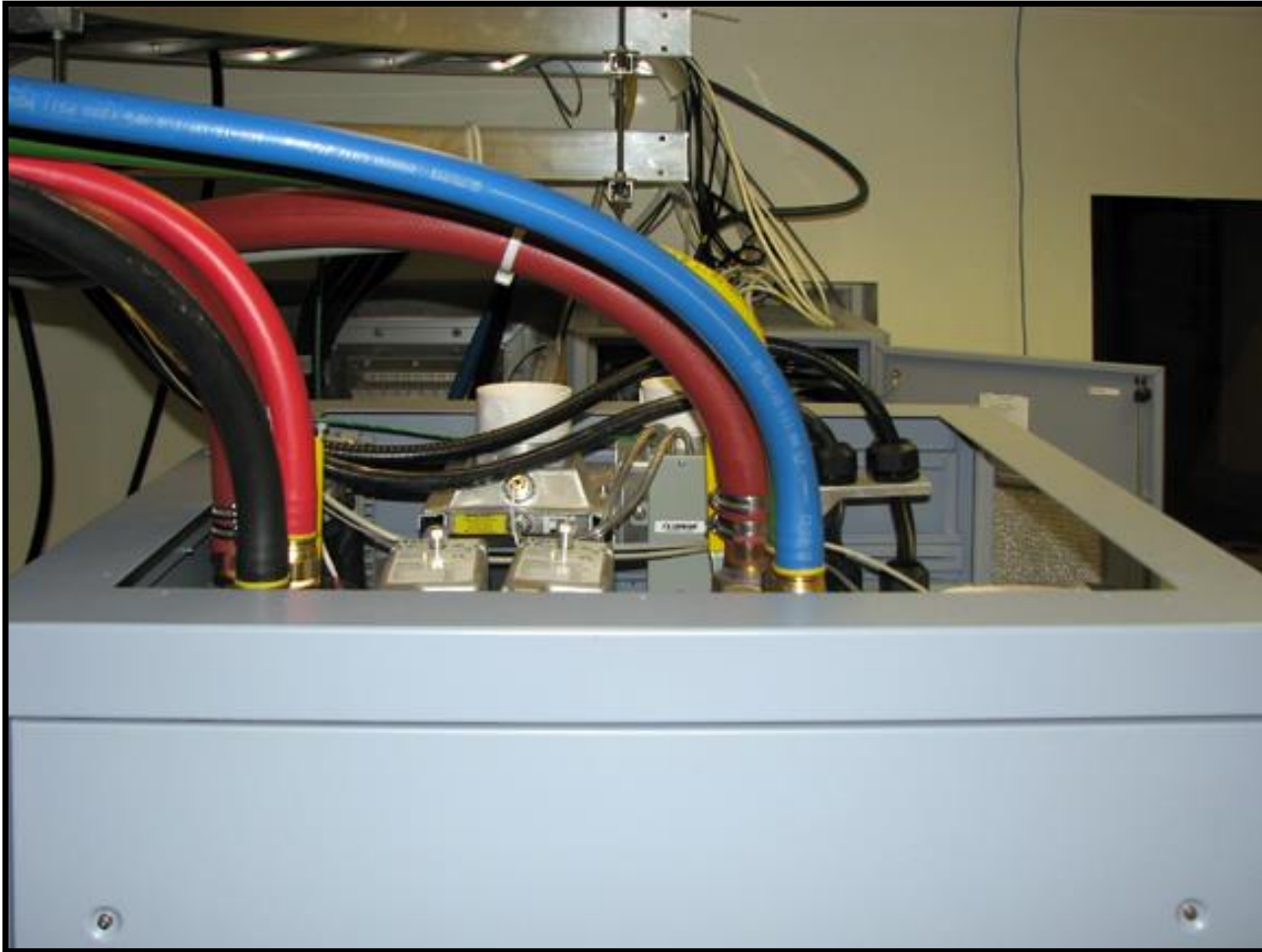
Heat Exchanger Cabinet (HEC)







- All cables and hoses penetrate from top.



Top view of HEC

- Bolt holes and space provided on either side of HEC for FE to affix MM3.



Penetration (Pen) Cabinet



Sited cabinet – Front view



Un-sited cabinet – Rear view

Pen Cabinet

- This cabinet is always sited adjacent to Scan Room wall.



Side View

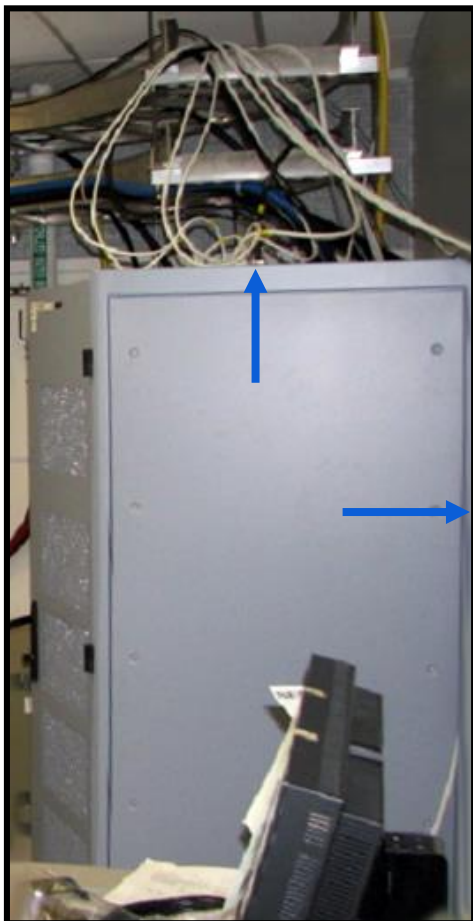


At installation
cabinet rear panel
is removed...



...and permanently affixed
to Scan Room wall to
create Penetration Panel

Pen Cabinet



Cables enter Pen Cabinet
from top and rear



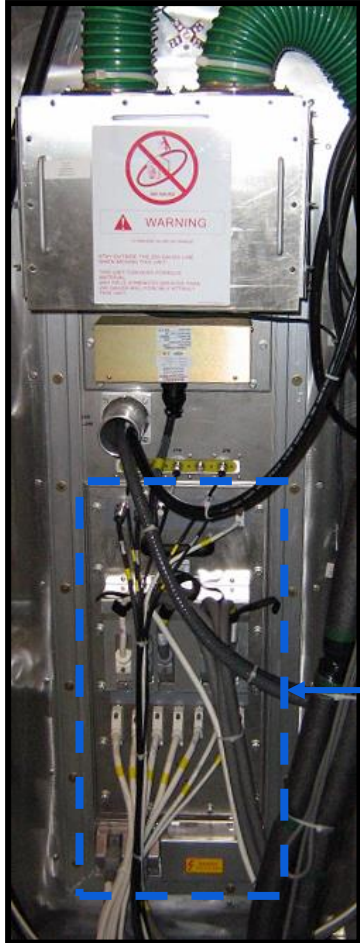
View through front of cabinet
to Penetration Panel



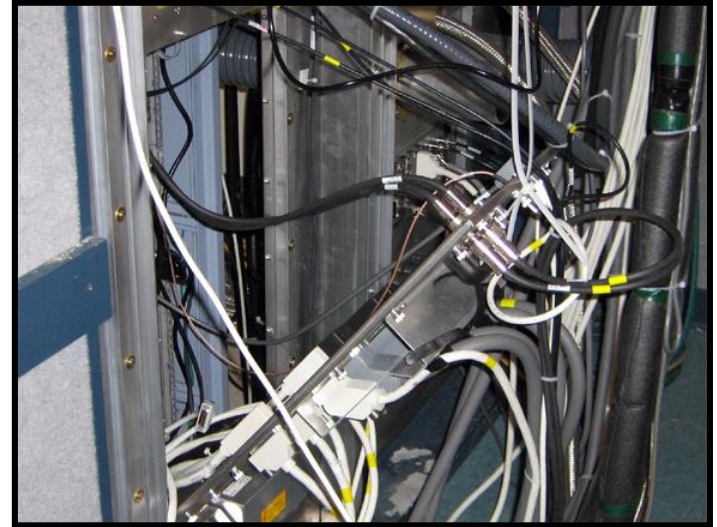
Scan Room view
of Pen Panel

Pen Cabinet

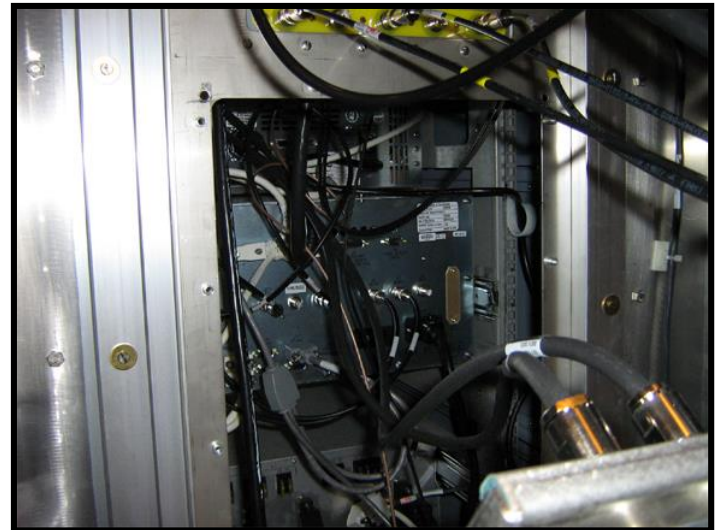
- Captured screws on panel
- Hand-hold on panel
- Remove or pivot out at base



Highlighted
removable service
access panel



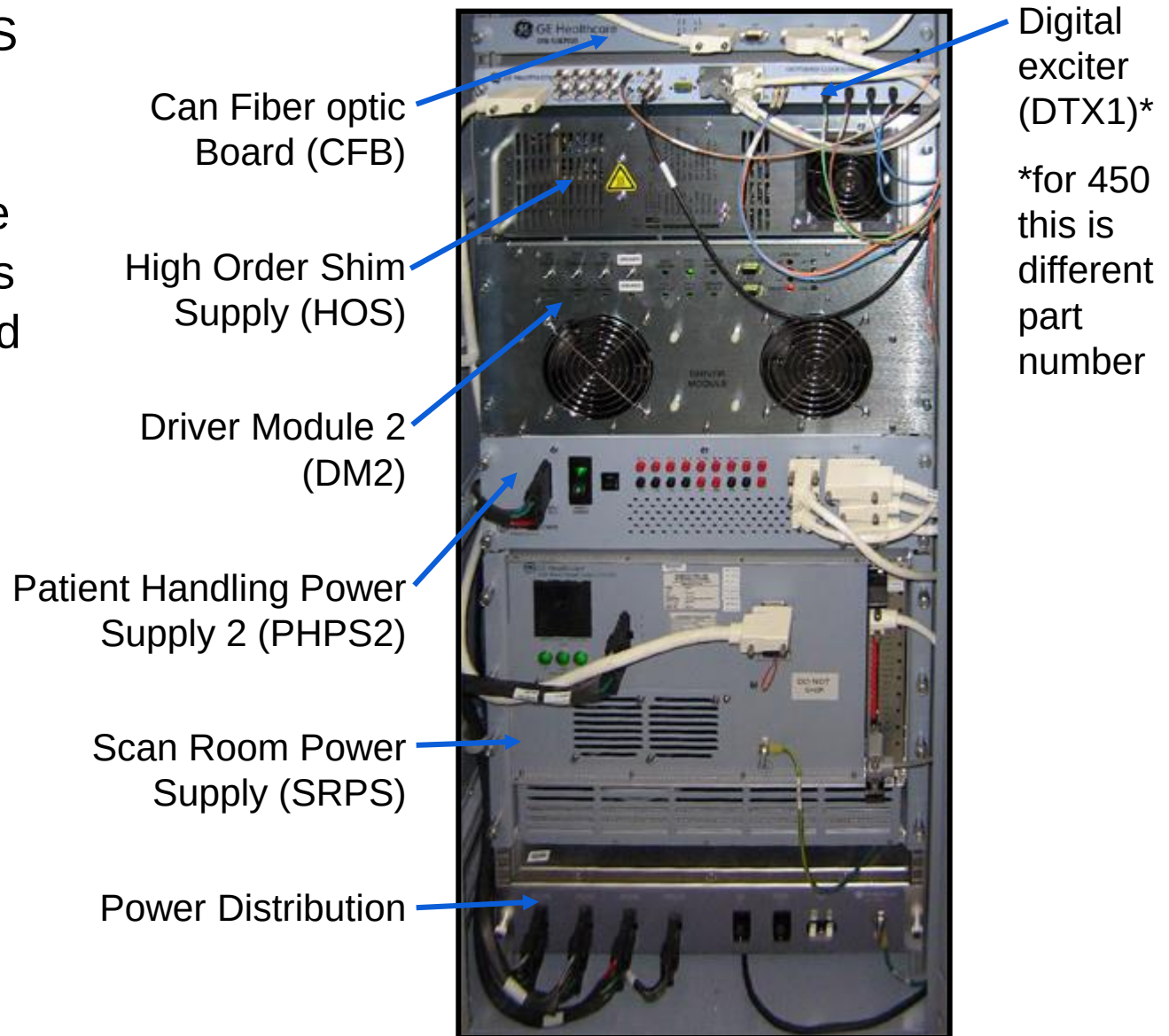
Side view- Service Access open



Front view- Service Access open

Pen Cabinet

- Sites with MNS option would have a DTX2 Exciter module (looks same as DTX1) installed above CFB.



Secondary Penetration Wall (SPW)



View from Scan Room

RF Transmit
Switch(s)



Gradient
Filter

For HEC chilled
air supply and
return for XRMB
and XRF Body
Coils

View from Equip. Room

Global Operator Console (GOC)



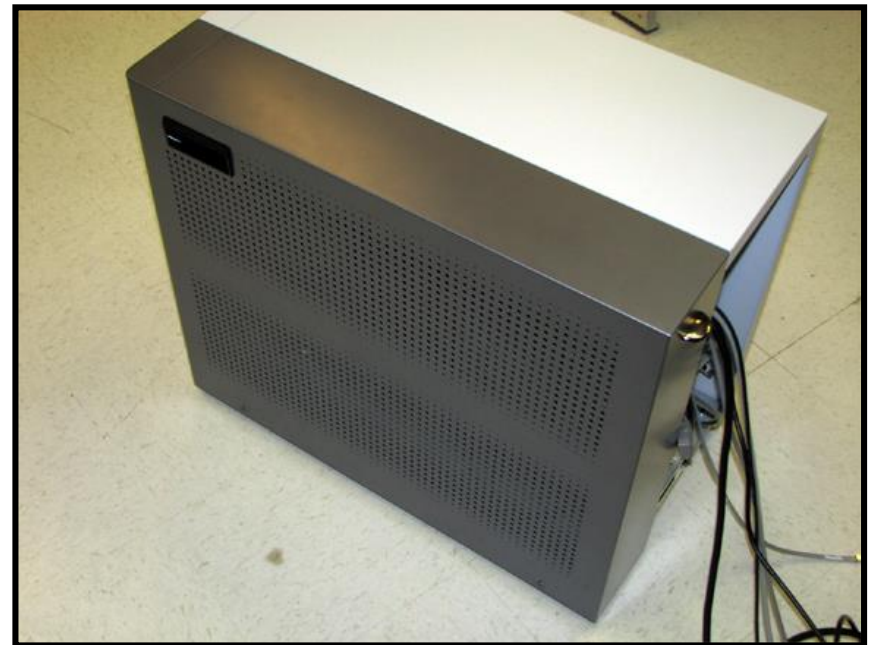


GOC open in rear; no cover plate.

Rear View

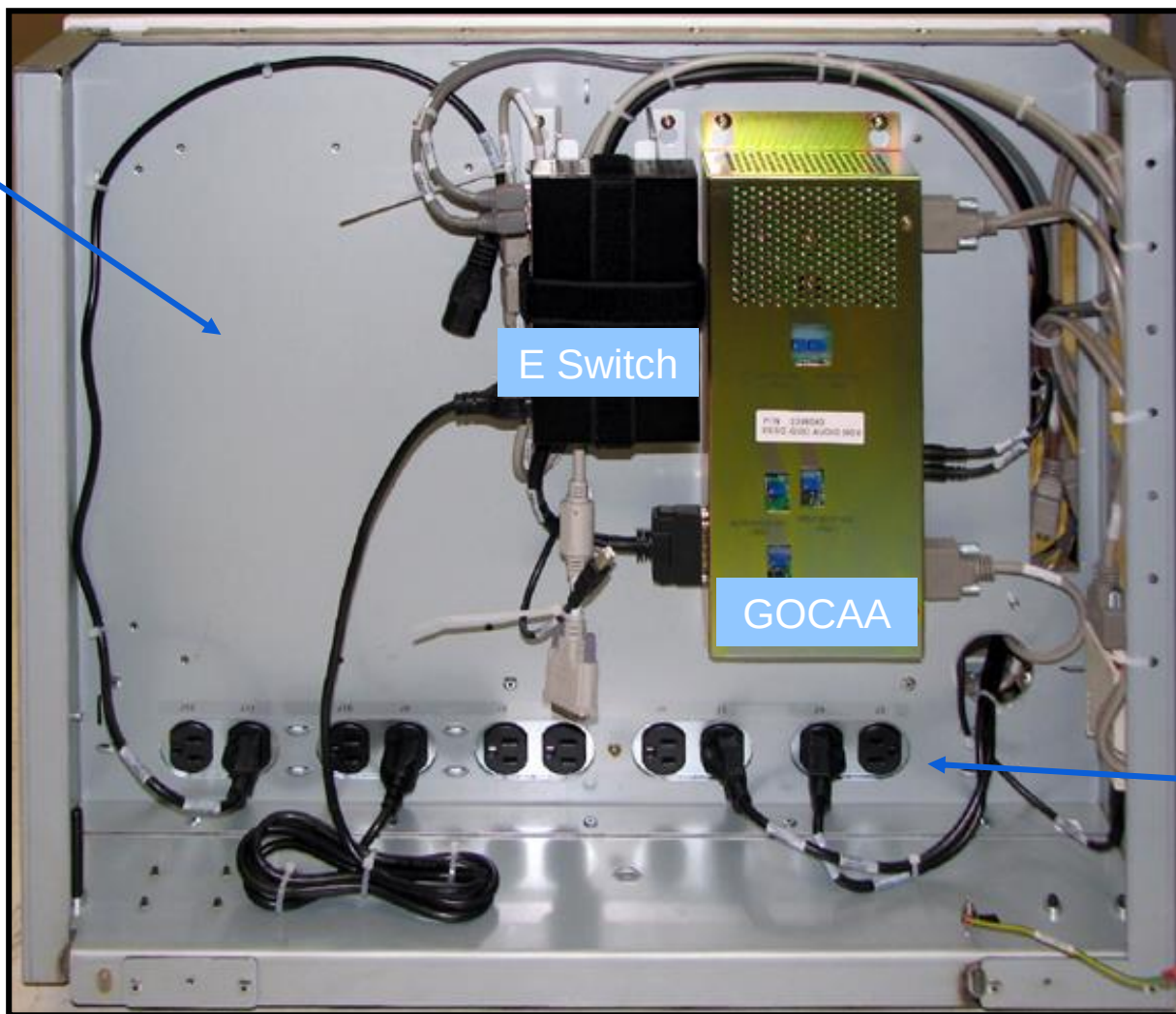


GOC CB1



Right Side View

Location for
optional
IRDM



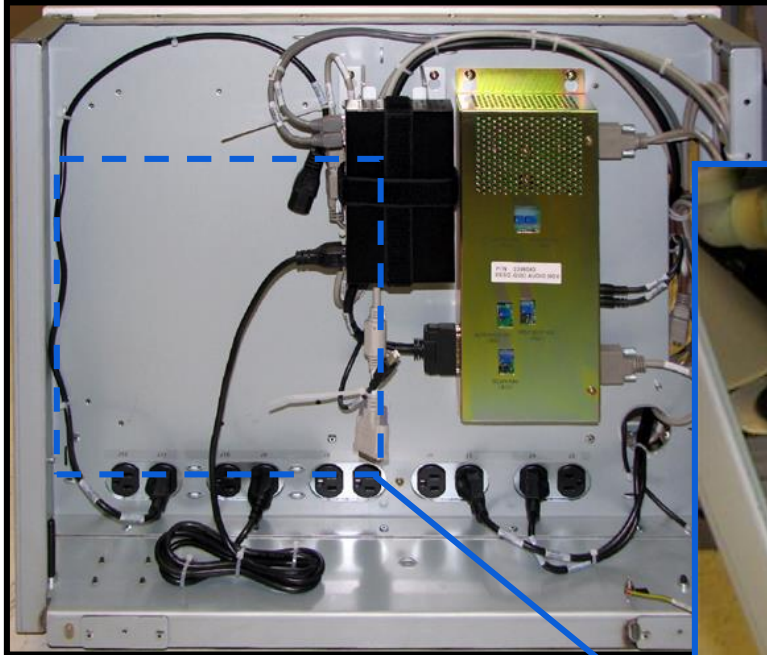
E Switch

GOCAA

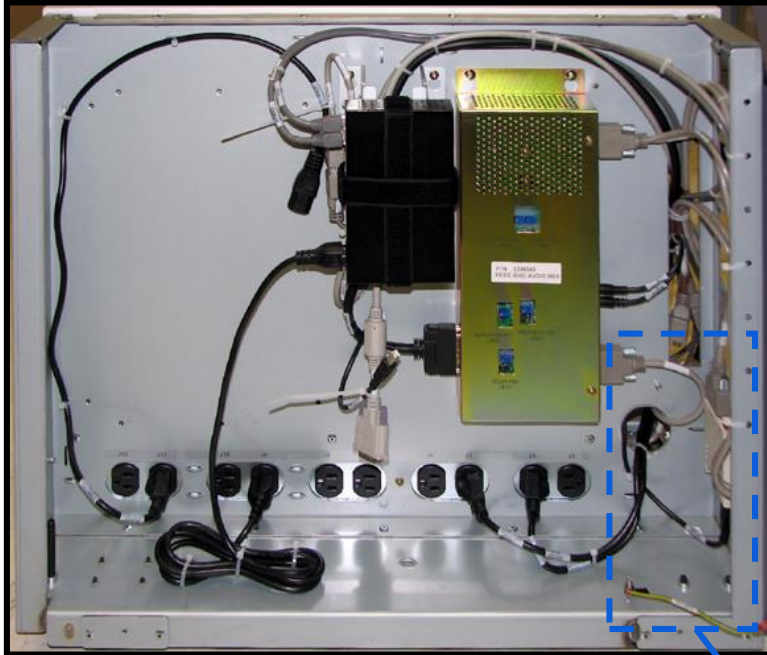
Power
distribution

View with right side cover removed.

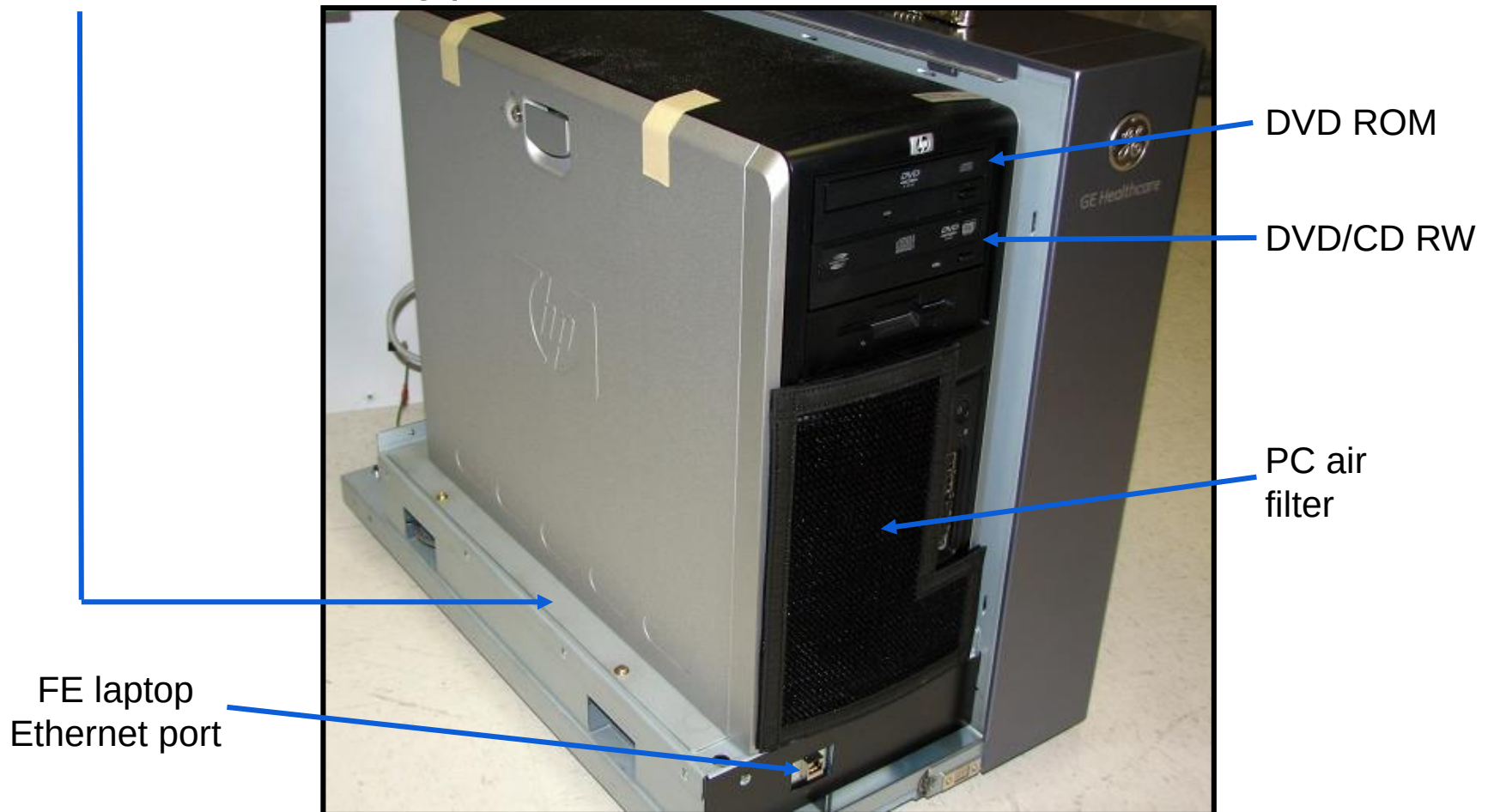
- Optional In-Room Display Module (IRDM) controller.



- USB to RS422 converter interconnects 1-Wire network between Host PC USB port to Master 1-Wire controller inside PHPS2.



- PC is secured to removable plate that attaches to floor of chassis. Removal is sliding plate with PC out of front of chassis.



View of PC with left side cover removed



- Customer OW Table is optional.
- Keyboard and mouse interconnect to Host through USB.





GE Optima/Discovery MR450 & MR750

Objectives

- Understand the hardware and signal differences between 16 and 32 channel systems
- Locate and identify electronic components in the scan room for RF, gradient, and table
- Identify location of magnet airflow sensors
- Identify major details appropriate for removal of components in the RF Hub / Rear Pedestal.

Overhead Cable Troughs

- 2 troughs
- Cables and hoses segregated



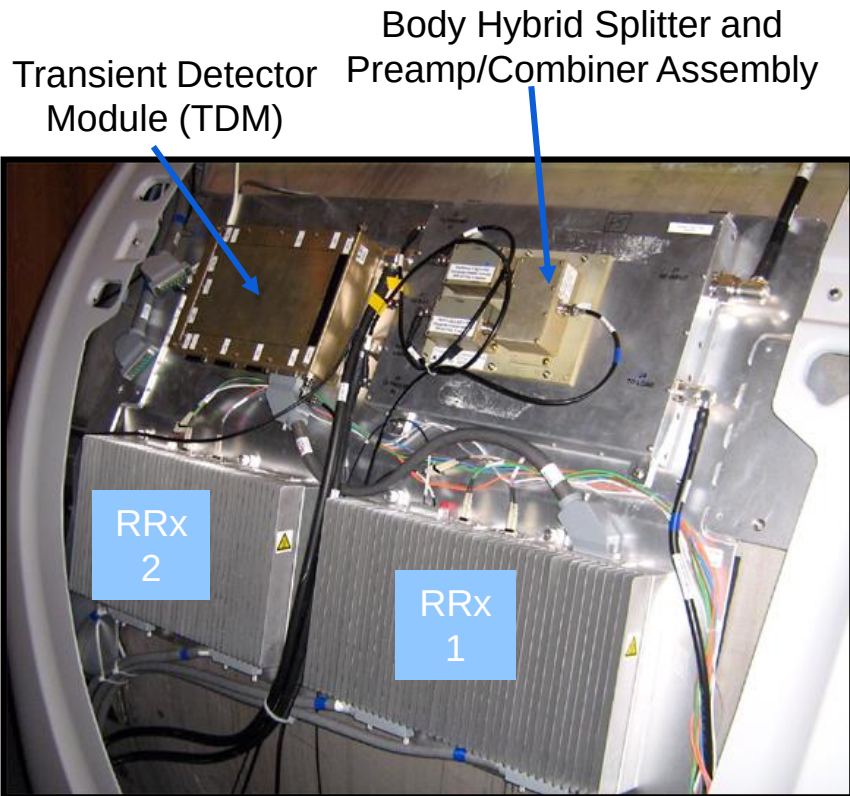
View from rear of magnet in scan room



View from front of magnet in scan room

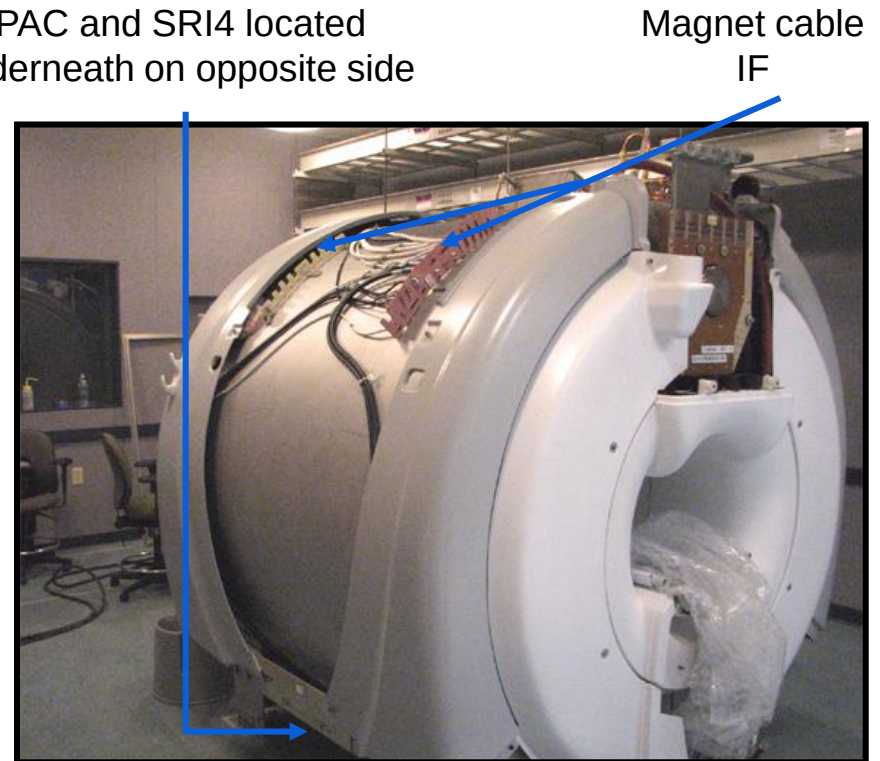
Forward Production Systems

- The magnet arrives from the magnet plant with all electronics pre-installed, except, if upgrade.



Left side from front

PAC and SRI4 located underneath on opposite side



Left side from rear

Magnet Upgrade

- Only the LCC300 magnet is supported
- During upgrade, all passive shim trays are removed and discarded prior to the installation of the gradient coil
- The gradient coil (XRMB) now contains the passive shims

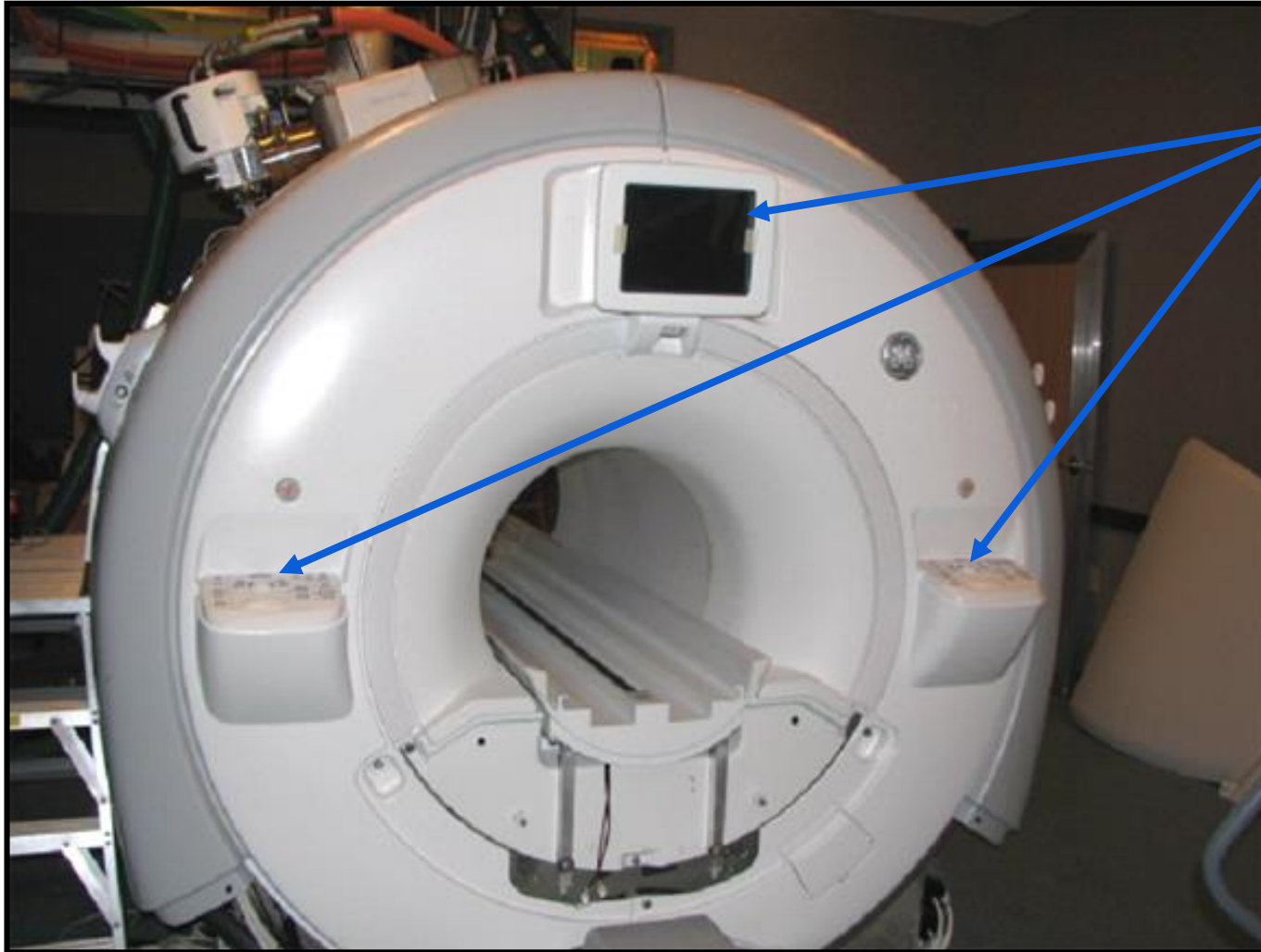


Magnet – 32-channel dock

- Dock connector – only present in 32-channel systems



Magnet Enclosure with IRM



- Shown with optional In-Room Monitor (IRM) and trackball/keypads installed
- For 16 channel system, the 7-segment display is standard (without IRM)

Magnet Enclosure with IRM



LCD Monitor



Trackball keypad

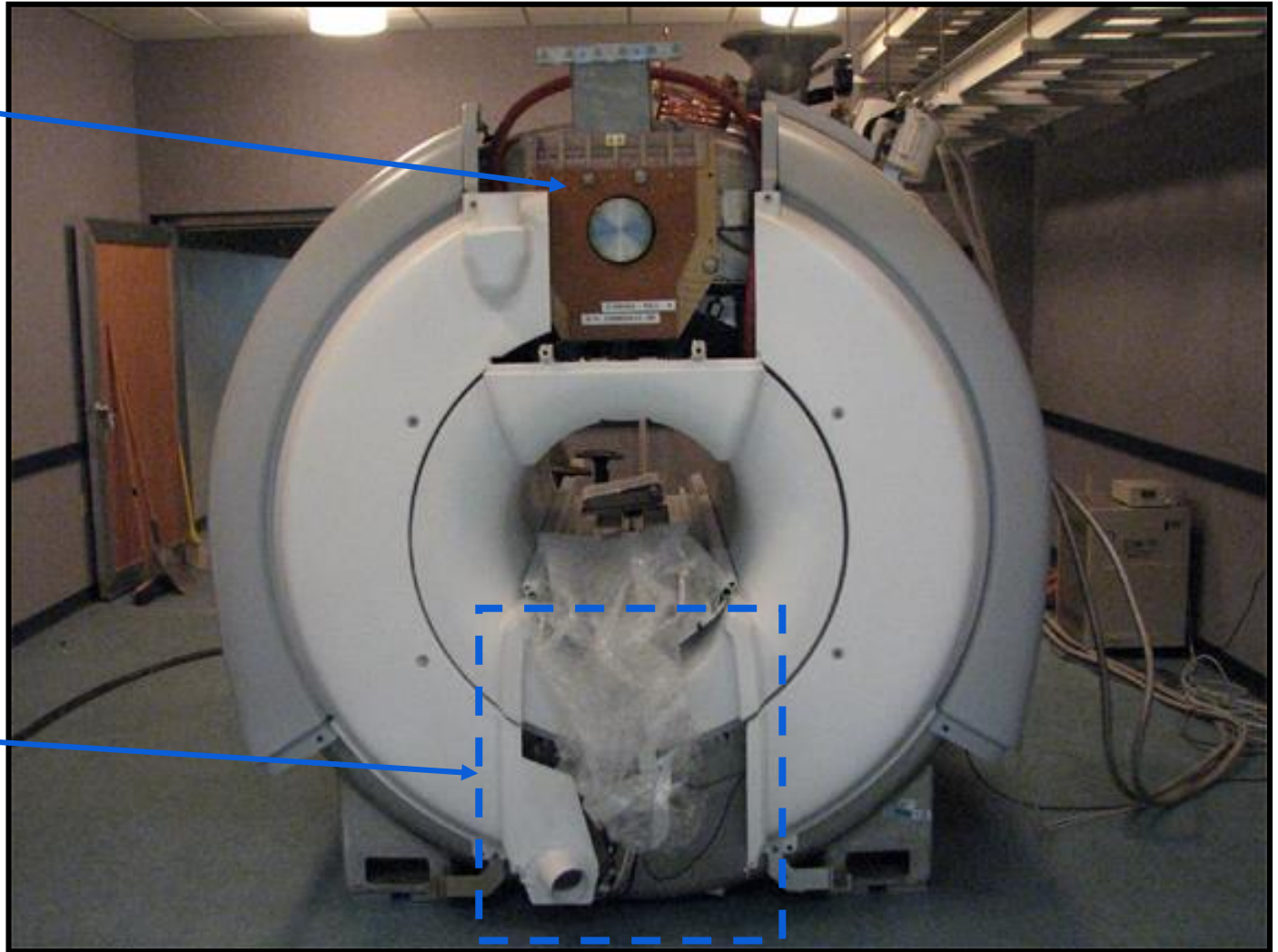
Features:

- Single Piece FRU
- Backlit buttons

Magnet Enclosure

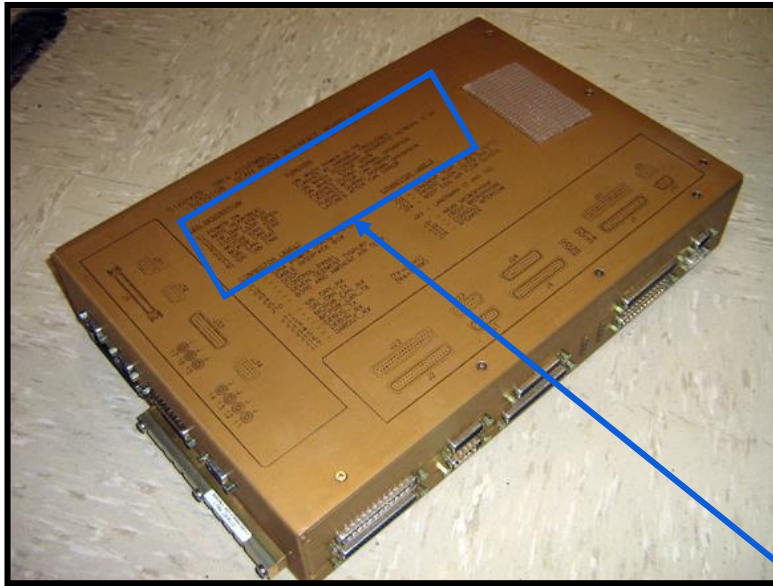
- XRMB Buss-bar Assembly

- Location for Rear Pedestal



Magnet Enclosure

- The 16-channel system may be equipped with IRM and trackball keypads but these are not standard
- All 32-channel systems are equipped with IRM and trackball keypad
- Standard Magnet Enclosure equipped with existing, HDx style, 7-segment, 4-digit LED display for indicating landmark location
- Standard Magnet Enclosure may have a keypad without a trackball



• SRI4 – New features

- Contains Remote Intercom Board (RIB) for patient communication.
- Does not manage Coil ID; this done by RFCB inside RF Hub.
- No DIP switches to configure.
- No internal access necessary.

SRI4 – Familiar features

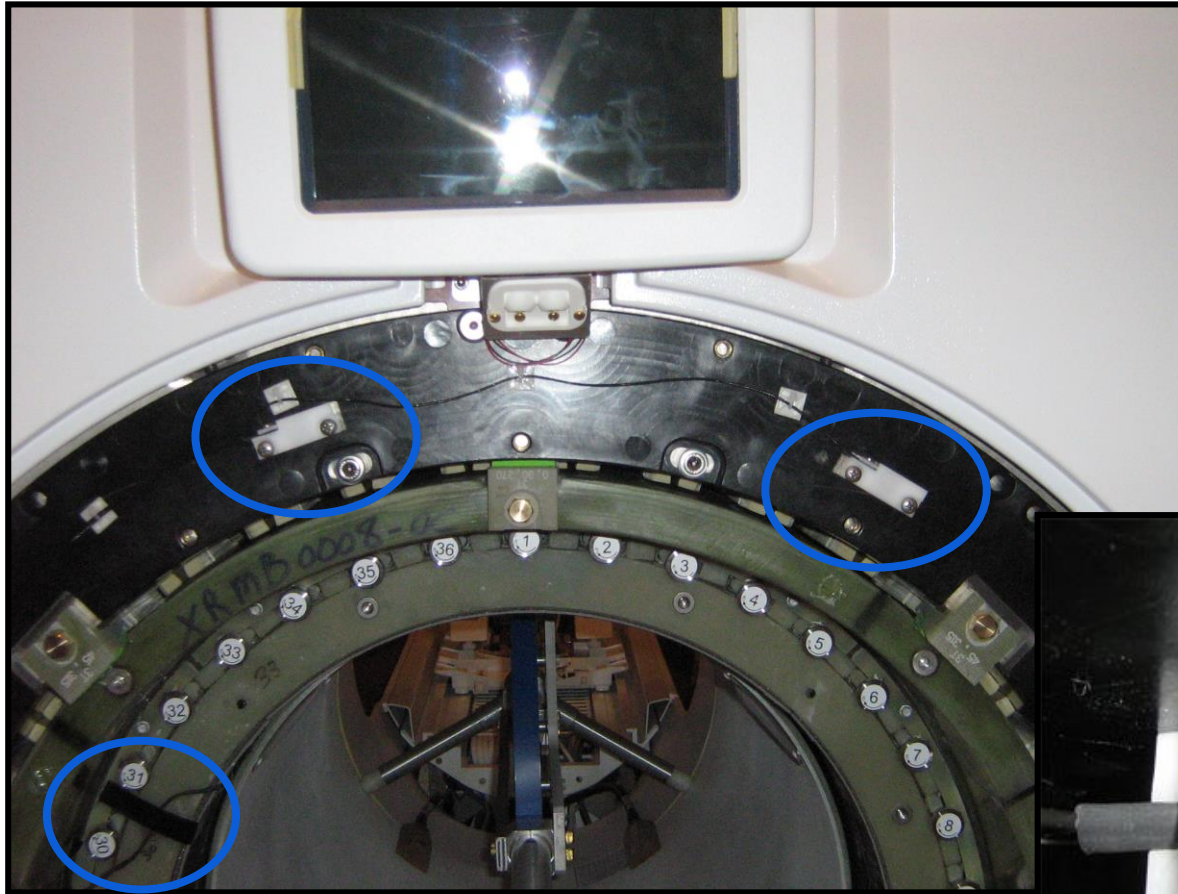
- Similar form and fit to SRI3
- Still located under the magnet on the shelf with the PAC
- Provides enclosure 7-segment display and keypad IF for standard magnet enclosure ONLY

<u>LED DESCRIPTION</u>	<u>FUNCTION</u>
1) POWER OK	ON WHEN POWER IS ON
2) NIOS HEARTBEAT	FLASHES AT VARIABLE FREQUENCY
3) 1W DIAG STATUS	ON WHEN SECONDARY DIAGNOSTIC NETWORK IS OK
4) ESTOP LOOP OPEN	ON WHEN ESTOP OPEN
5) MCPDB CAN STAT	FLASHES DURING NORMAL OPERATION
6) MCPDB CAN ERR	LATCHES AFTER CAN ERROR
7) SRI CAN STAT	FLASHES DURING NORMAL OPERATION
8) SRI CAN ERR	LATCHES AFTER CAN ERROR

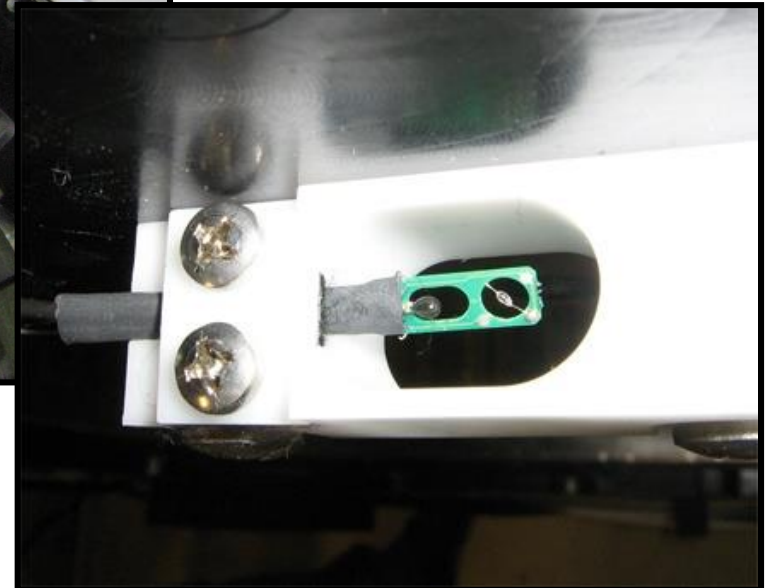
LED Descriptions

Patient Handling

Air Flow Sensors and Bore Temp



- Both the air flow sensors and the bore temperature are connected to SRI4. Air flow connected through a (silver) transducer box.



Patient Handling Ambient Air Temp Sensor



- Temperature from this sensor is reported to the customer in degrees Celsius in the small SAR window at bottom of scanning area
- The sensor is connected to SRI4

PHILIPS 32-Channel Patient Transport Table



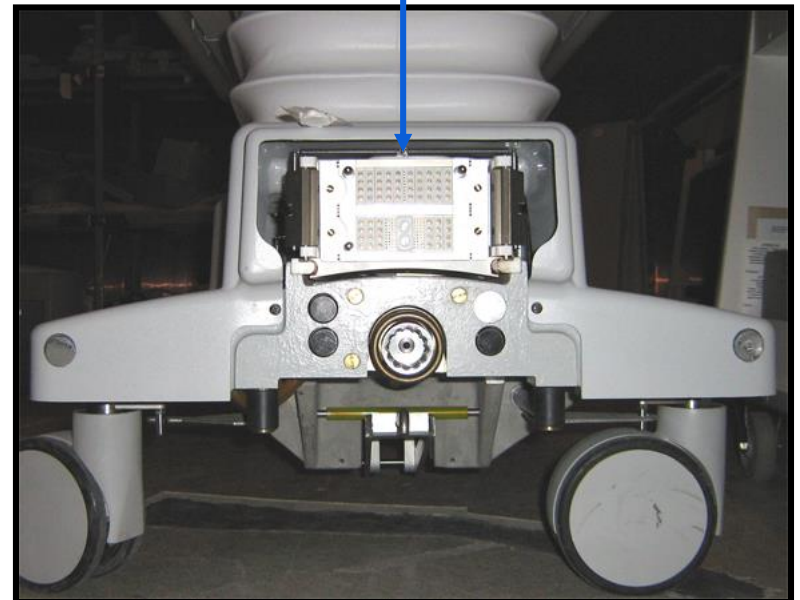
32-Channel Patient Transport Table

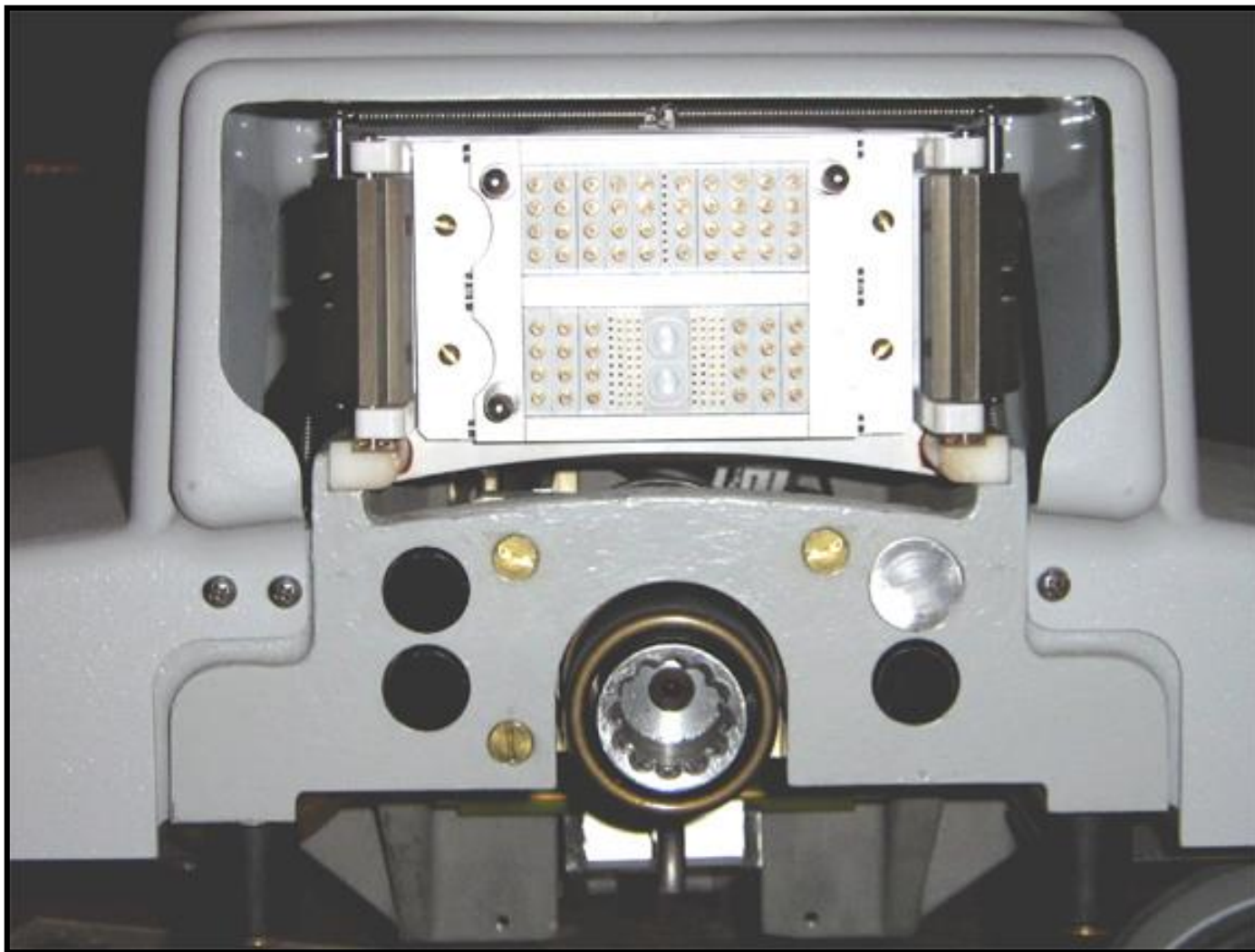


32-channel dock and connector

The connector floats on four spring-tensioned stand-offs, thus permitting freedom for repeated proper alignment with the table connector

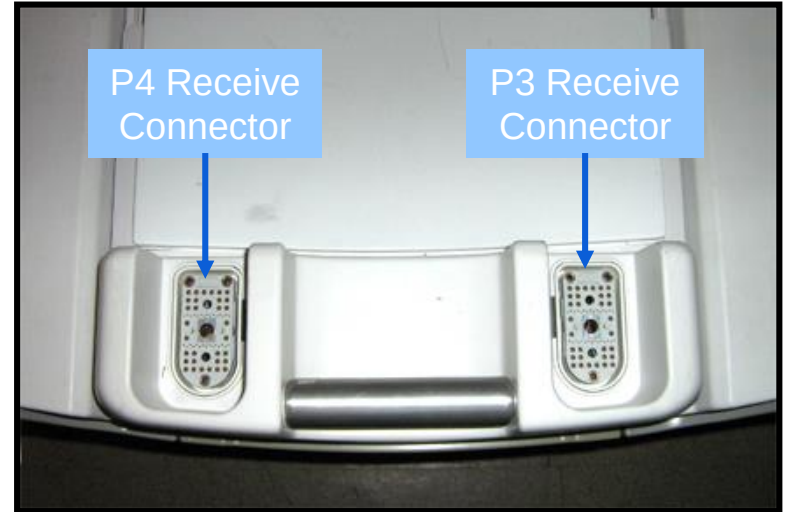
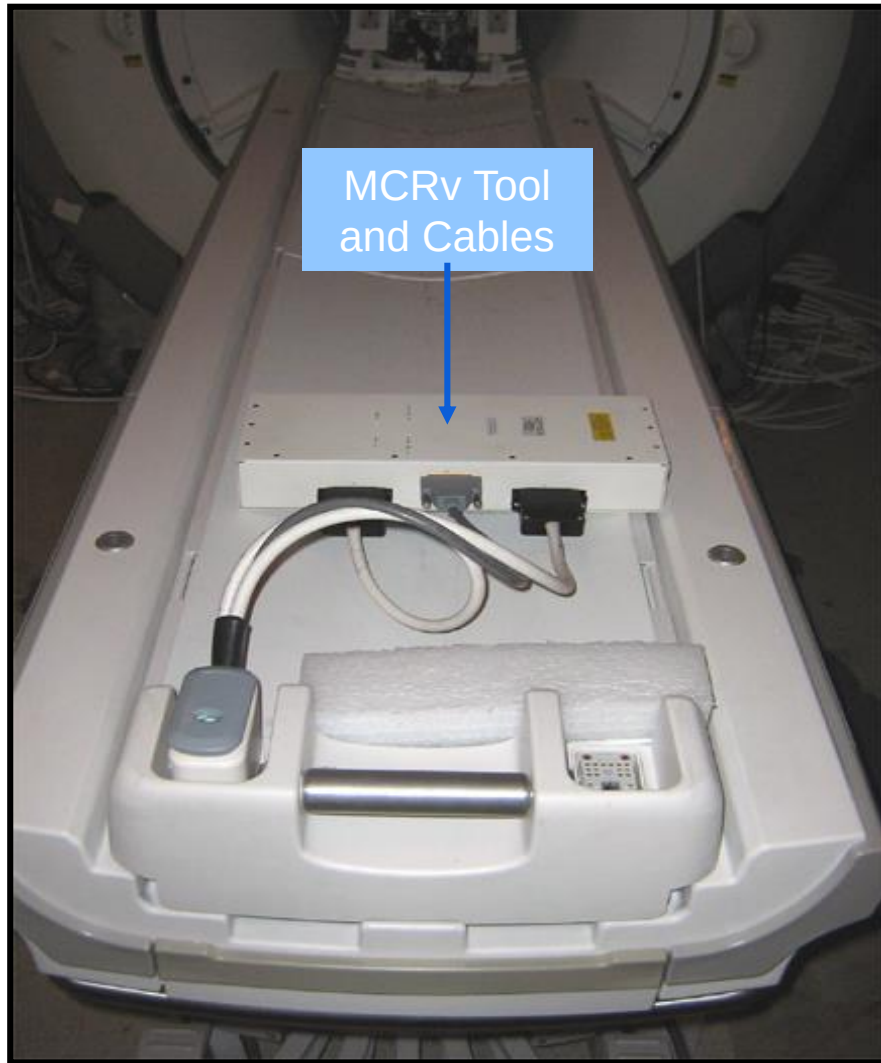
32-channel patient table dock connector



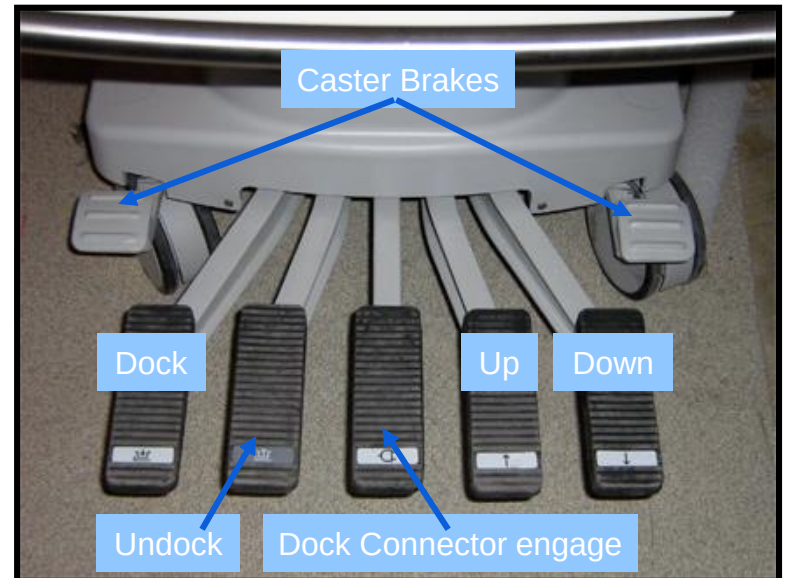


Close-up of retractable table dock connector

32-Channel Patient Transport Table

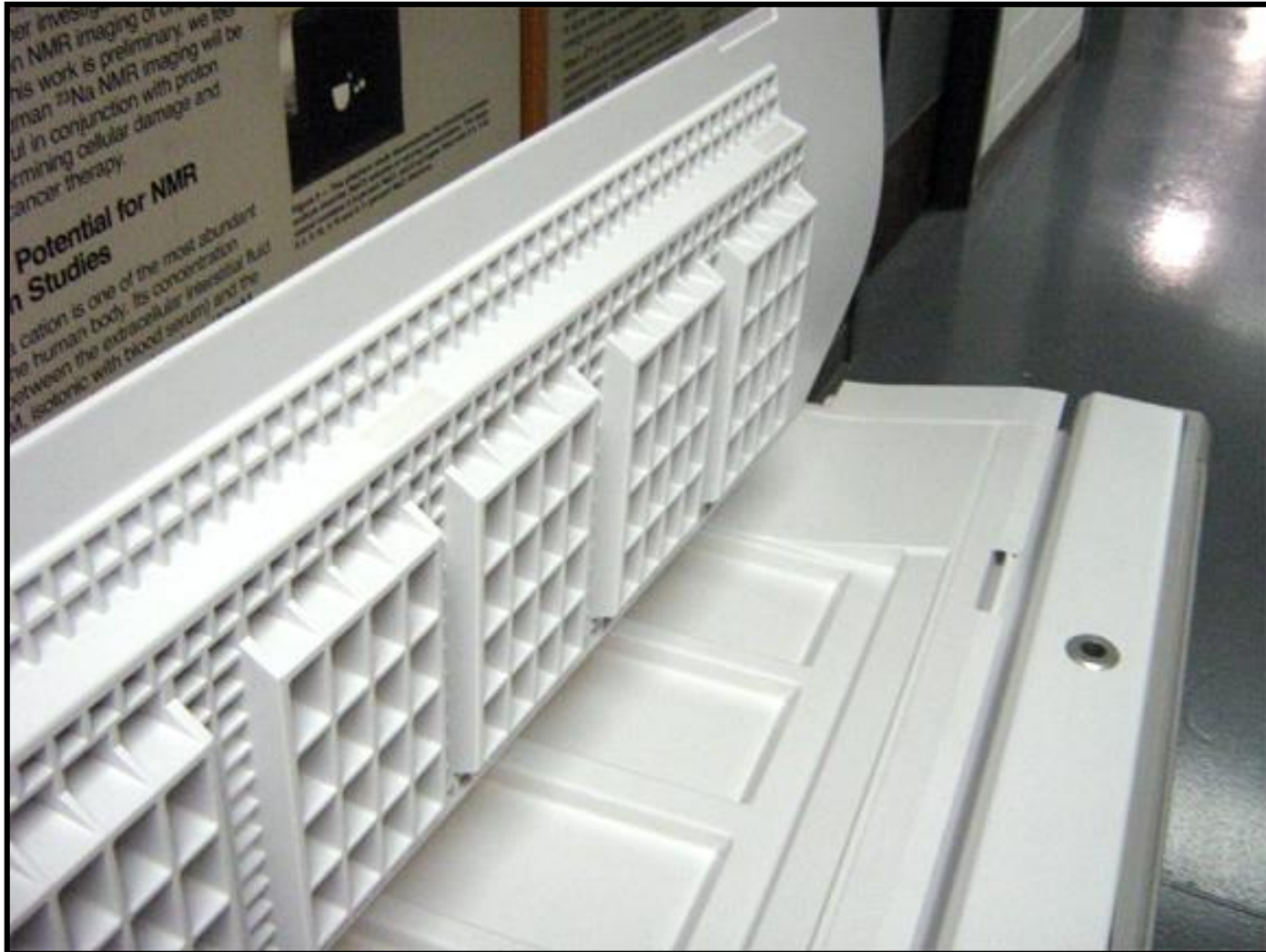


Each P connector supports 32 channels

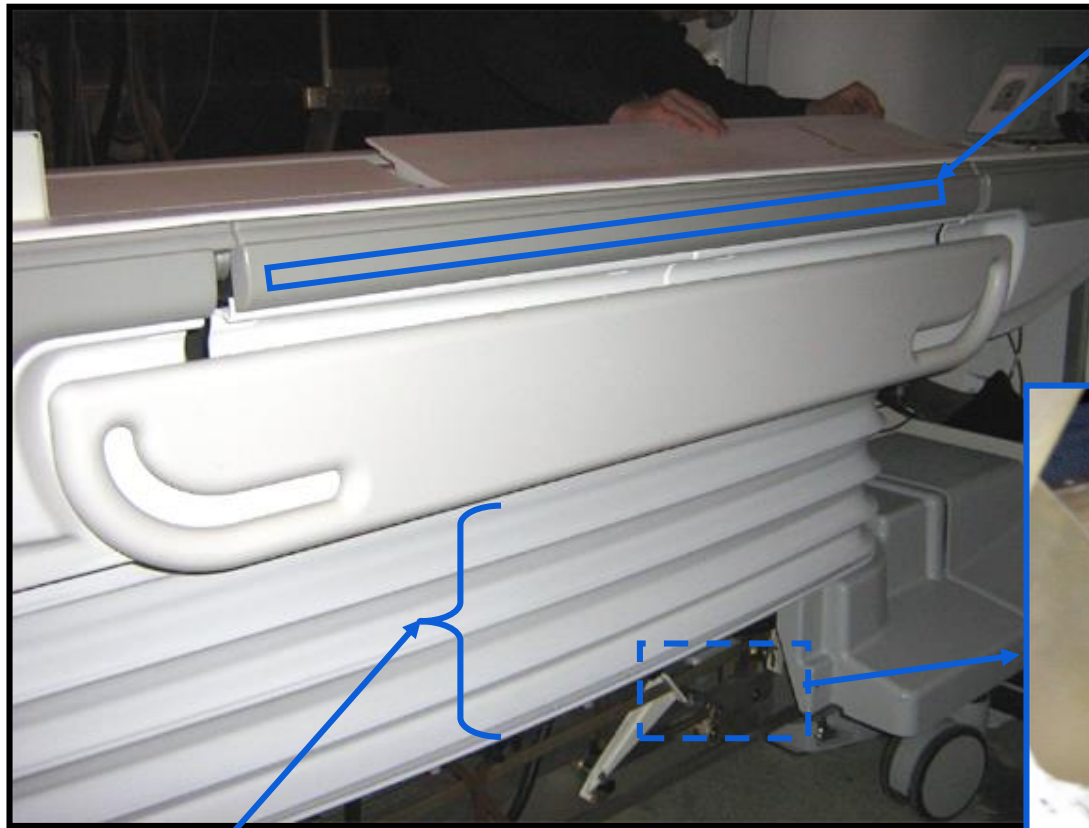


32-Channel Patient Transport Table

Space for UNO coil (future)

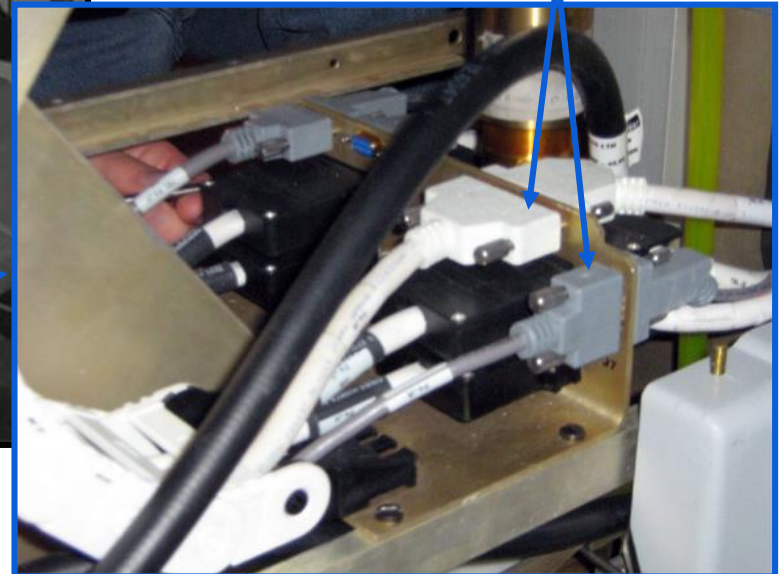


32-Channel Patient Transport Table



Touch-and-go landmarking strip
Buttons underneath bumper pad

Internal table IF for RF and data
cables from P3 and P4



Bellows skirt

- Bellows skirt can be raised and secured in place with Velcro for easy Service access

32-Channel Patient Transport Table

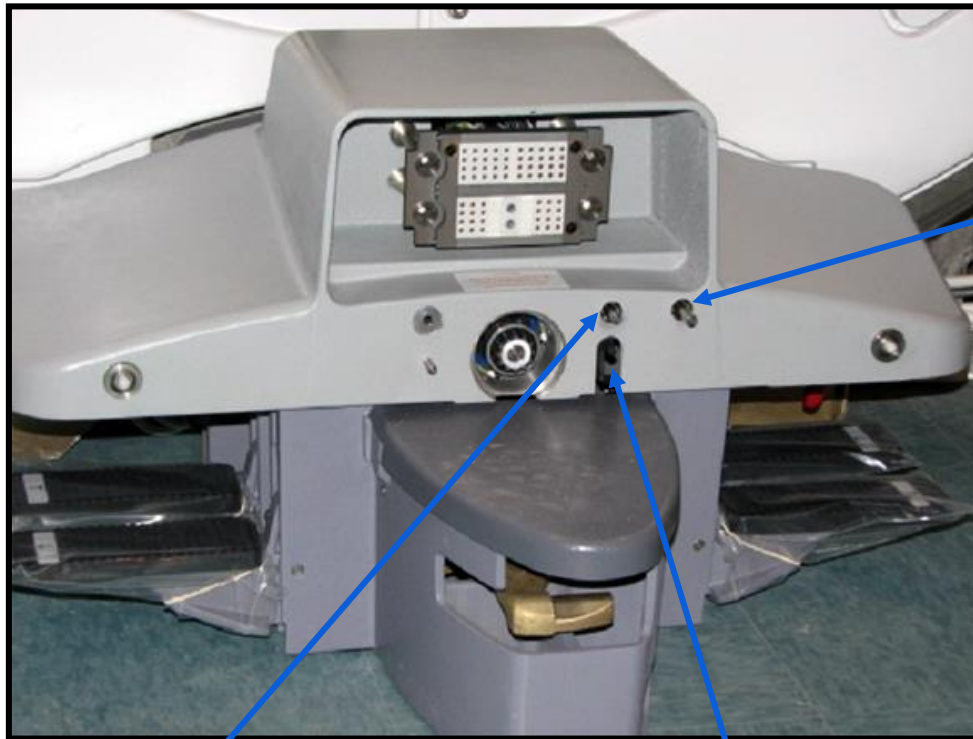
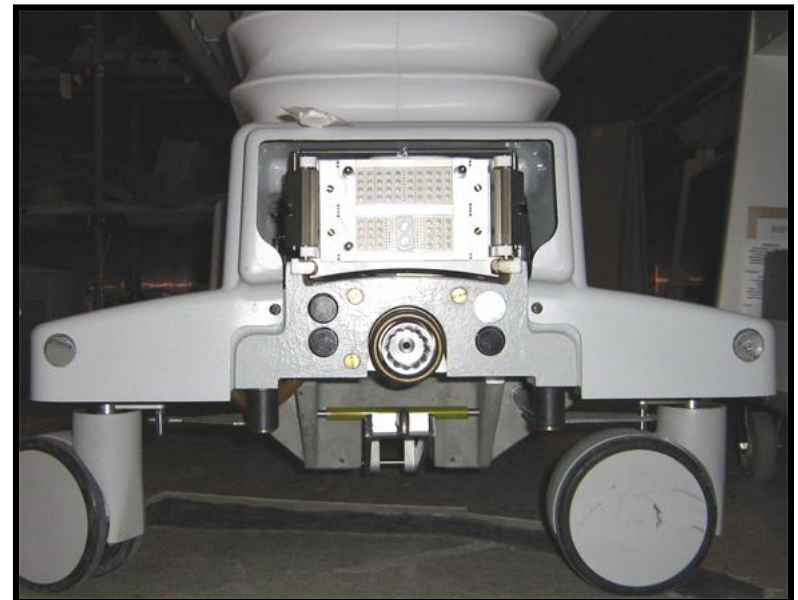


Table Up-limit
Switch

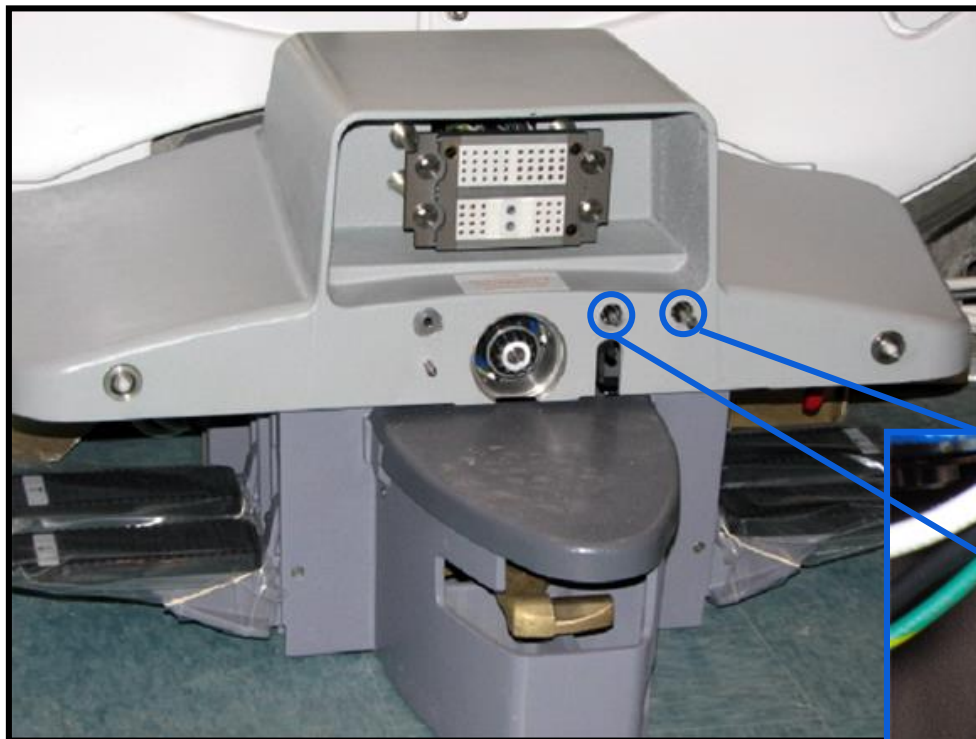
Dock Down-Pedal
activator

2-stage Table Detection Switch

- 1st stage – table detect (any table)
- 2nd stage – 16-channel table detect

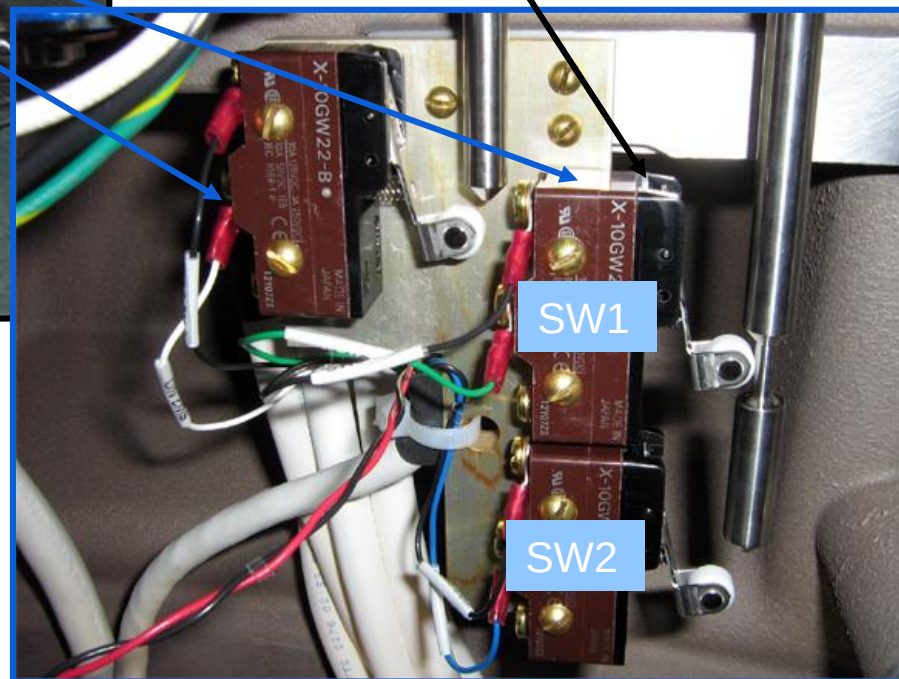


32-Channel Dock



2-stage Table Detection Switch

- 1st stage – table detect (any table)
- 2nd stage – 16-channel table detect



Upper dock casting removed –
view inside of up-limit (left) and
2-stage switch assembly (right)

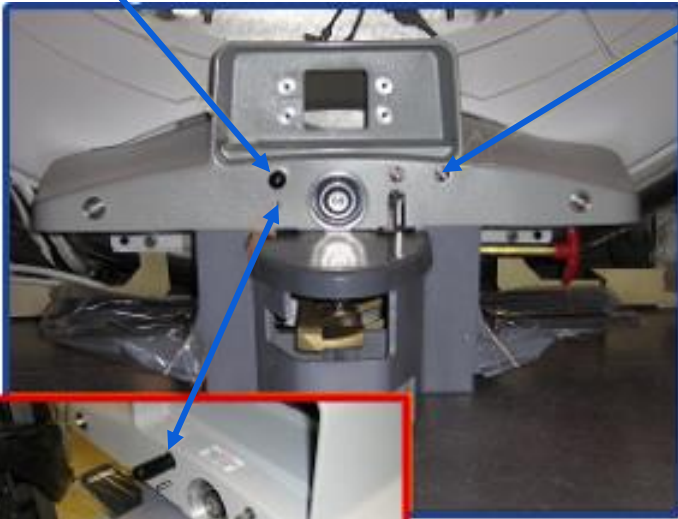
PHILIPS 16-Channel Patient Transport Table and Dock

16-channel table key; extends out about 7cm from dock face

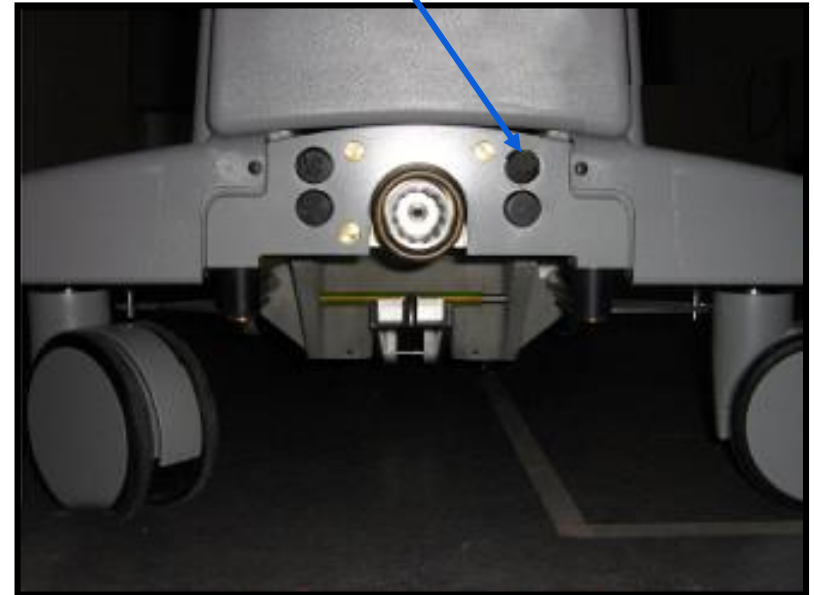
2-stage table detection switch

- 1st stage – table detect (any table)
- 2nd stage – 16-channel table detect

16-channel table key recess (no plug present; unfilled)



16-channel table key; designed to prevent a 32-channel table from docking to a 16-channel system



32-Channel Patient Transport Table

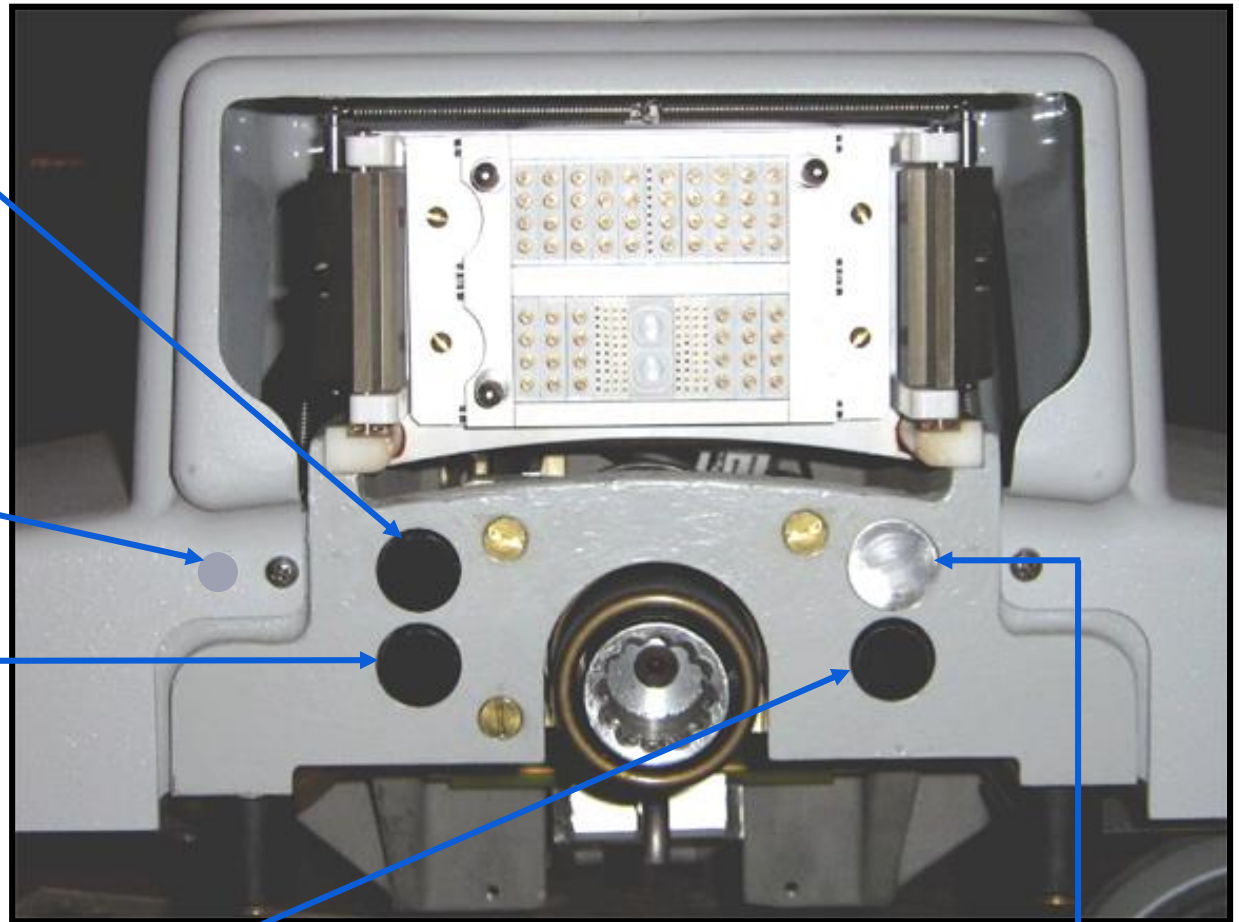
Table up-limit plunger
(*out* permits the system to move the carriage)

Table docked recess hole;
note that there is no plunger
(it is present on a 32-
channel table, not on a 16-
channel table)

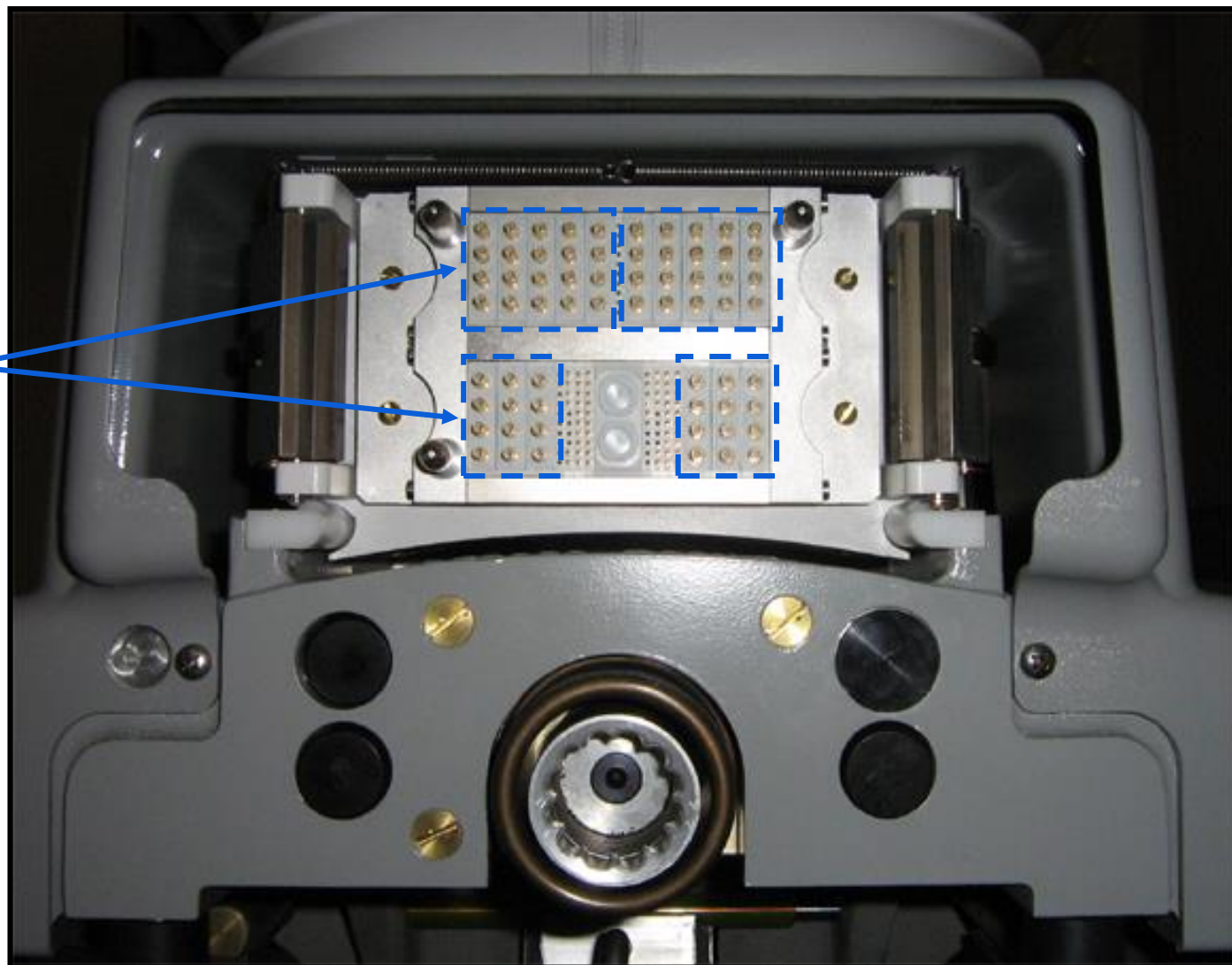
Dock down-pedal plunger
(*in* lowers table)

Cradle movement enables
plunger (*in* permits cradle
advancement off the table)

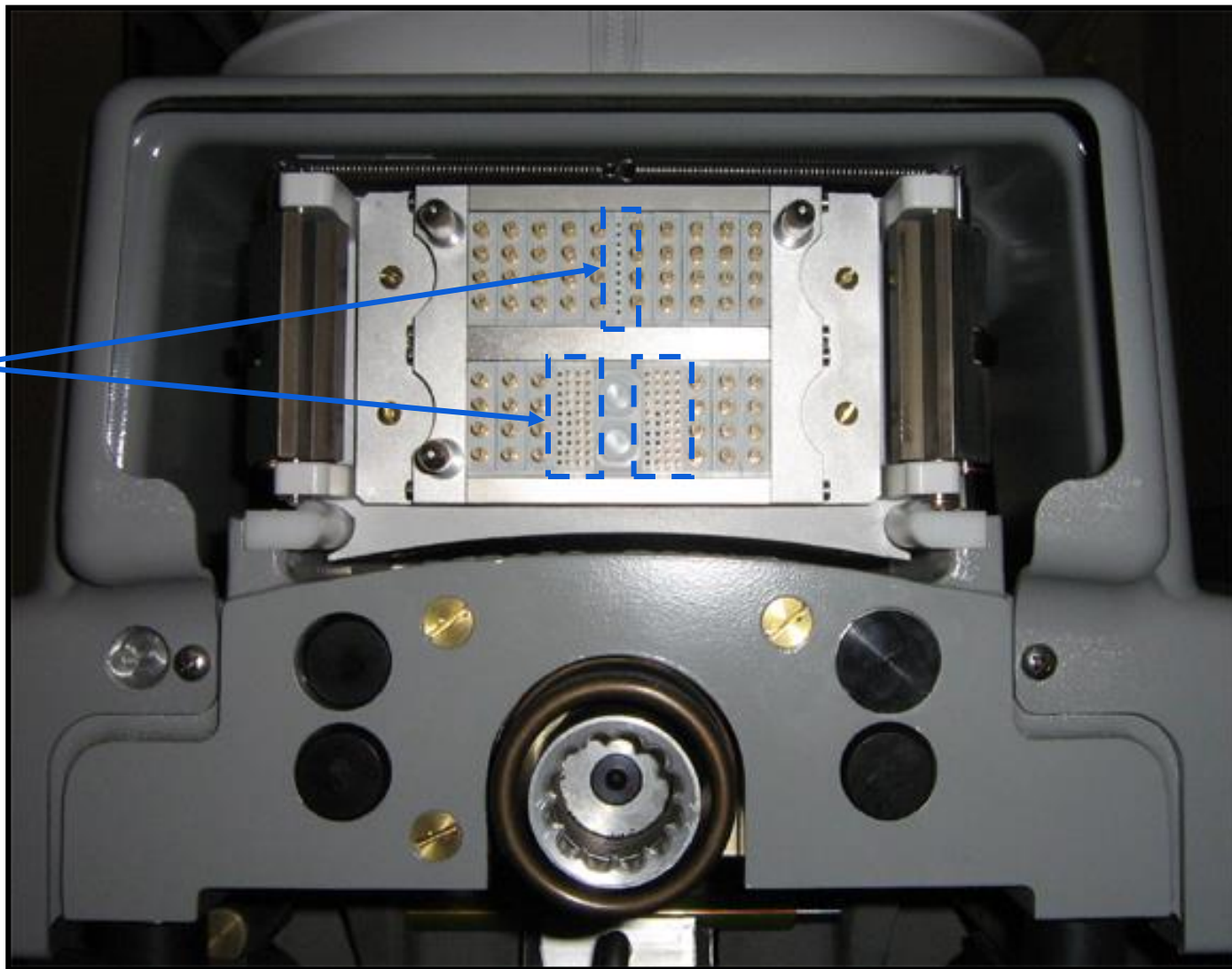
16-channel dock key; solid-filled plug
as shown to prevent a 32-channel
table from docking with a 16-channel
system



- Male coax receive connectors



Male
status/control
pins



PHILIPS 32-Channel System Scanning Matrix

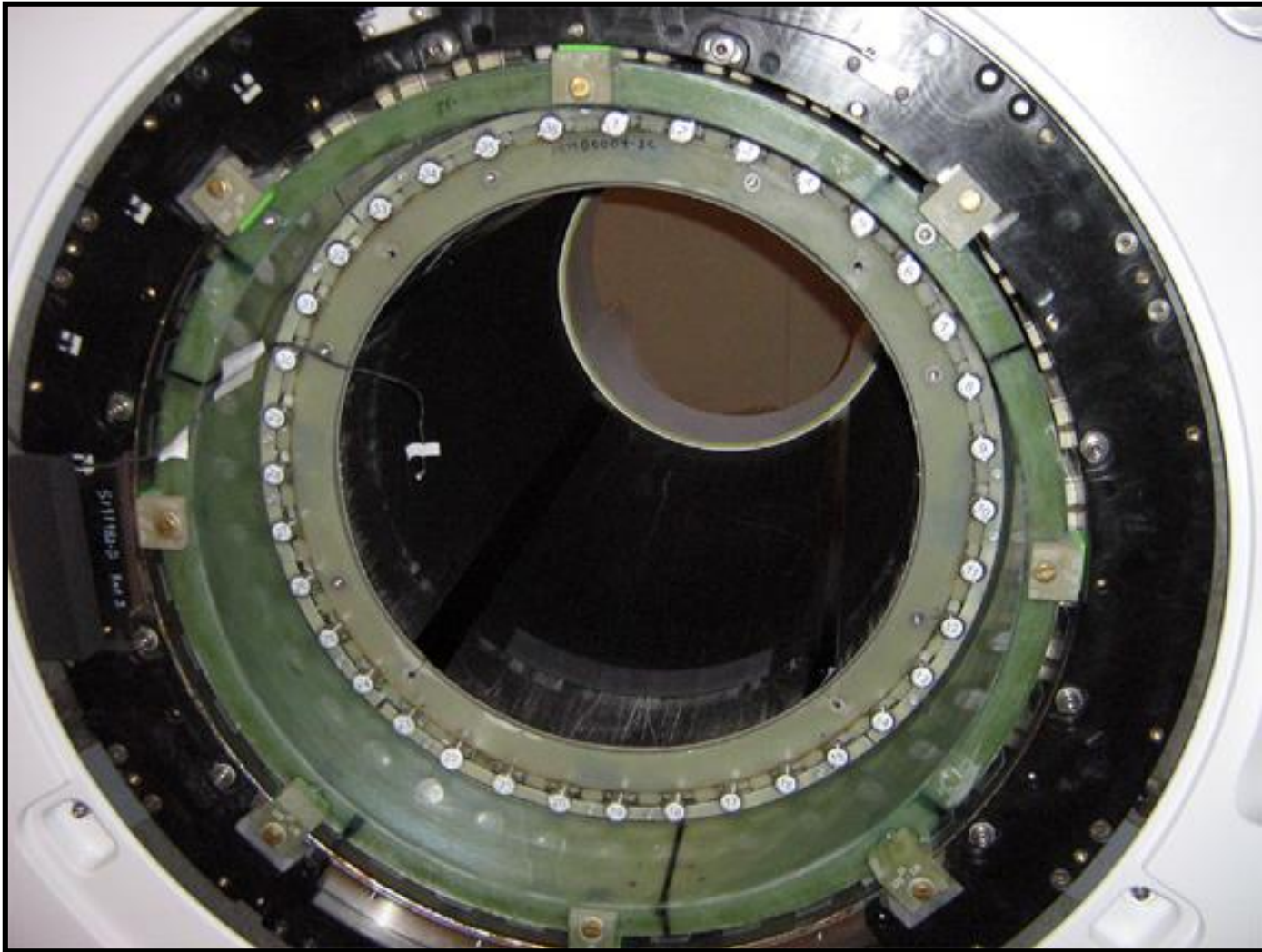
Dock Switch Signals and HART ID				
Table Status	Table Up Switch	Table Dock Switch	HART ID OK?	Scan Permitted?
No table attached	0	0	Don't Care	No
16-channel table	0	1	Don't Care	No
32-channel table	1	0	No	No
32-channel table	1	0	Yes	Yes
16-channel table	1	1	Don't Care	Yes

- Where:
 - 0 indicates switch is *not* pressed in
 - 1 indicates switch is pressed in
 - Table up-limit switch: 0 = not up, 1 = up
 - Table dock switch: 0 = No table or 32-channel table docked, 1 = table docked
 - 16-channel table does not have a 32-channel recess hole
 - 16-channel table does not contain any electronics (no HART ID, no baluns)

PHILIPS Signa MR750 Patient Transport Table Notes

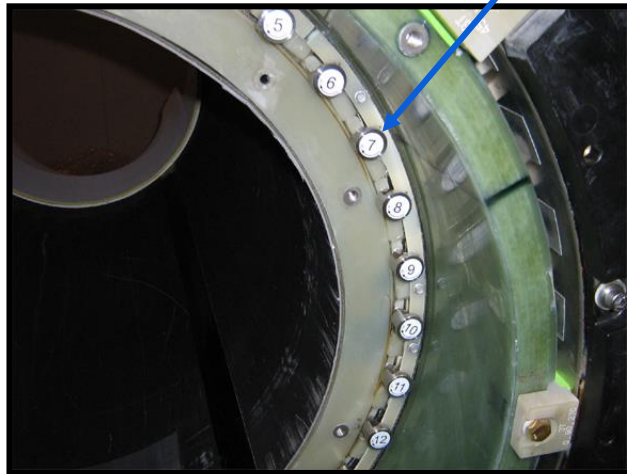
- Rated for 500 lbs. (to support, not to lift)
- 32-channel table is equipped with HART ID
- 32-channel table supports Touch-and-Go landmarking that requires the optional IRM feature
- 16-channel patient transport table looks similar but it does not have P connectors and does not have a dock connector
- 16-channel and 32-channel docks have the same switches
- Only the 32-channel dock has a multi-coil connector
- 16-channel system will **not** support a 32-channel patient table
- 32-channel system **will** support a 16-channel patient table

PHILIPS eXtreme Resonance Module Body (XRMB)



Front view

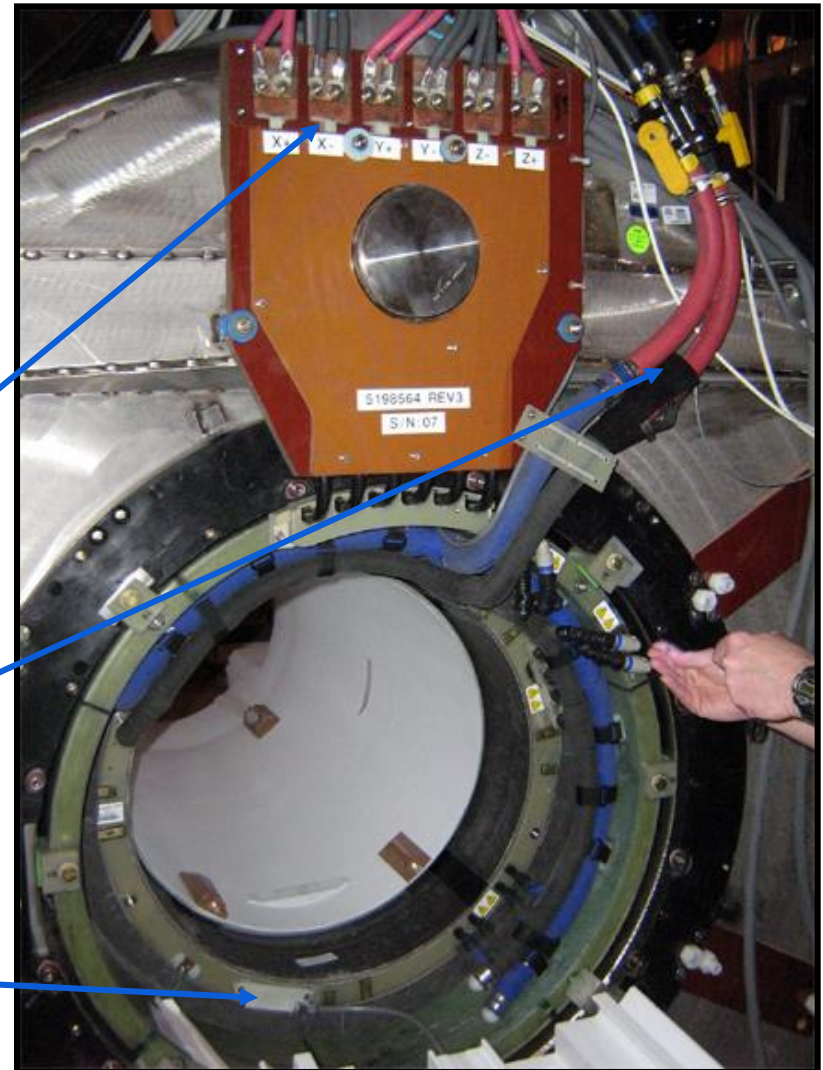
- Z-winding inside coil is hollow for chilled coolant flow
- Coil has embedded temperature sensors
- Coil has passive shim trays



X, Y, and Z
gradient
connections

Chilled water
supply and
return from HEC

Embedded
gradient
temperature
sensors

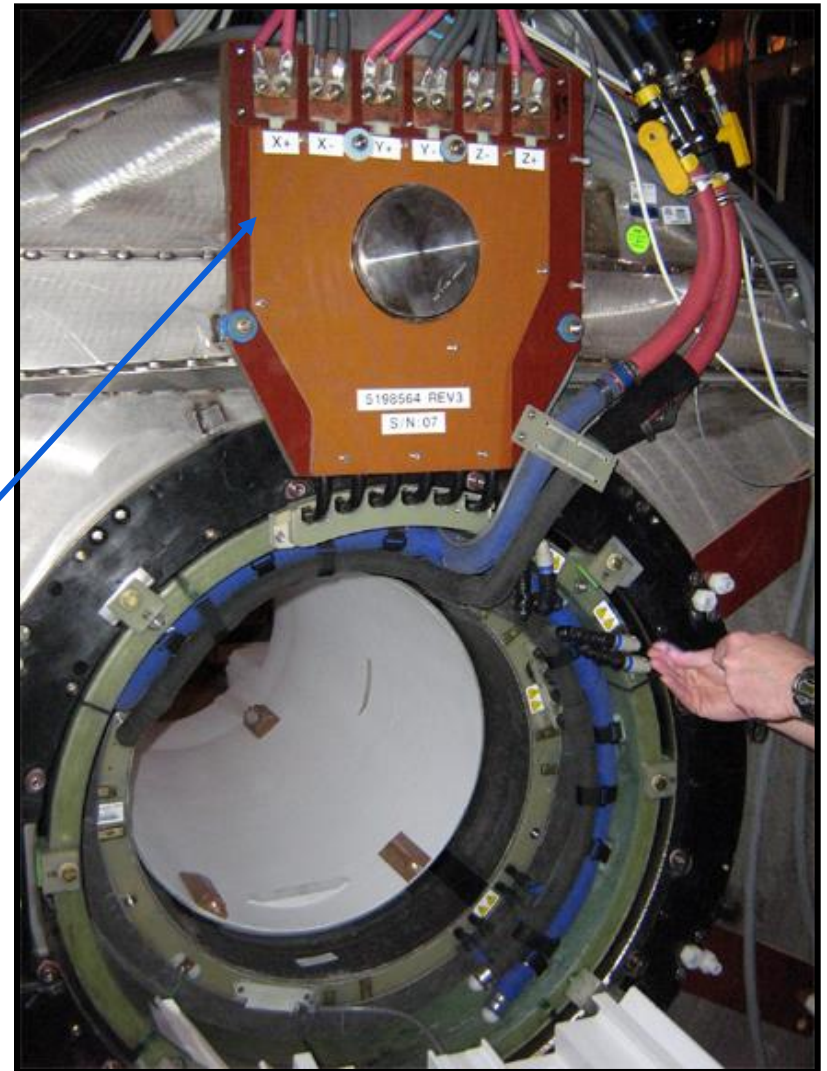


Rear view

PHILIPS eXtreme Resonance Module Body (XRMB)

- Utilizes existing coil cart
- Utilizes existing removal tool
- Gradient cables attach to XRMB through the buss-bar assembly

XRMB Buss
bar Assembly



Rear view

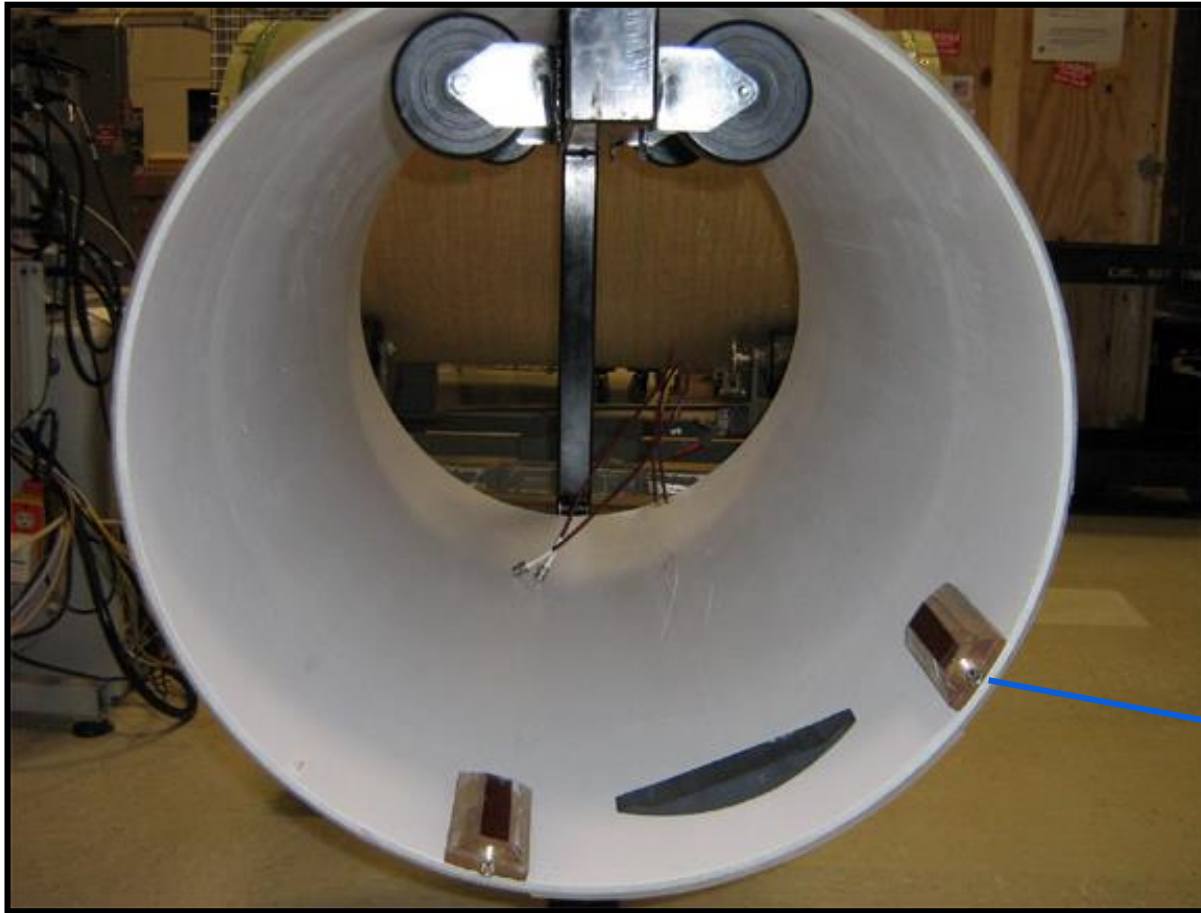
PHILIPS eXtreme Resonance Module Body (XRMB)

- Buss-bar assembly attaches to the XRMB using cable and studs
- Alignment of the buss-bar assembly to XRMB during installation/ replacement is *critical*
- Buss bar (cable style) lifespan is approximately 10 years

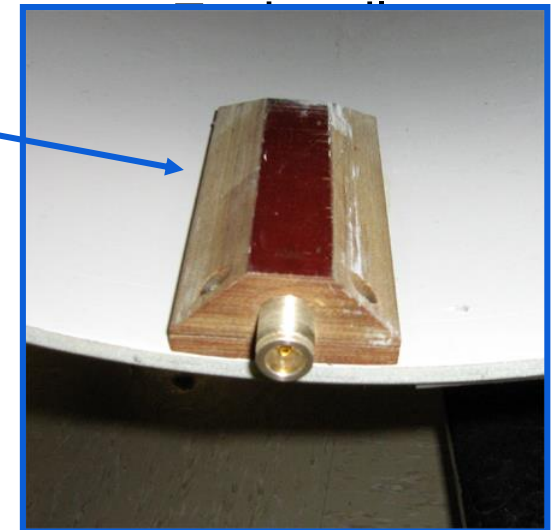


PHILIPS eXtreme RF Body Coil (XRF Body Coil)

- BP coil design
- 2 DD bias lines on 750, 4 on 450
- I and Q ports
- No FRU DD parts; the coil is a FRU



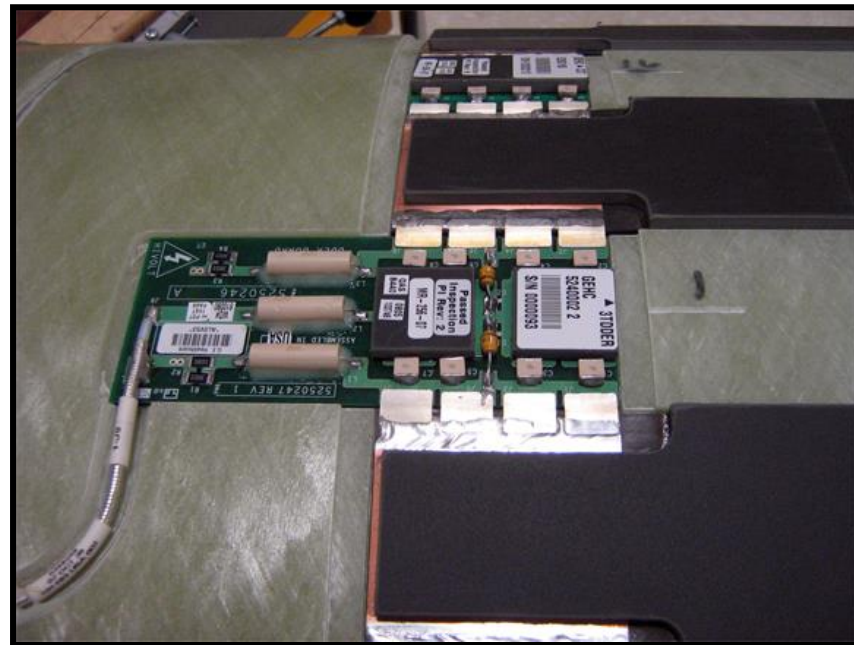
Rear view



PHILIPS eXtreme RF Body Coil (XRF Body Coil)



Exterior view



Exterior view –
closeup of DD coil circuitry

XRF Body Coil

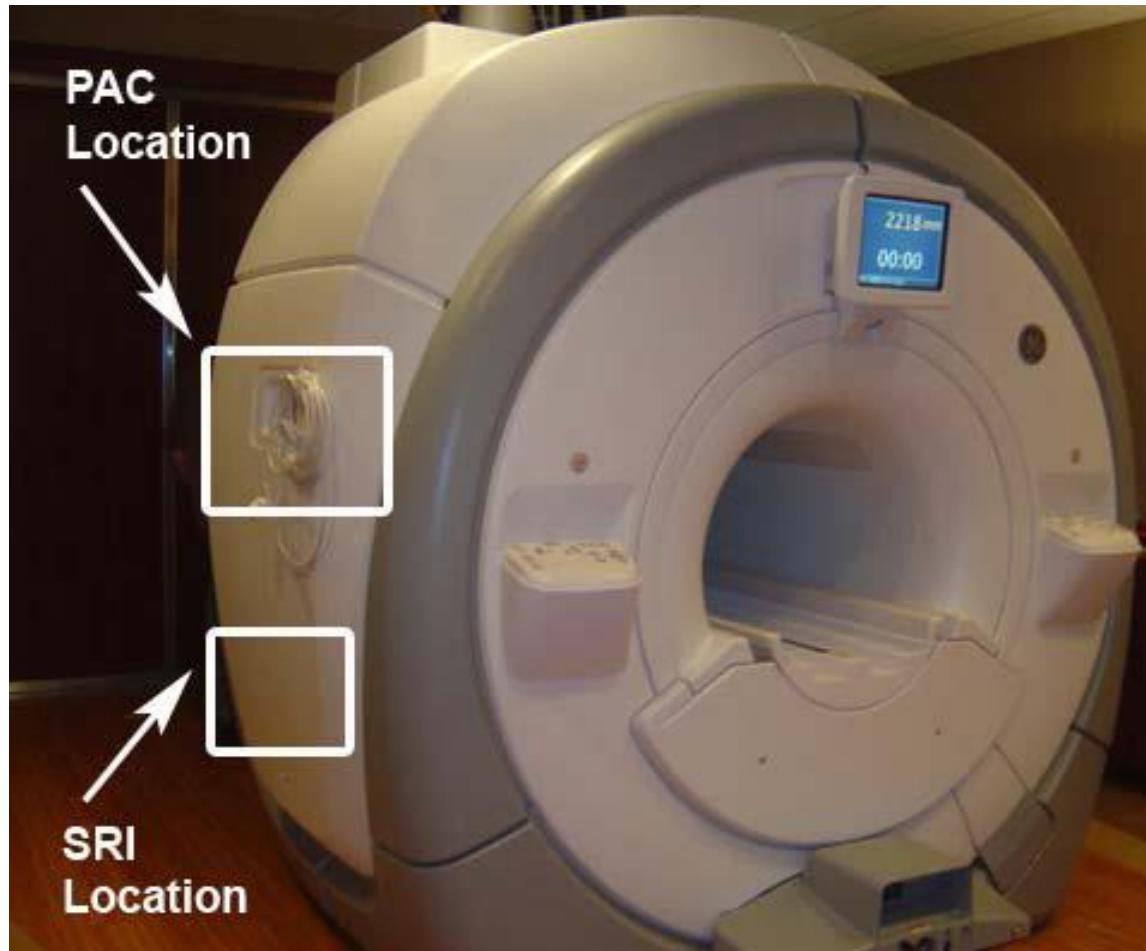
- I and Q RF drive connectors are built into the coil; no fixed pig-tail cables.
- Has fixed DD bias line pigtail cables
- Does not have DD Boards; diodes are built into the coil and are not FRUs
- Must be tuned to gradient coil; this is required anytime an XRMB or XRF Coil is replaced
- GE Magnet Factory in Florence is responsible for tuning.

Optima 450w Body Hybrid

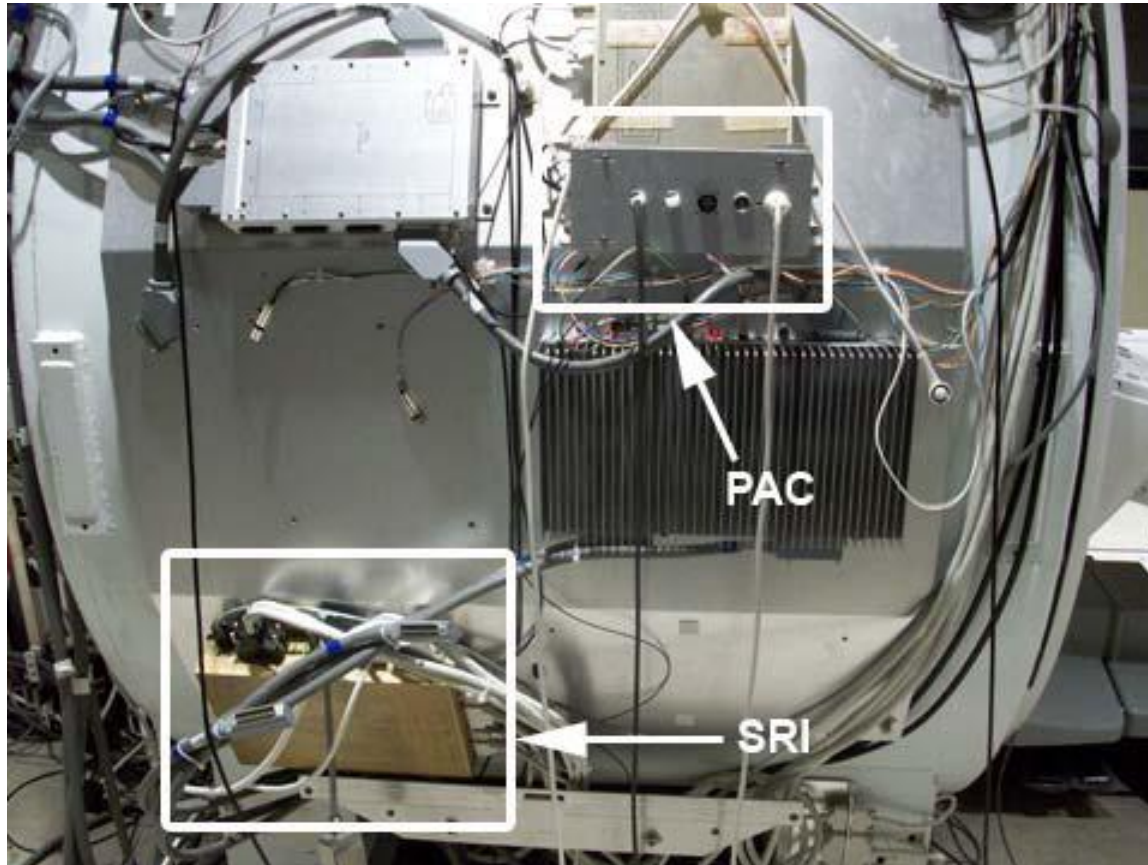
- As per replacement procedure, skirt and side covers need to be removed to service Hybrid.



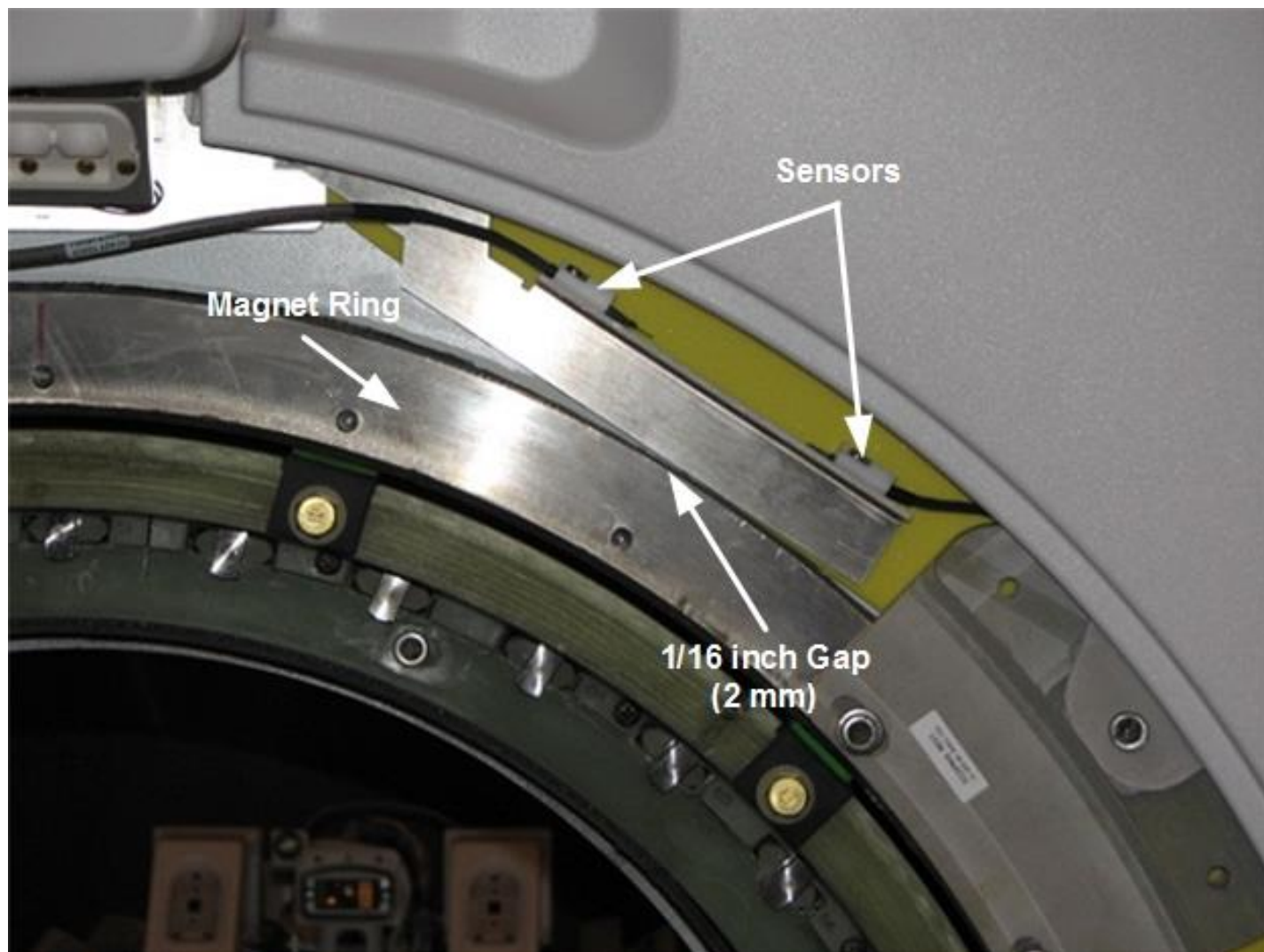
PAC & SRI Location



PAC & SRI Locations



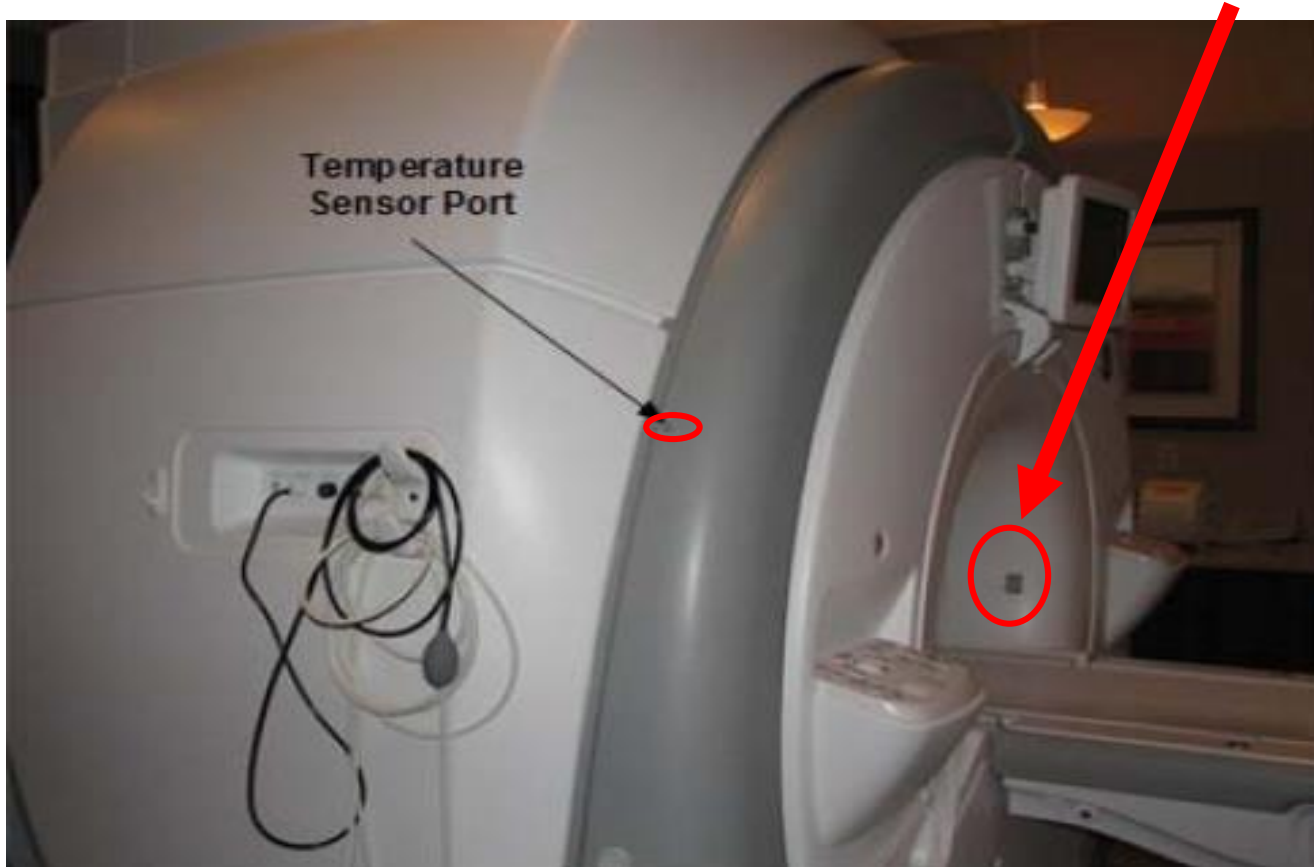
RF Coil Airflow Sensors



Note that the sensors are mounted differently than those on the Discovery

Ambient room temperature sensor

New style patient (bore) LED (each side)



Body Coil Cooling (chilled air)



7-Segment LED Display







GE Optima/Discovery MR450 & MR750

Objectives

- Understand the RF theory
- Identify and understand components of the transmit chain
- Complete RF calibrations
- Exercise RF chain diagnostics

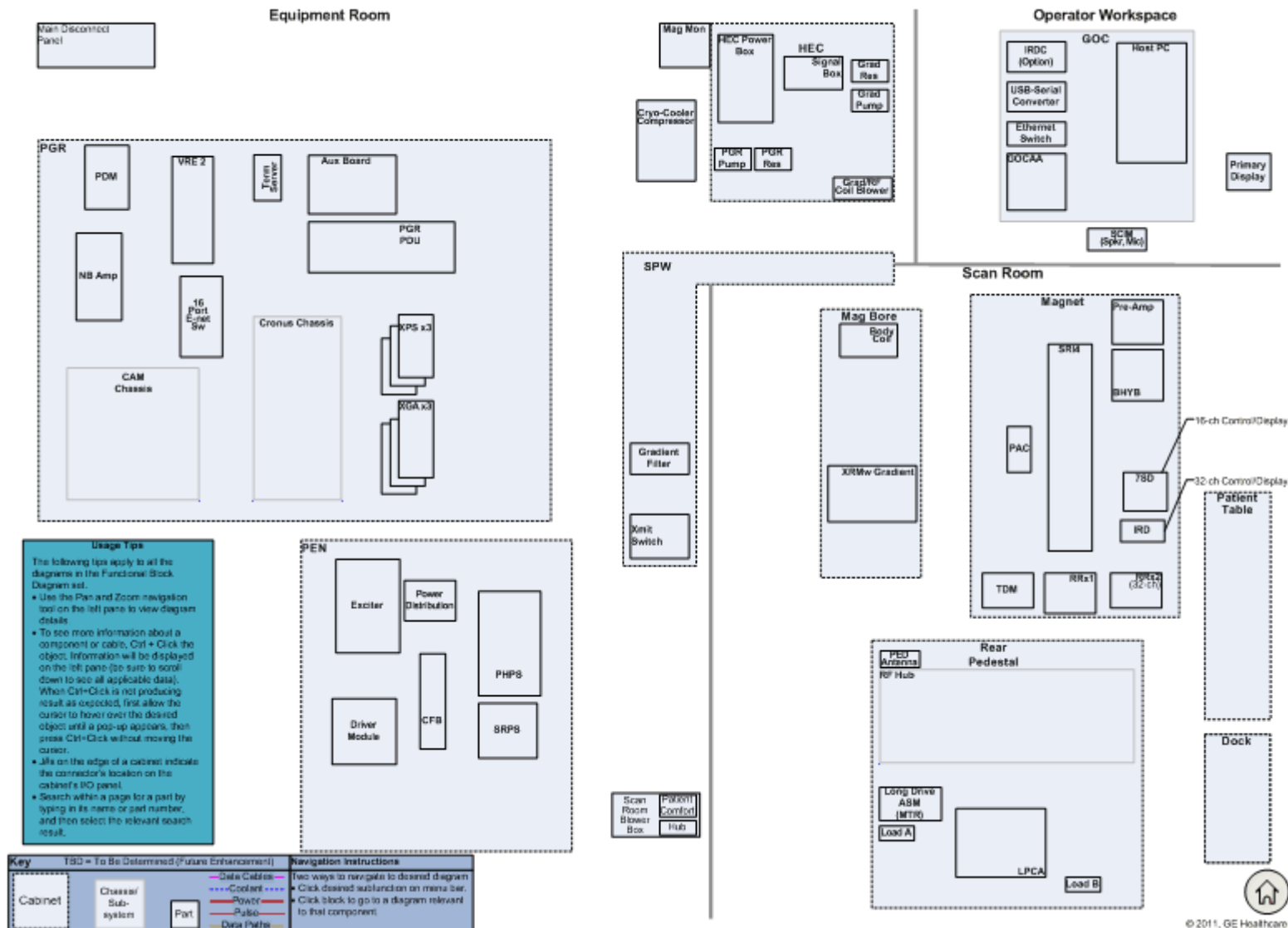
Functional Diagrams

MR Service Engineering – Functional Block Diagrams – MR450w

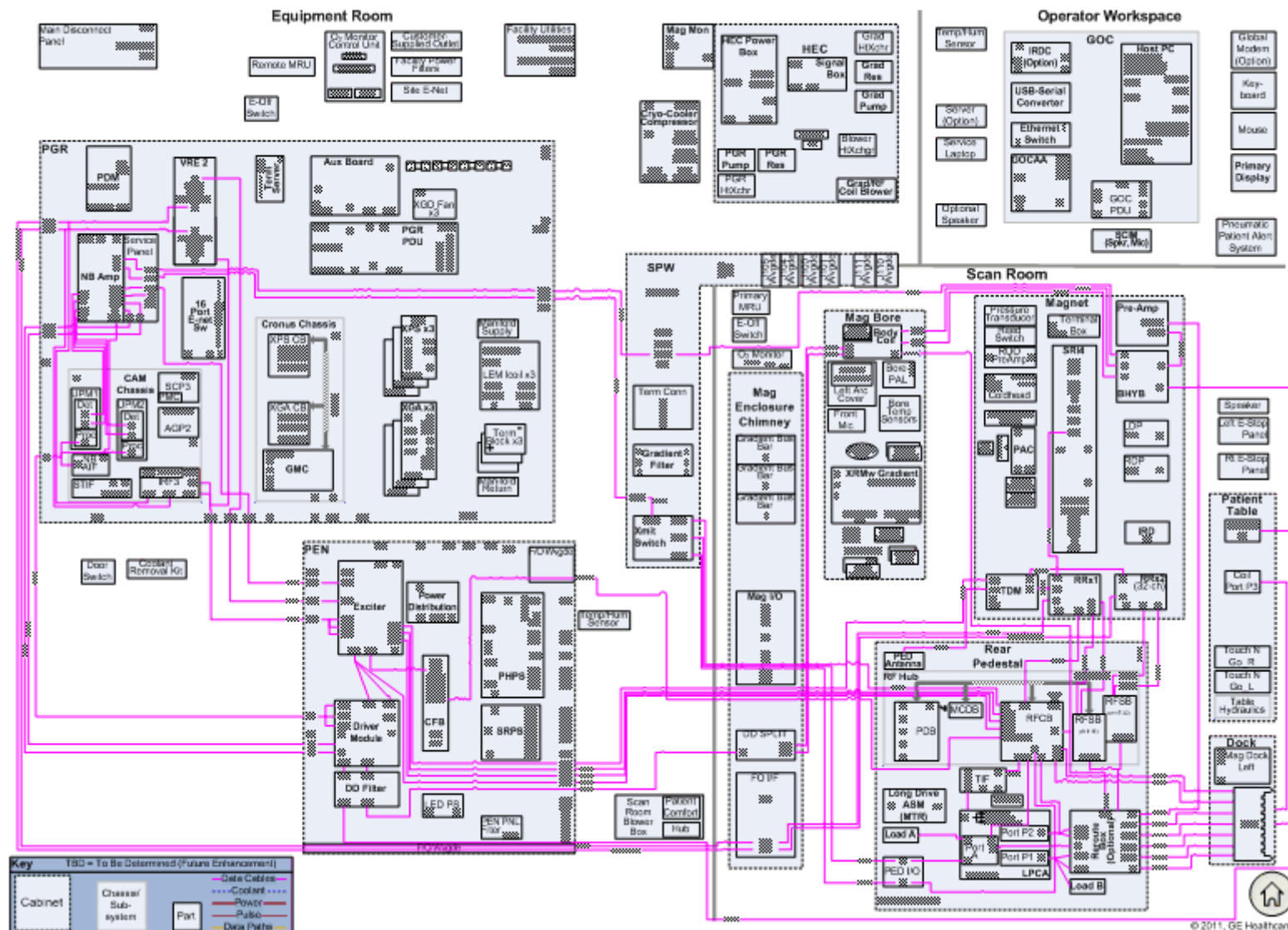
Subsystems | GRADIENT | POWER | POWER – SW RELEASE 22.x+ | GOC | COMMUNICATION | RF | PATIENT HANDLING | MAGNET | COOLING | DIAGNOSTICS | REPLACEMENTS

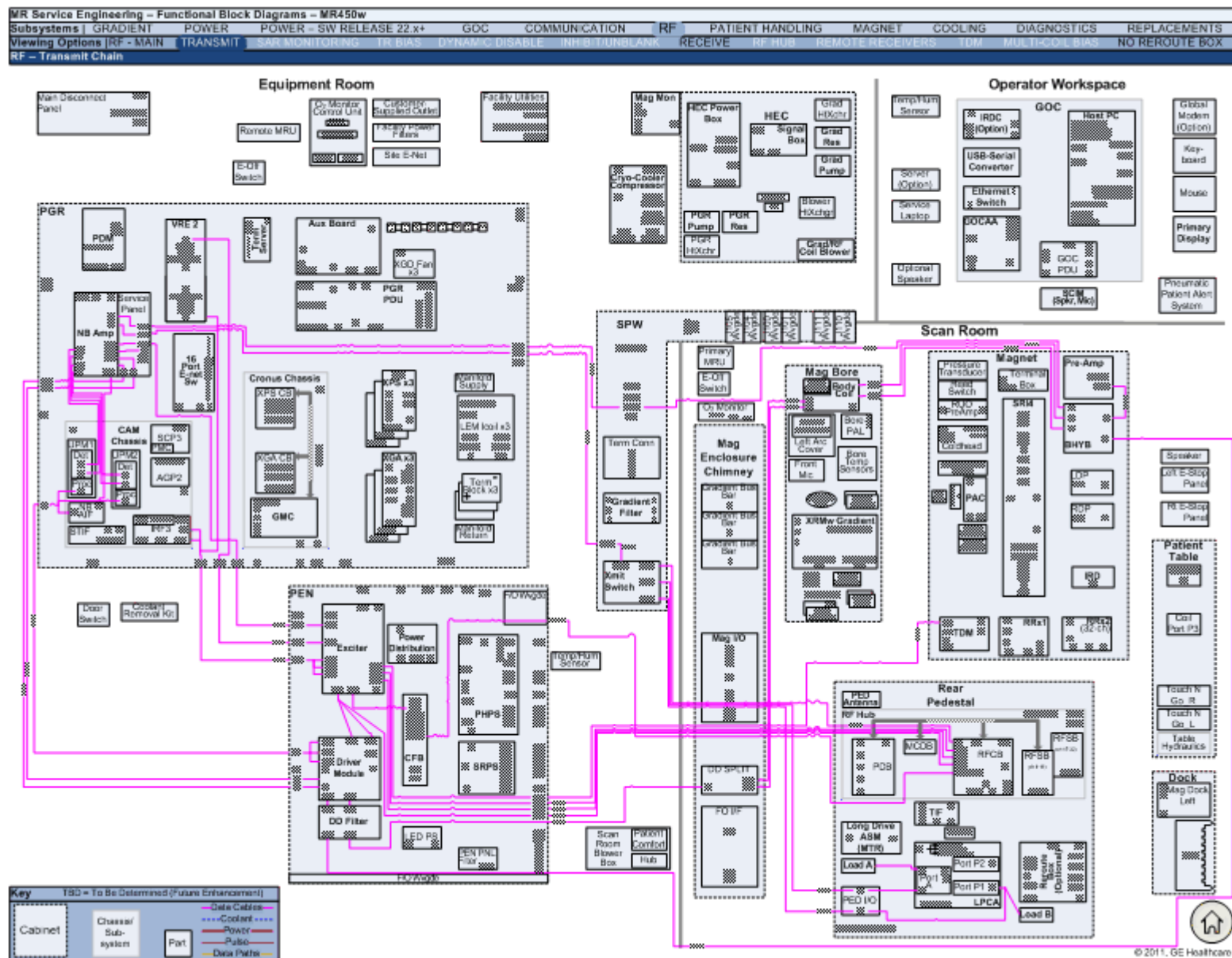
Viewing Options |

Block Diagrams Home Page



MR Service Engineering – Functional Block Diagrams – MR450w



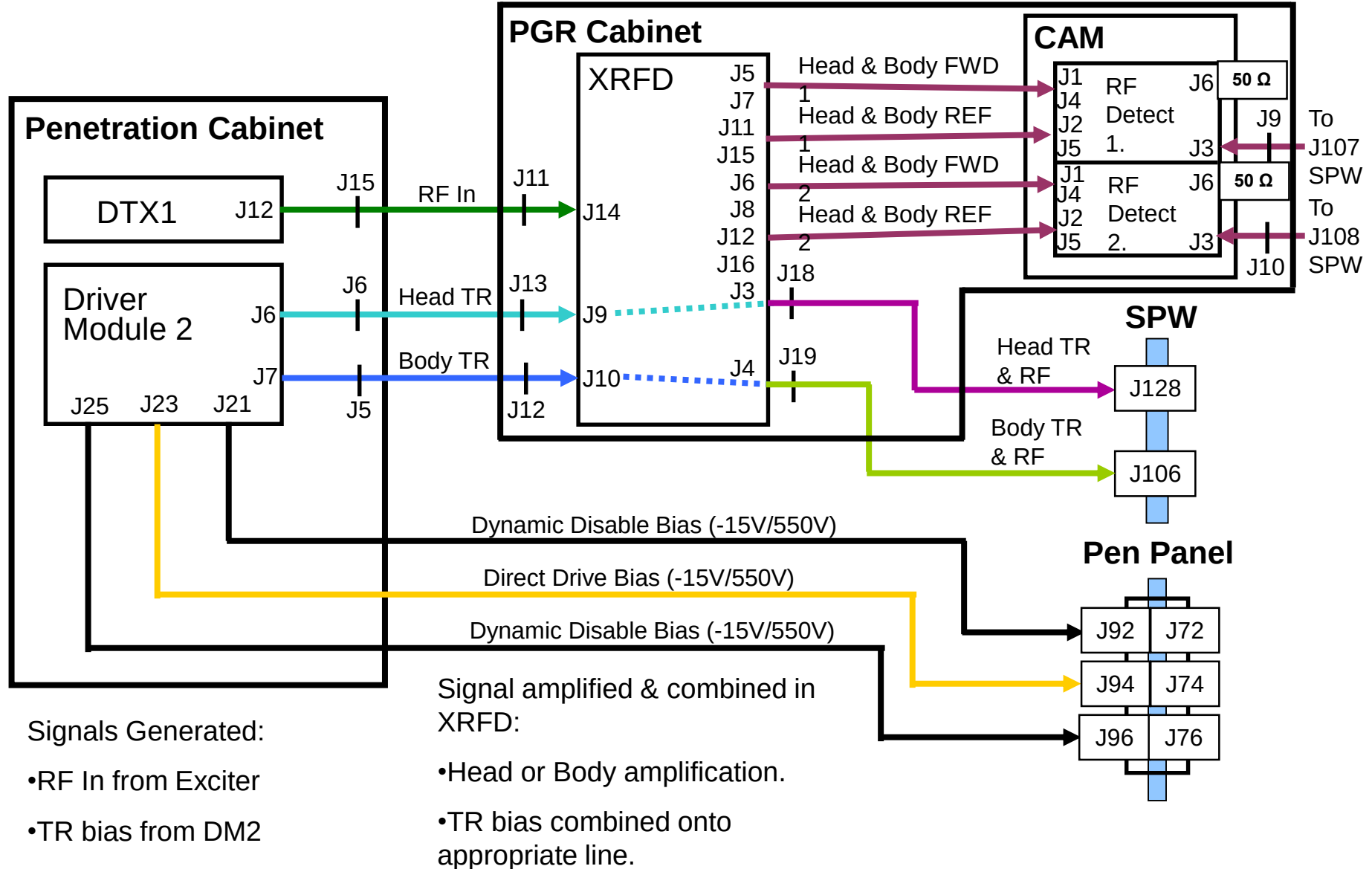




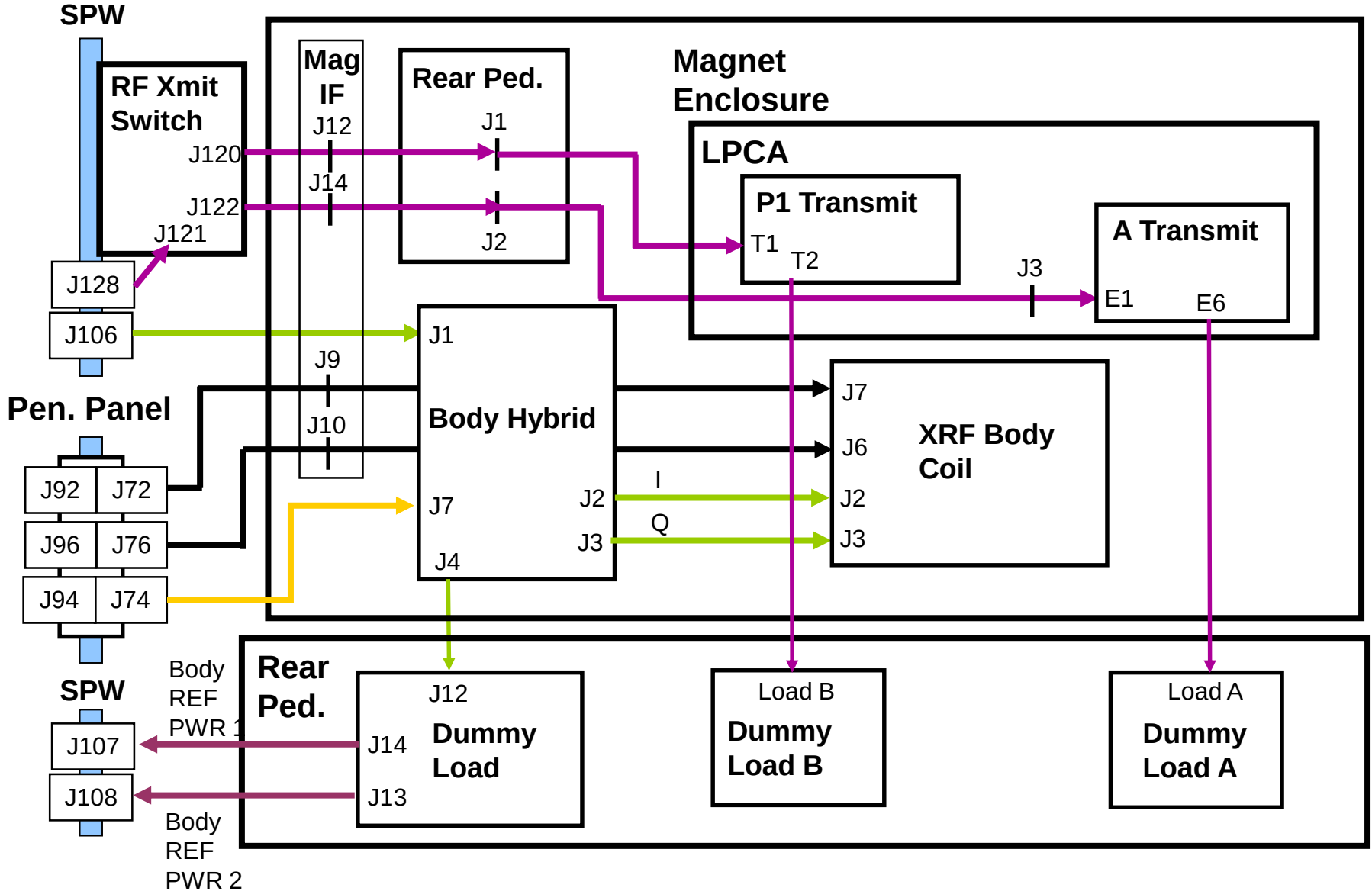




Equipment room



Scan room





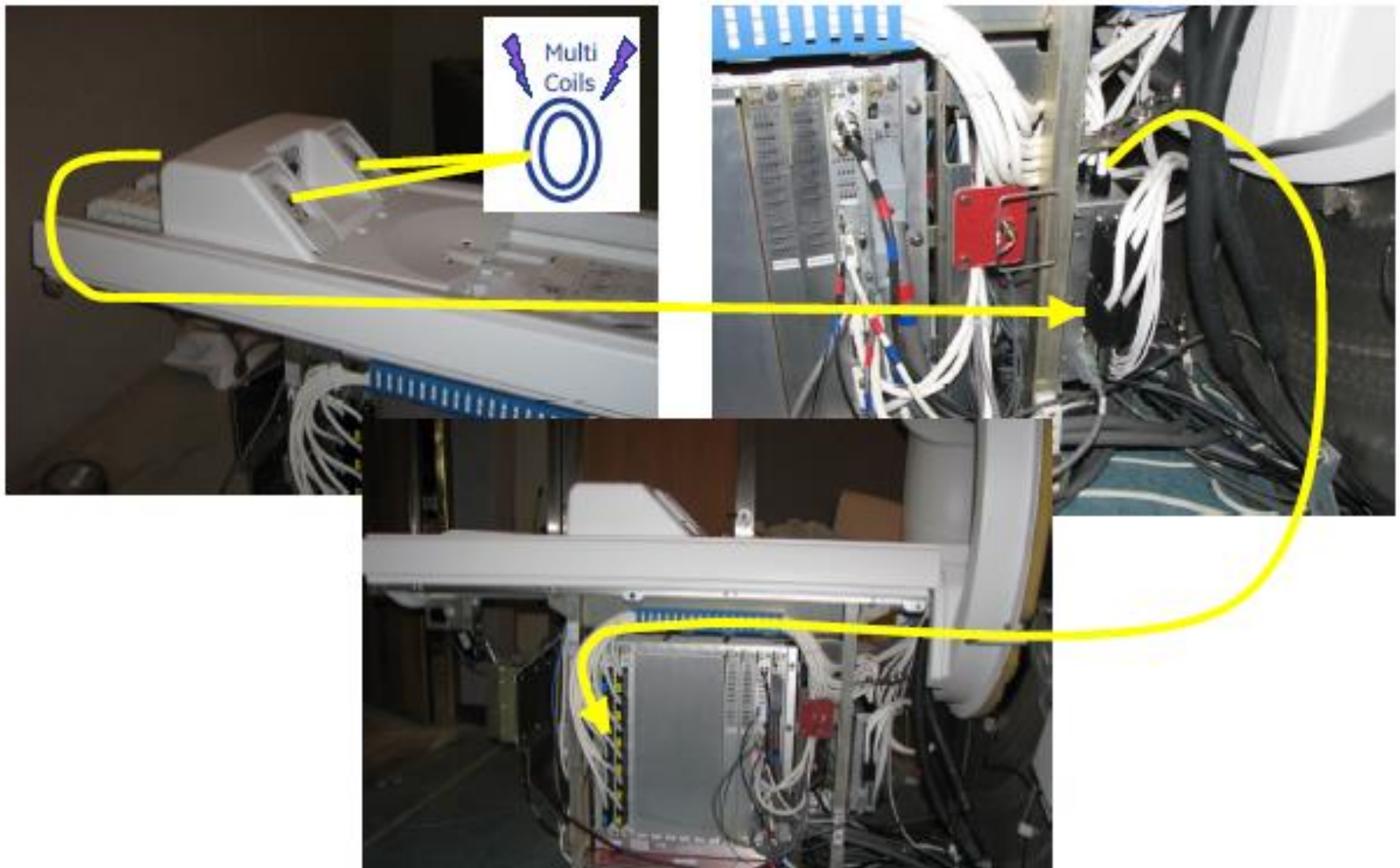


GE Optima/Discovery MR450 & MR750

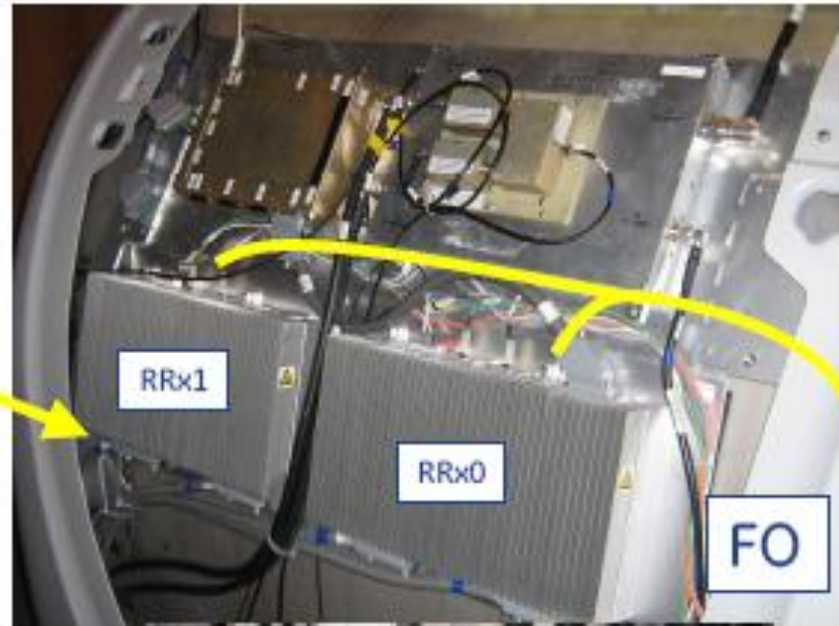
Objectives

- Identify and understand components of the Receive Chain
- To understand the Theory of operation of Receive Chain components and signal paths associated with Scan Room (Table connectors, Rear Pedestal, Receivers) and Equipment Room (ICNs, network connections to GOC).
- To understand troubleshooting and diagnostic tools available.

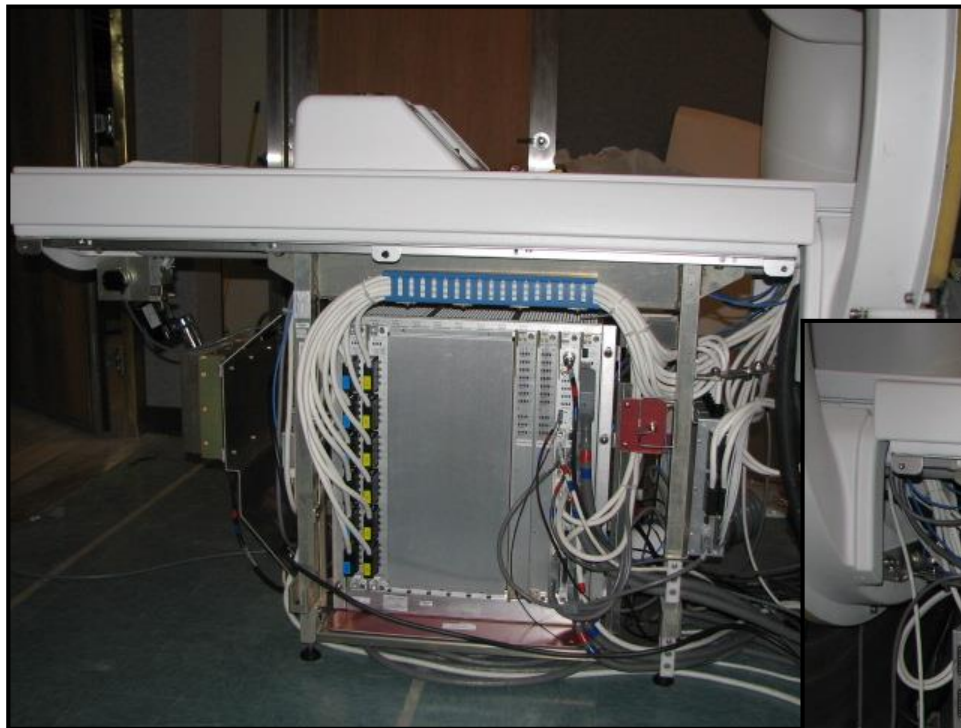
Receive chain



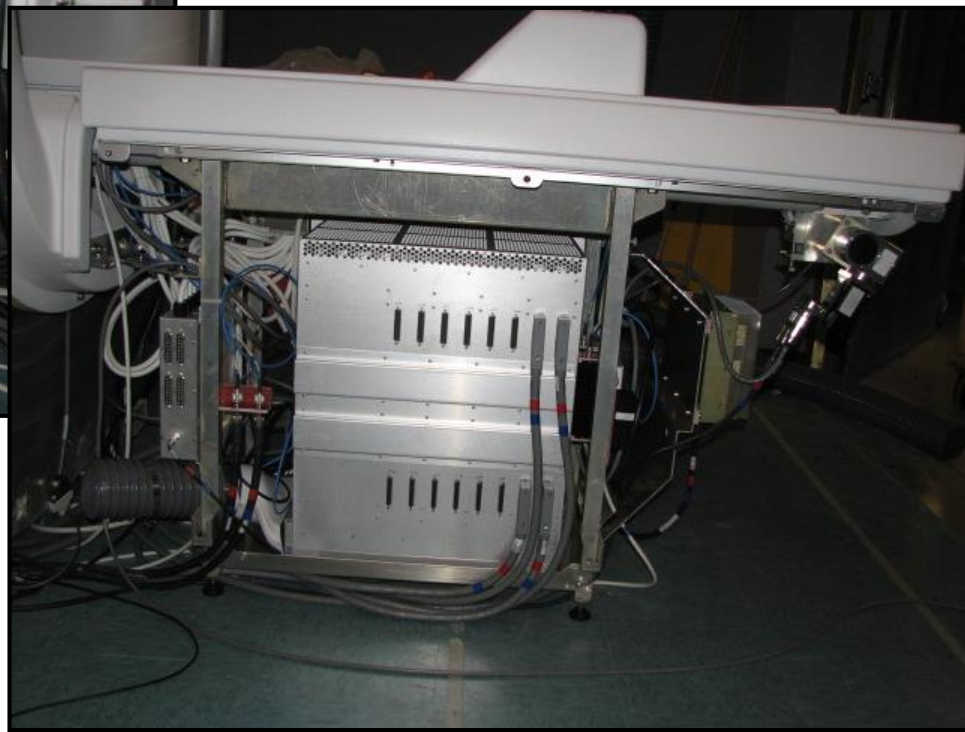
Receive chain



Rear Pedestal



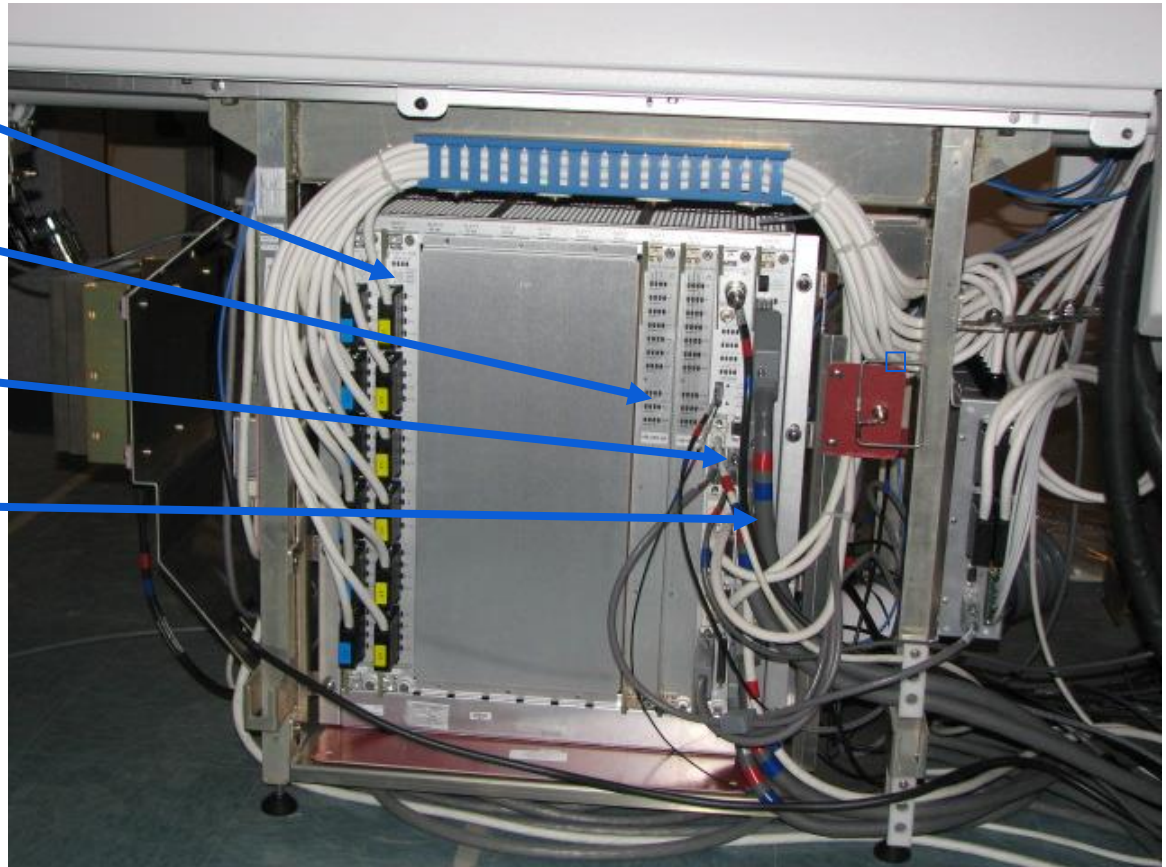
Left side from front



Right side from front

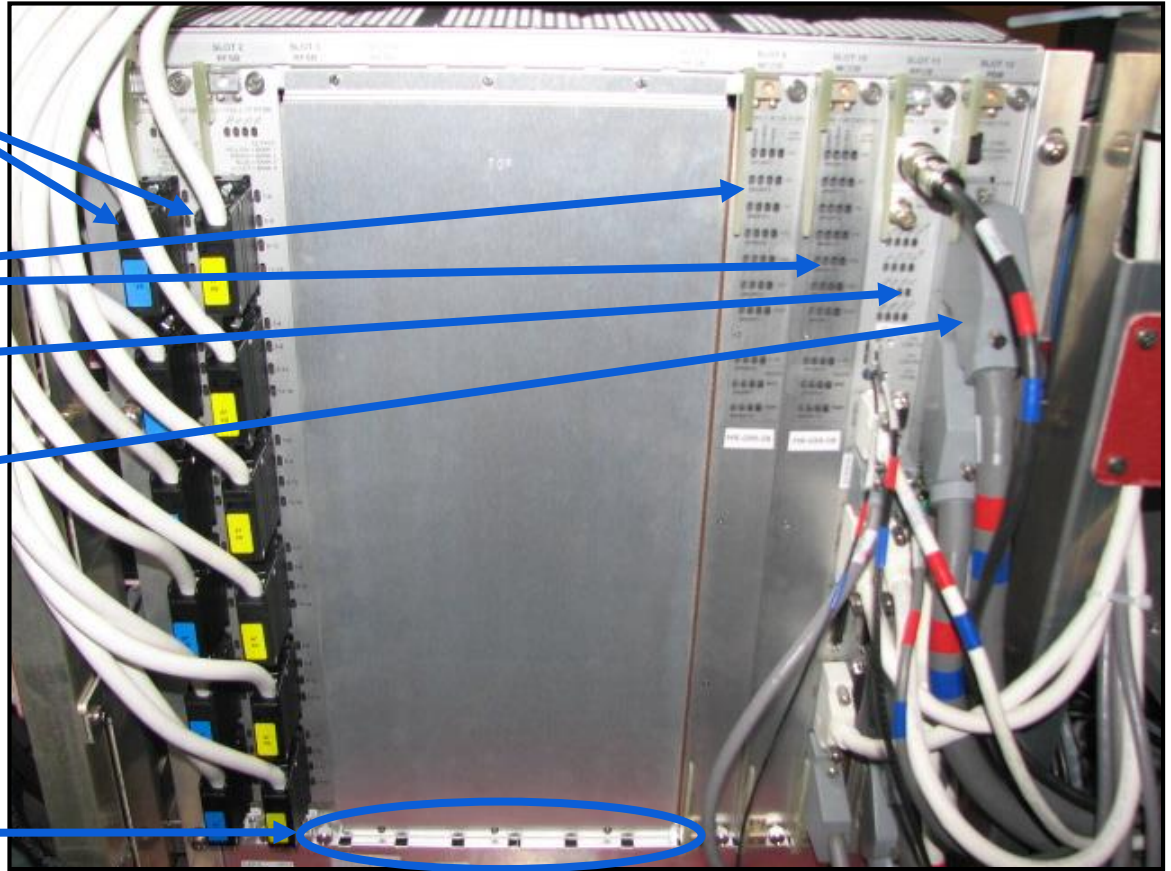
The **16 channel RF Hub** contains one of each:

- RFSB* (2 Shown)
(RF switch board)
- MCDB (2 Shown)
(Multi-coil driver board)
- RFCB*
(RF control board)
- PDB
(Power distribution board)
- *Different p/n for 450



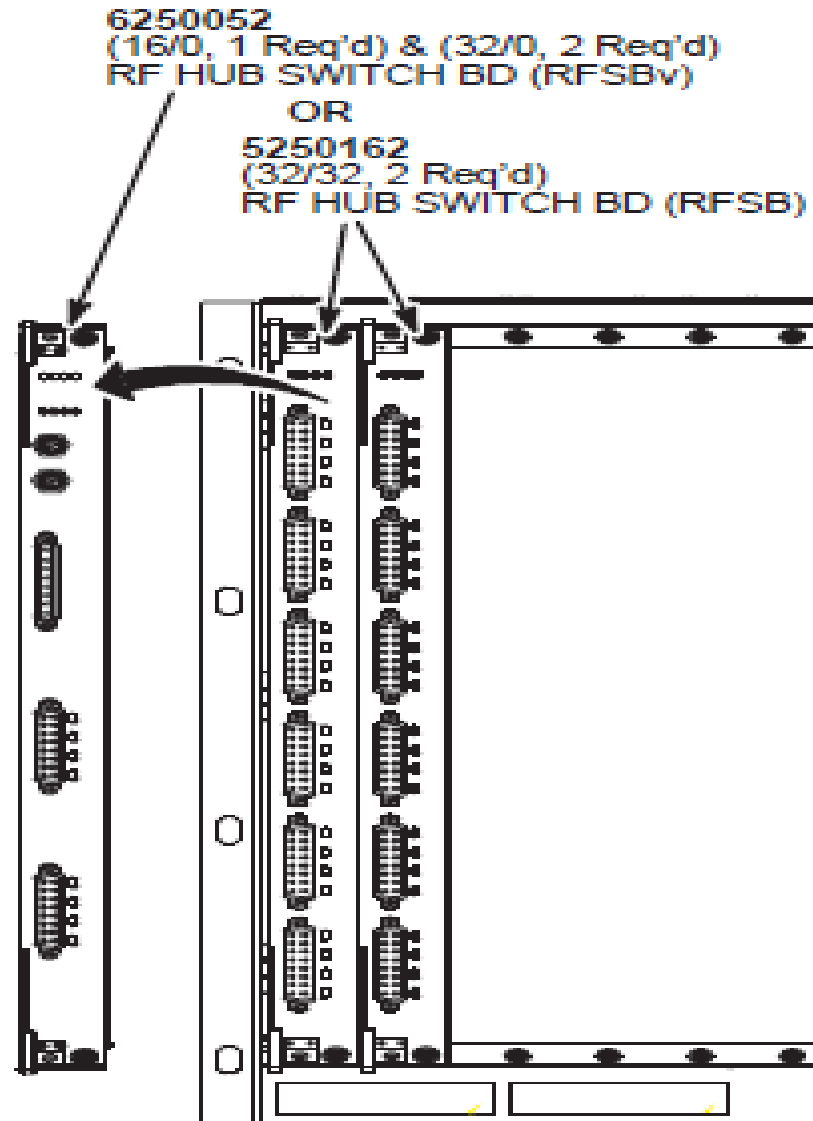
The **32 channel RF Hub** contains:

- RFSB*
(RF switch boards)
- MCDB
(Multi-coil driver boards)
- RFCB*
(RF control board)
- PDB
(Power distribution board)
- Six unused slots for future expansion to 128 channels
- * different for 450



PHILIPS Optima 450w has 16/0 or 32/0 configuration

RF HUB for 16/0, 32/0 and 32/32

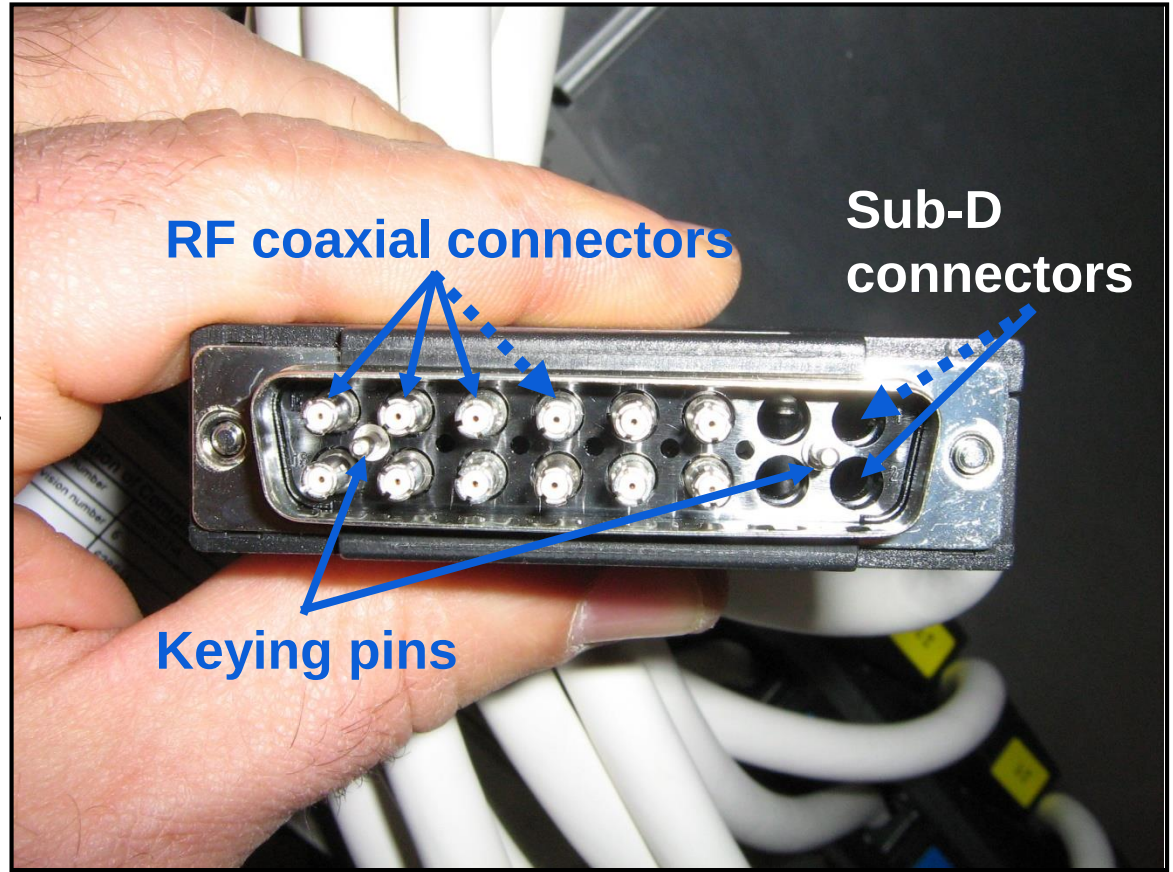


Rear Pedestal

The 16W16 receive connector of the **32 channel RF Hub**:

Keying 16W16 connectors prevents vertical swapping (a mix-up) of cables. Cables can be swapped side-to-side (the same connector plane) between RFSB units in a 32-channel RF Hub.

Most sub-D connectors only partially populated with RF coax connectors. The location of open ports varies between connectors.

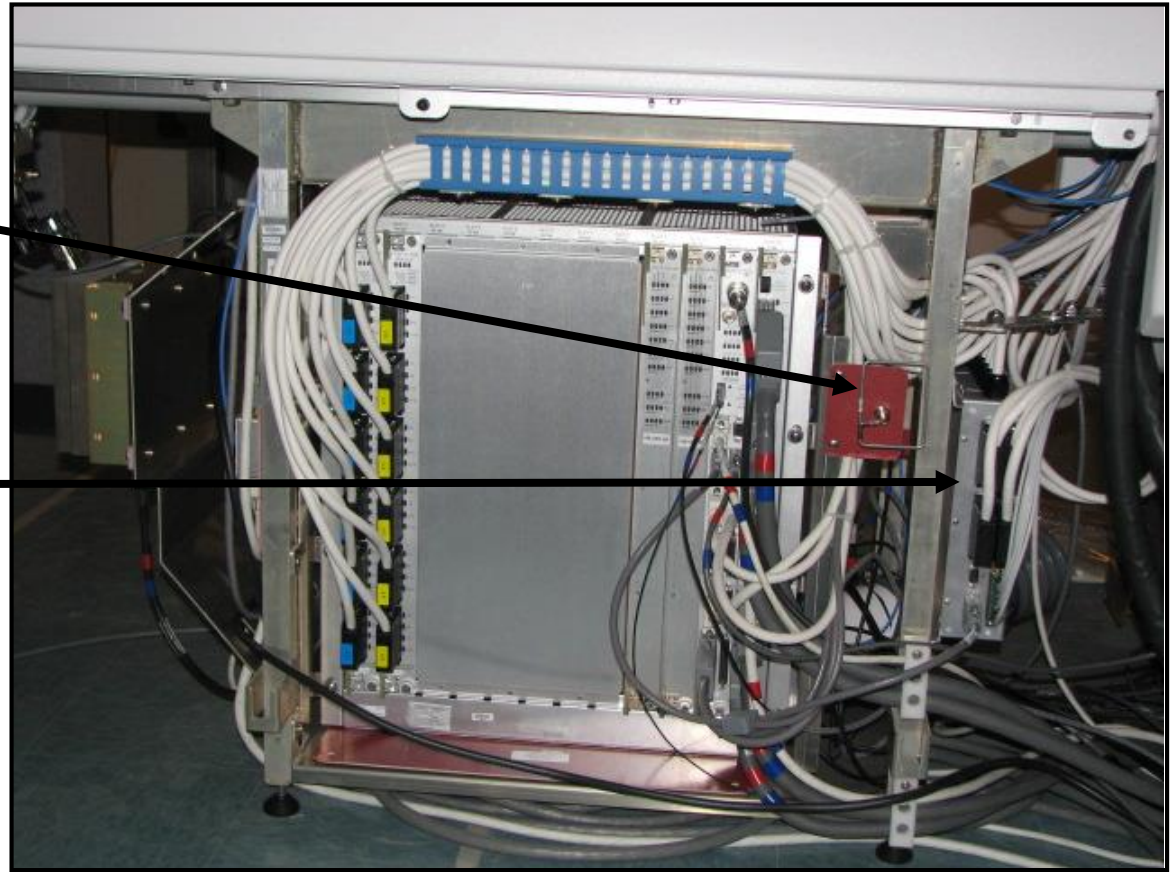


The **32 channel RF Hub:**

TDM antenna

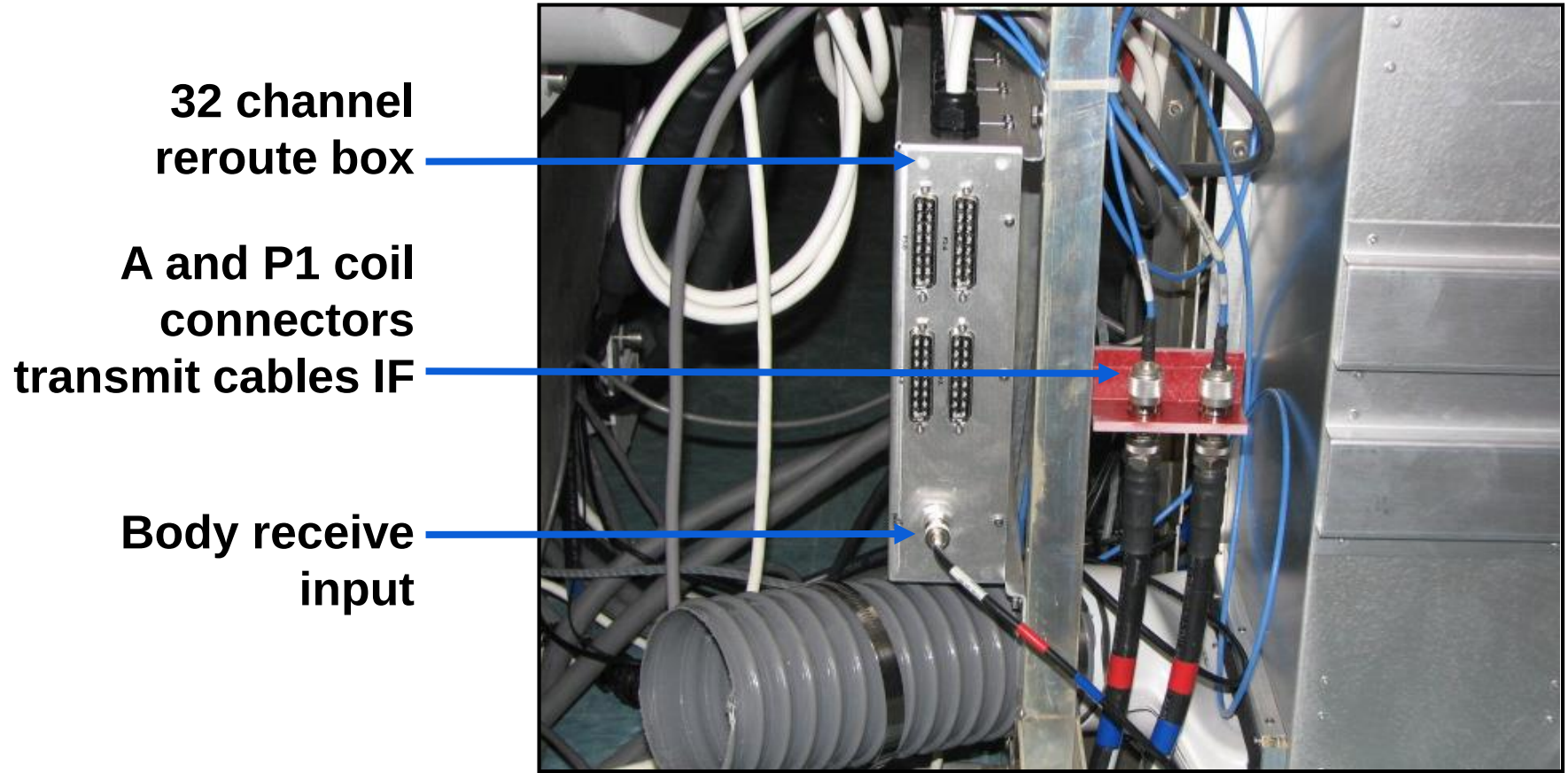
**32 channel
reroute box**

Note: 16 channel
systems contain a 16
channel reroute box.



Front view

The **32 channel RF Hub**:



Rear view

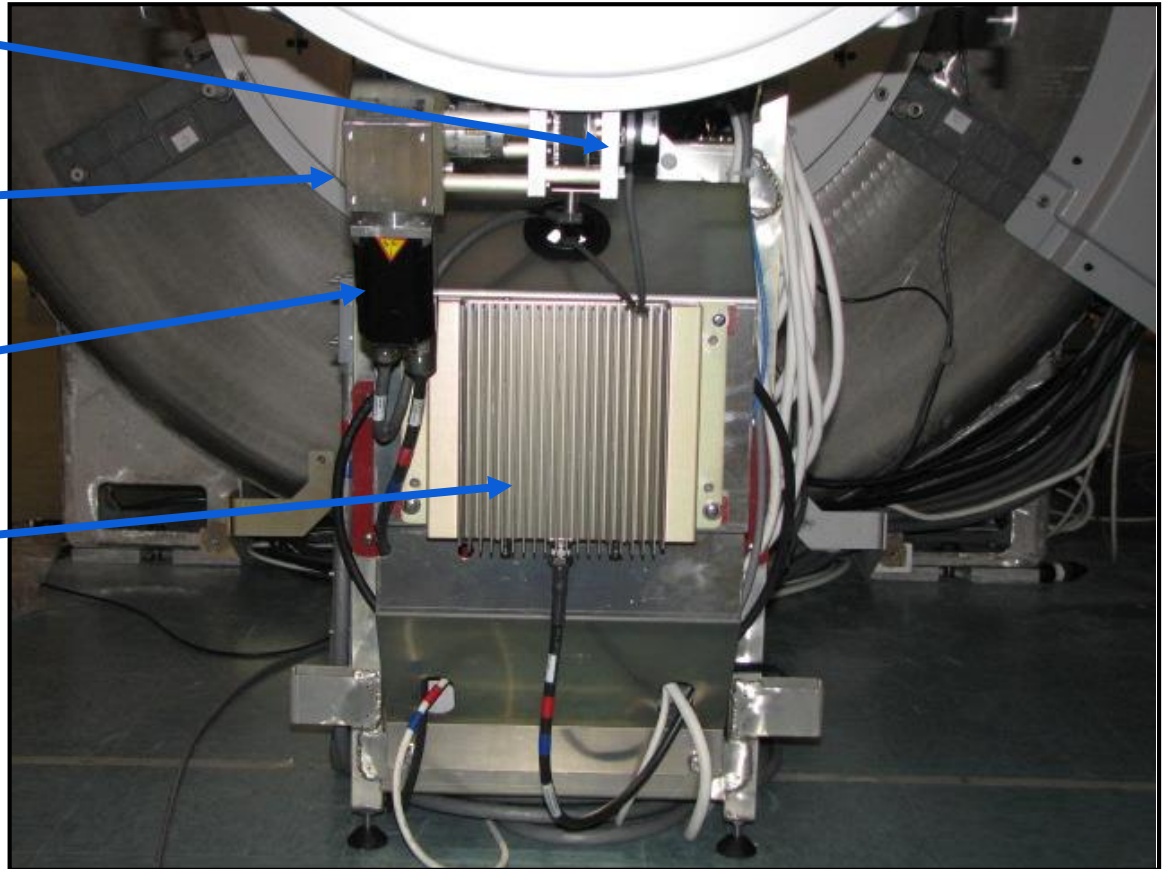
Rear Pedestal

Encoder

**Longitudinal
drive gear**

**Longitudinal
drive motor**

**Body hybrid
dummy load**



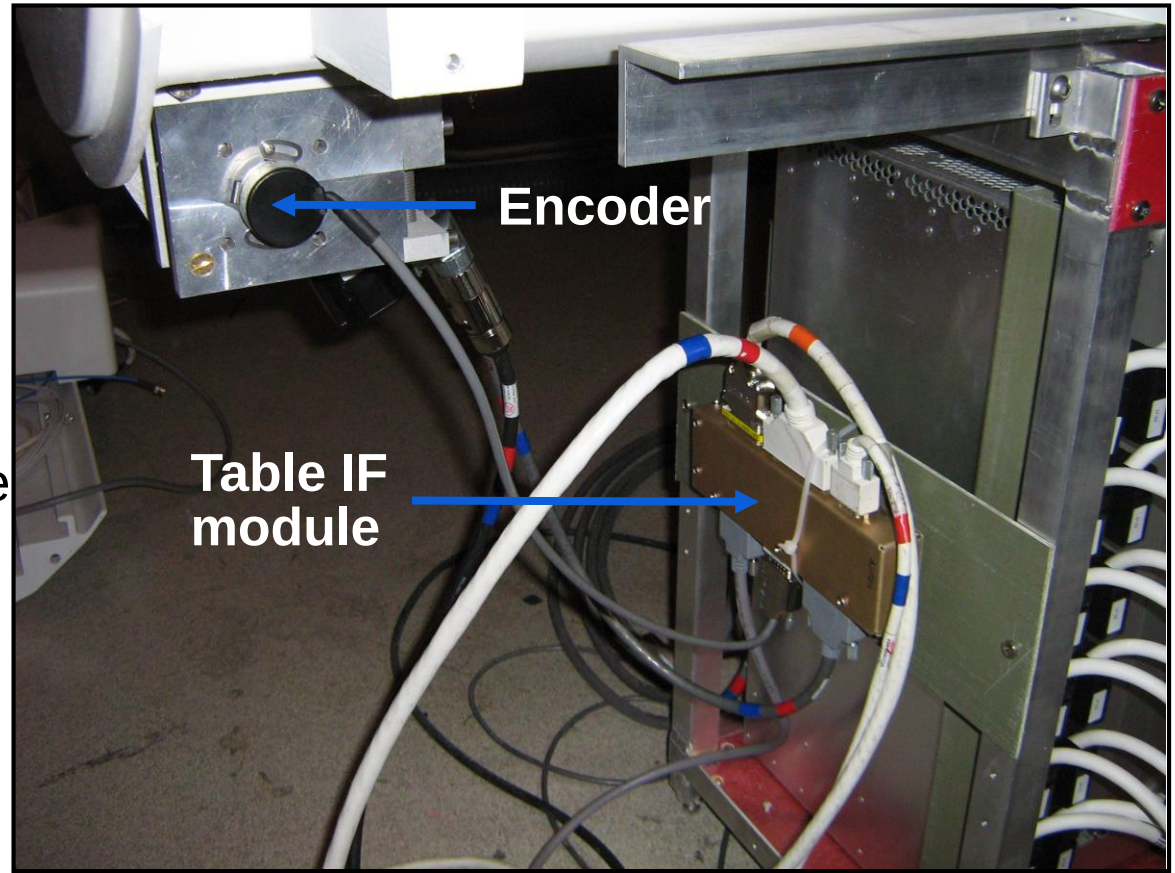
Note: The dummy load is only on a Discovery MR 750 (3.0T) system.

It absorbs reflected power from the body hybrid splitter during scanning.

End view

The **Table IF module**:

- Routes signals for
 - the LPCA
 - the motor and encoder
 - the PHPS2
 - the RF hub
 - the SRI4
- Supplies 5VDC to the longitudinal drive motor and the hall switches and encoder from 8VDC, supplied by the SRI4



Left side view (dummy load removed)

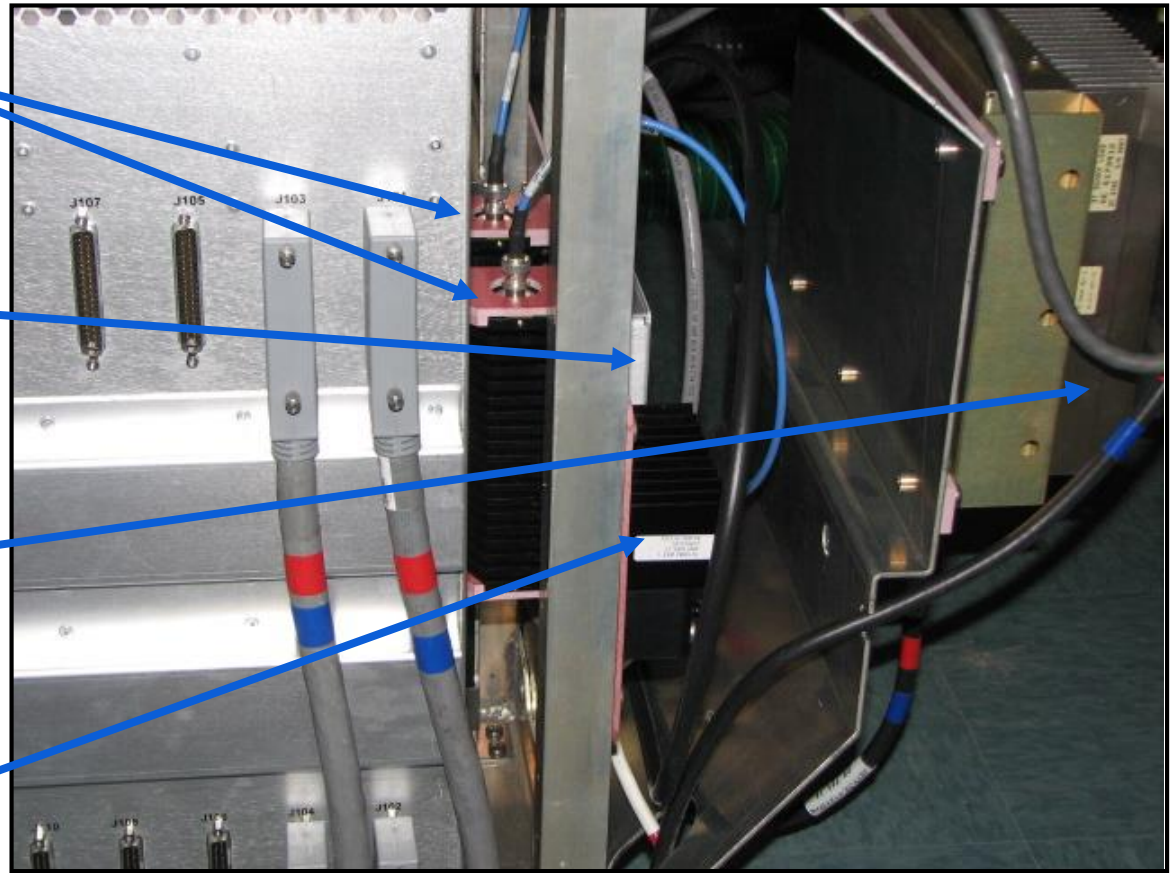
Rear Pedestal

**A and P1
connectors
reflected power
dummy loads**

**Table IF module
(obscured view)**

**Body hybrid
dummy load**

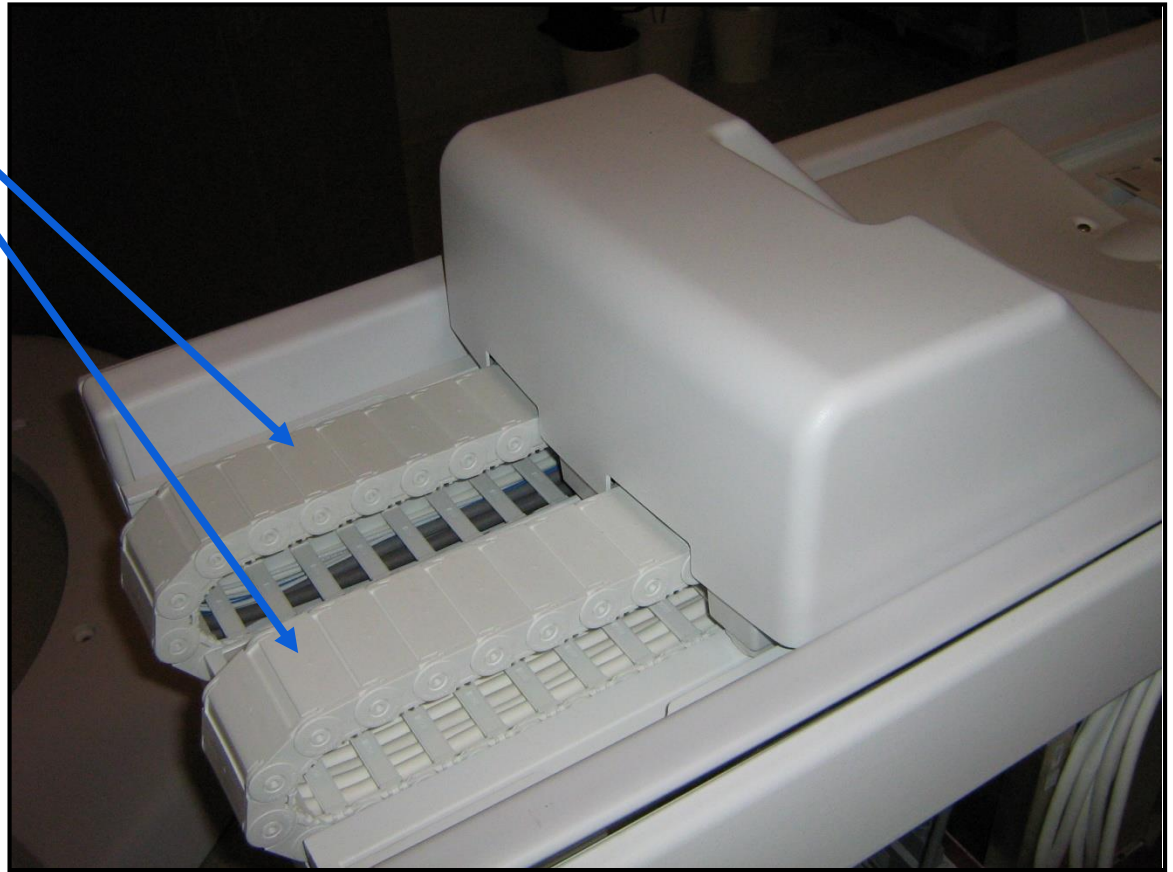
**Patient bore
light source**



Right side view (rear of RF hub)

Rear Pedestal - LPCA

Cable tracks



Rear view of the LPCA

Rear Pedestal - LPCA

- Each P connector in a 16-channel system supports 16 channels.
- Each P connector in a 32-channel system supports 32 channels.
- The Legacy A connector supports 8 channels.

**A connector
(supports transmit)**

**P1 connector
(supports transmit)**

P2 connector

**New carriage
hook design**



Front view of the LPCA

Rear Pedestal - LPCA

- There is no active circuitry inside the LPCA

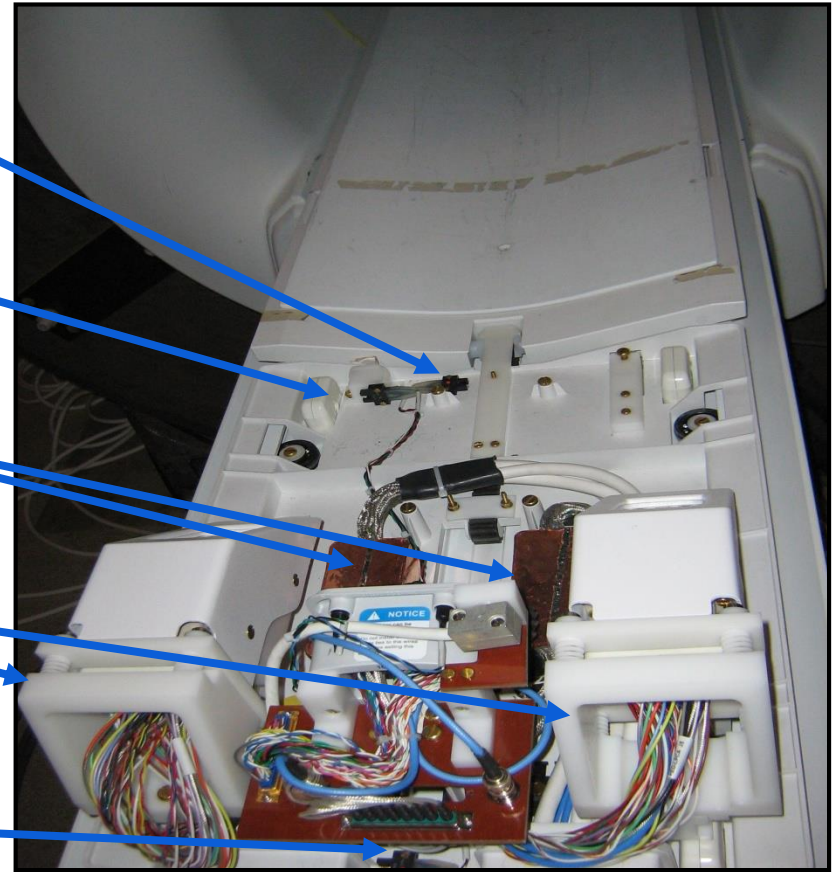
Front home sensor

**Cradle
engagement
sensor**

Cable track baluns

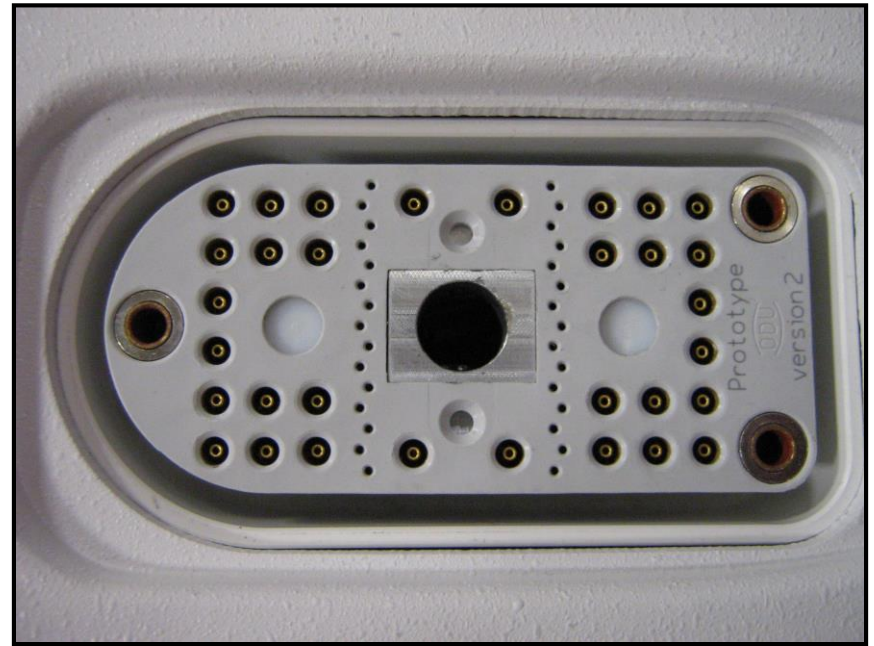
**P2 (left) and P1
(right) Quick-
disconnect**

Rear home sensor



**Rear view of the LPCA
(cover removed)**

- Each P connector in a **32**-channel system can support 32 receive channel ports.
- The P1 connector supports transmit and receive; all other P connectors are receive-only.
- MC bias is only provided on receive coax connections

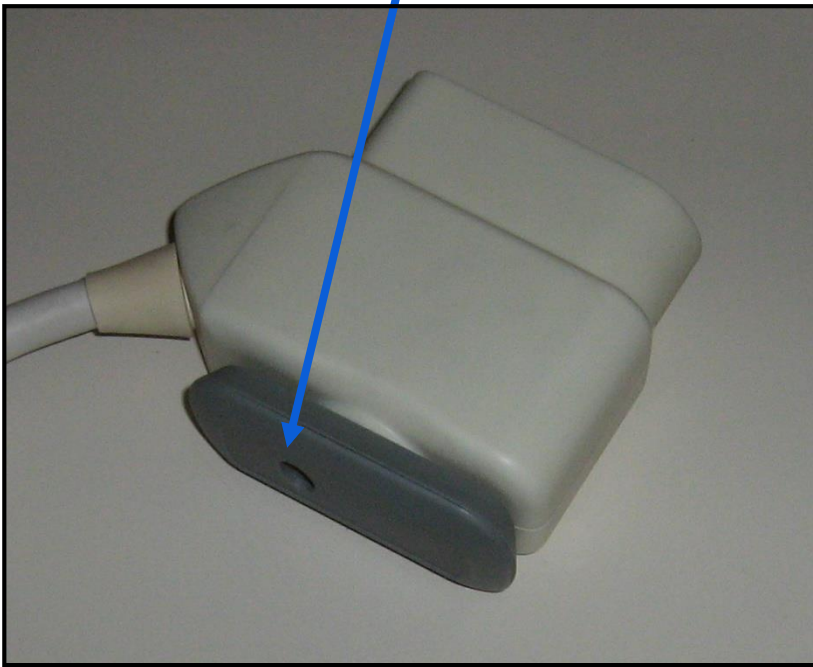


P3 connector (close-up view)

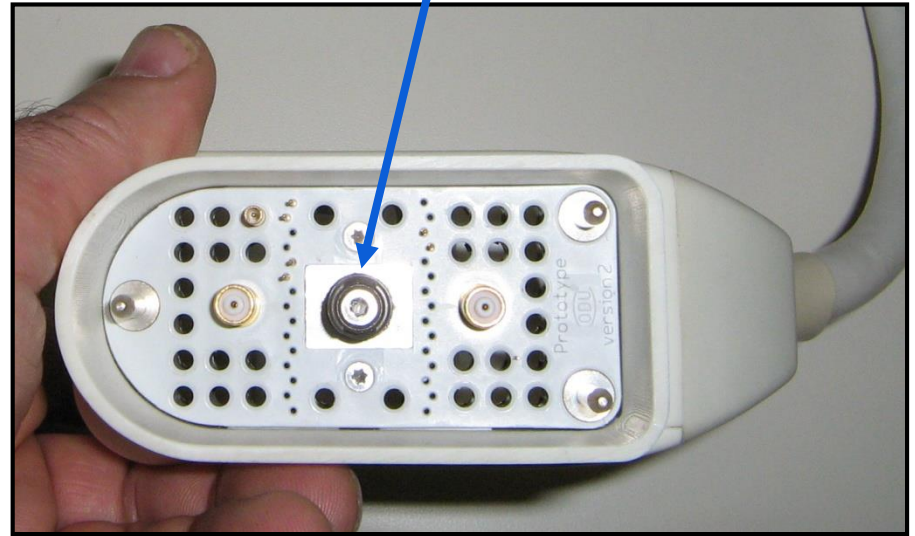
Key connector points

- The T-handle and threaded shaft design mechanically engage and disengage the coil and connector.
- Any customer can make a good coil connection.

T-handle



T-handle shaft



LPCA: 16-channel P connectors

Key connector points

- Notice that only **16** of the 32 ports are populated with a receive connector. The remaining 8 upper and 8 lower rows of ports are empty.
- Aside from possessing only 16 receive ports, the 16-channel P connector has the same form and fit as the 32-channel connector.

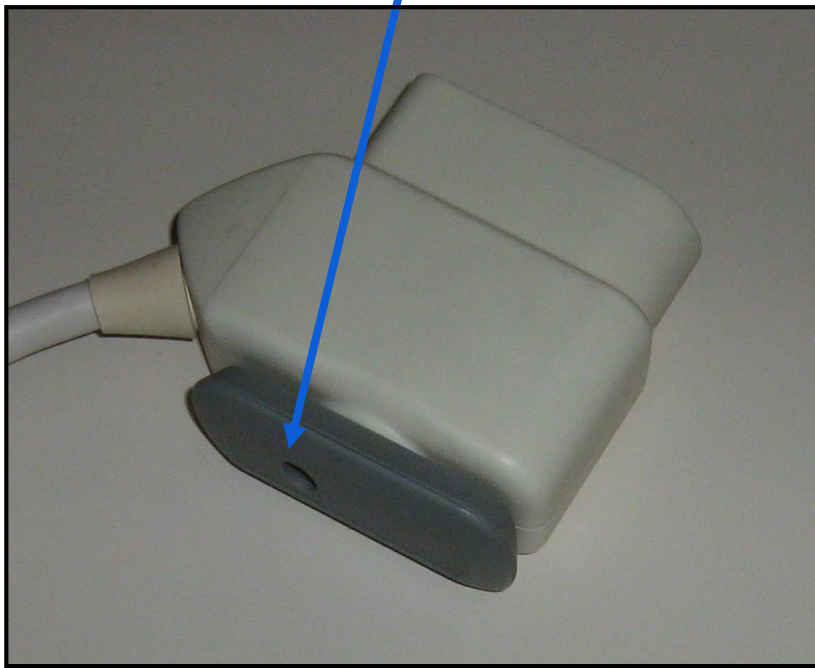


16 channel P1 connector (close-up view)

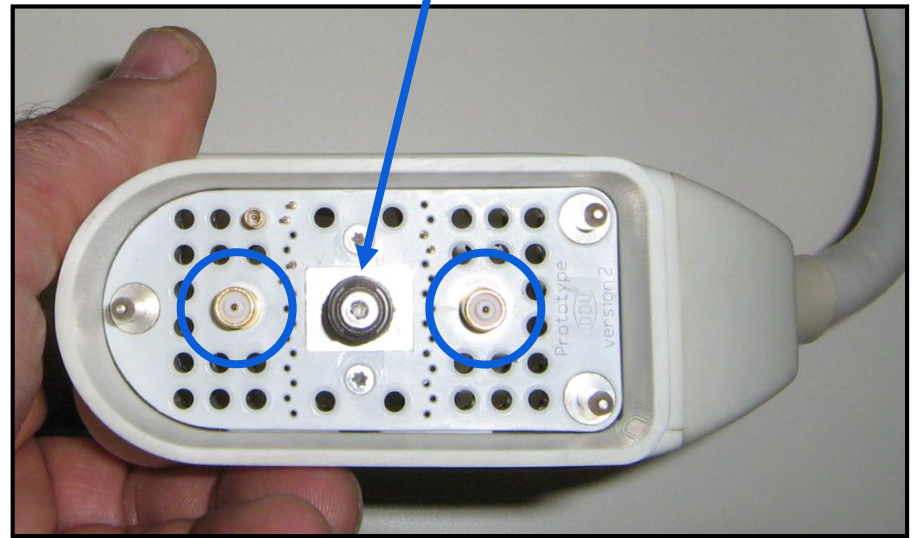
LPCA: 16-channel P connectors

- The large coax connection on the left is for transmit.
- The connection on the right is for reflected power from coil to load.

T-handle



T-handle shaft



LPCA: ODU connector

- The P connector attaches to the ODU emergency egress connector inside the LPCA.
- This permits quick egress of the patient from the bore during emergencies without risk of damage to the surface coil.
- This connector has a limited connection cycle life. It is part of the cable track FRU.

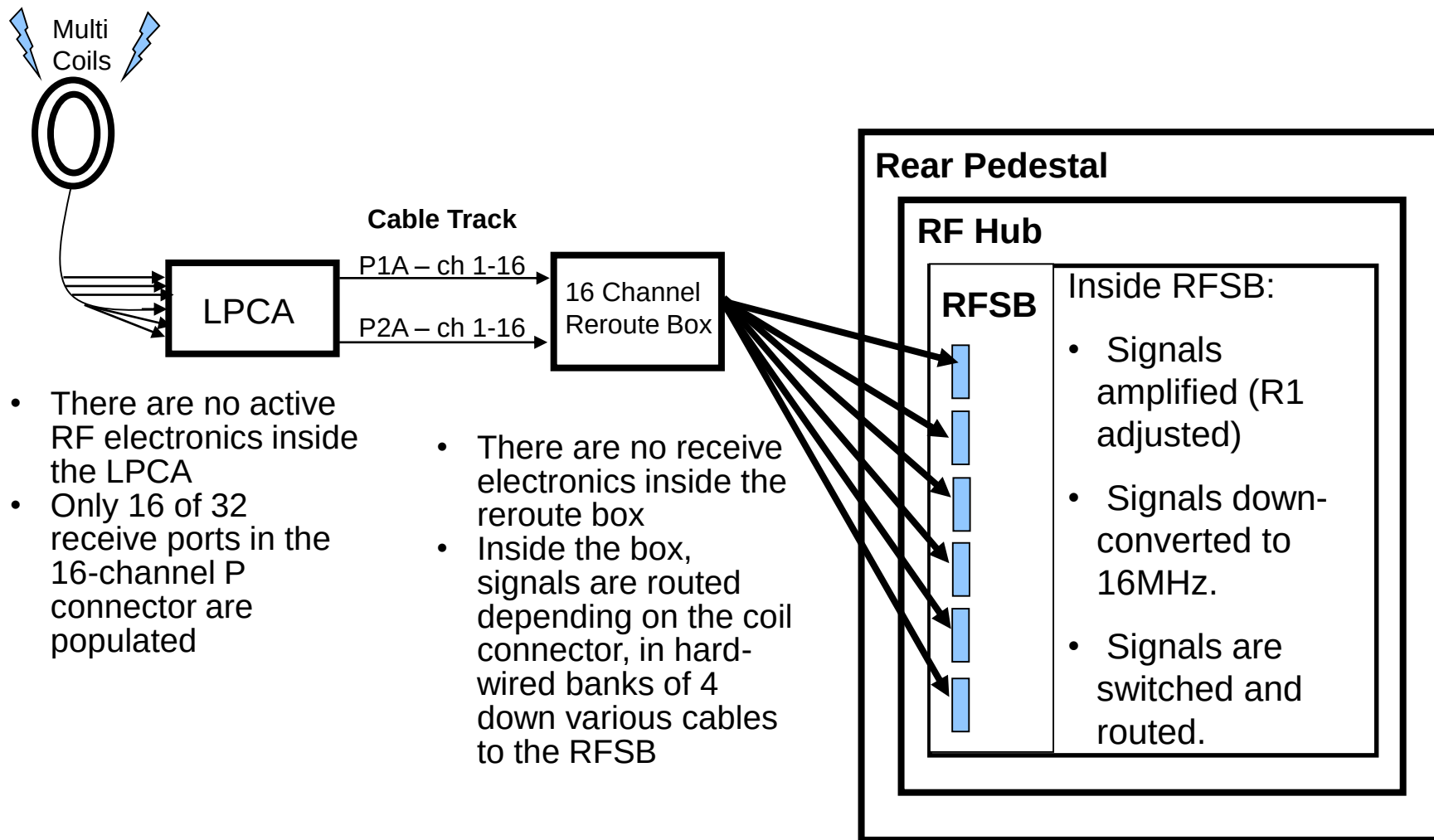


32-channel ODU emergency egress interface for P1



16-channel ODU emergency egress interface for P1

Receive chain theory: 16 channels

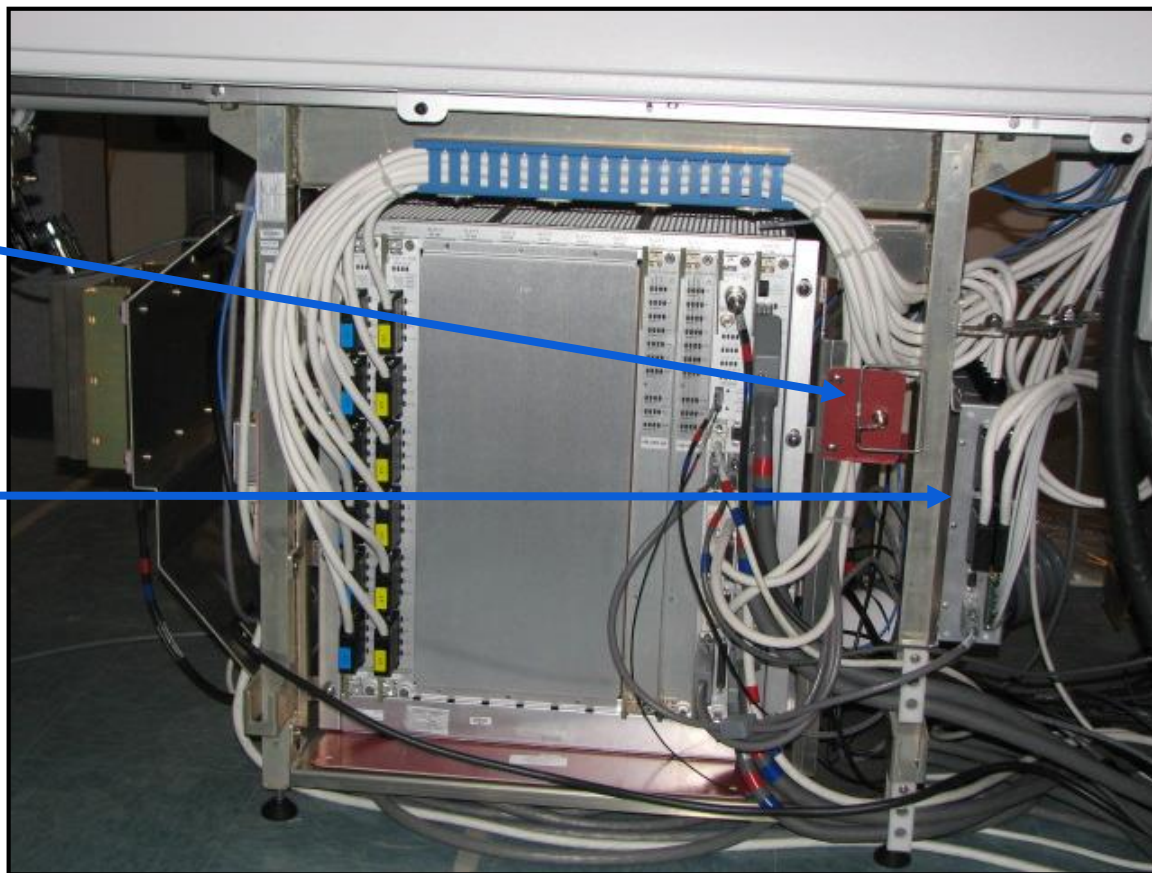


- There are no active RF electronics inside the LPCA
- Only 16 of 32 receive ports in the 16-channel P connector are populated

- There are no receive electronics inside the reroute box
- Inside the box, signals are routed depending on the coil connector, in hard-wired banks of 4 down various cables to the RFSB

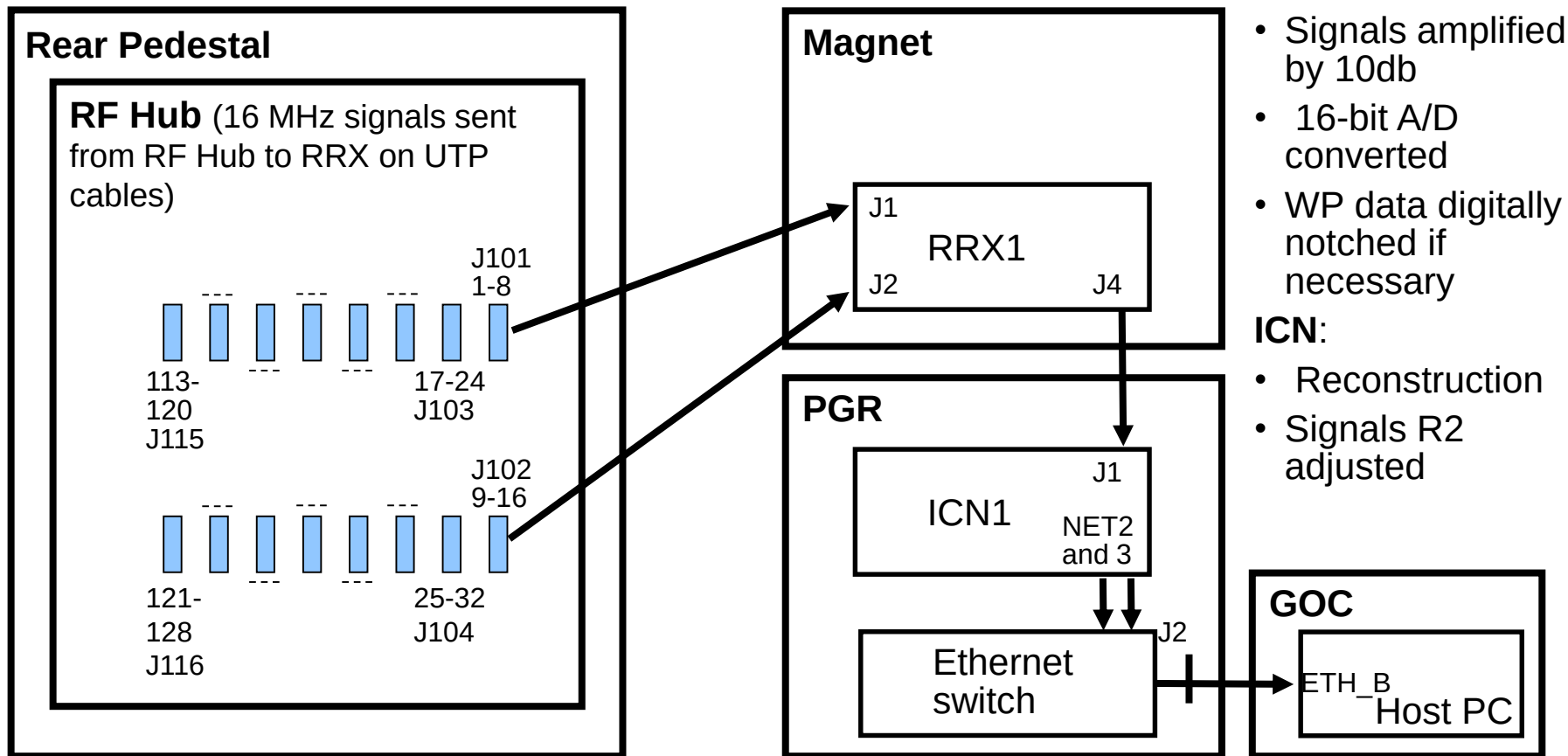
TDM Antenna

**32-channel
reroute box**



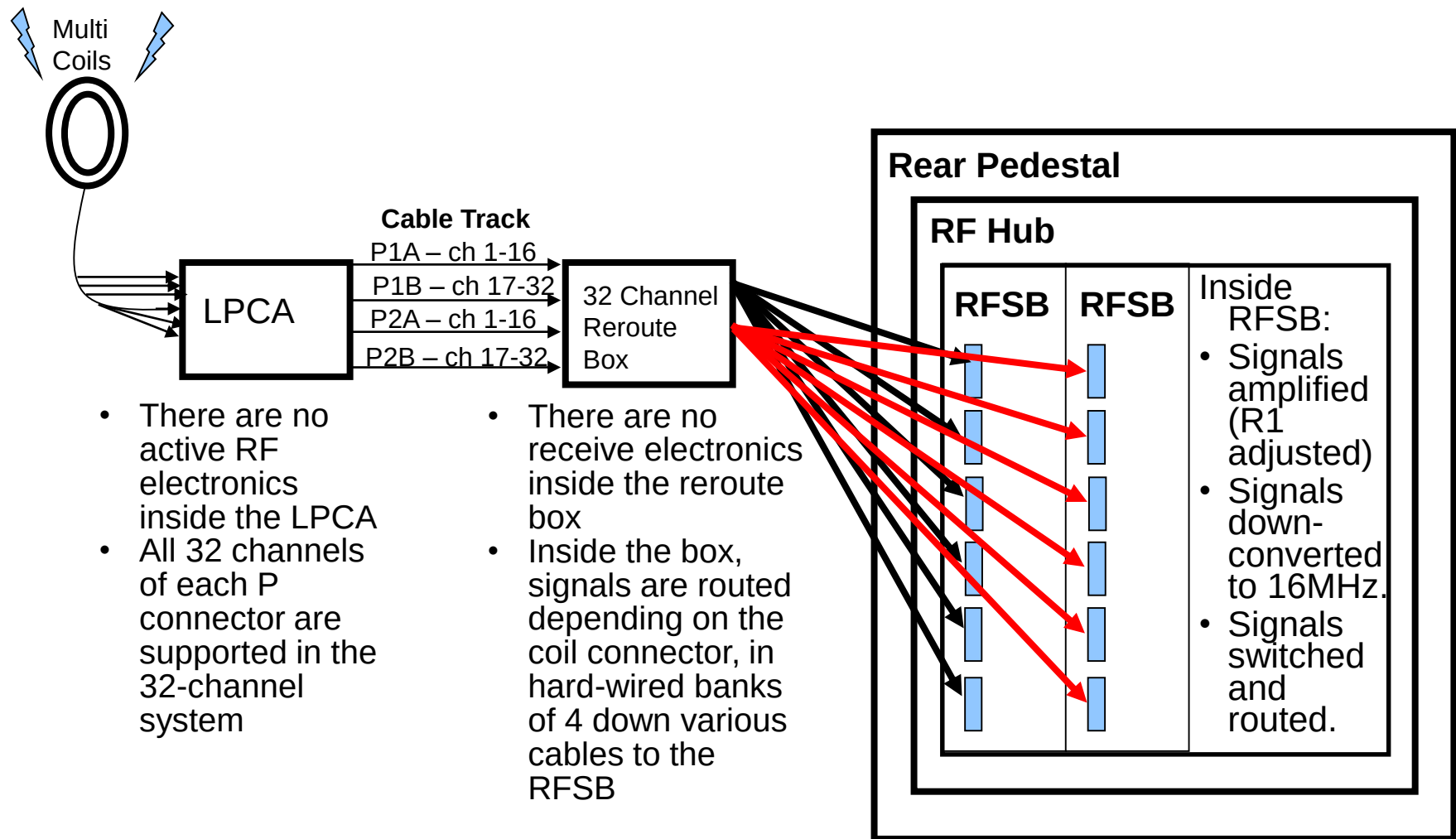
Front view of the RF hub

Receive chain theory: **16** channels

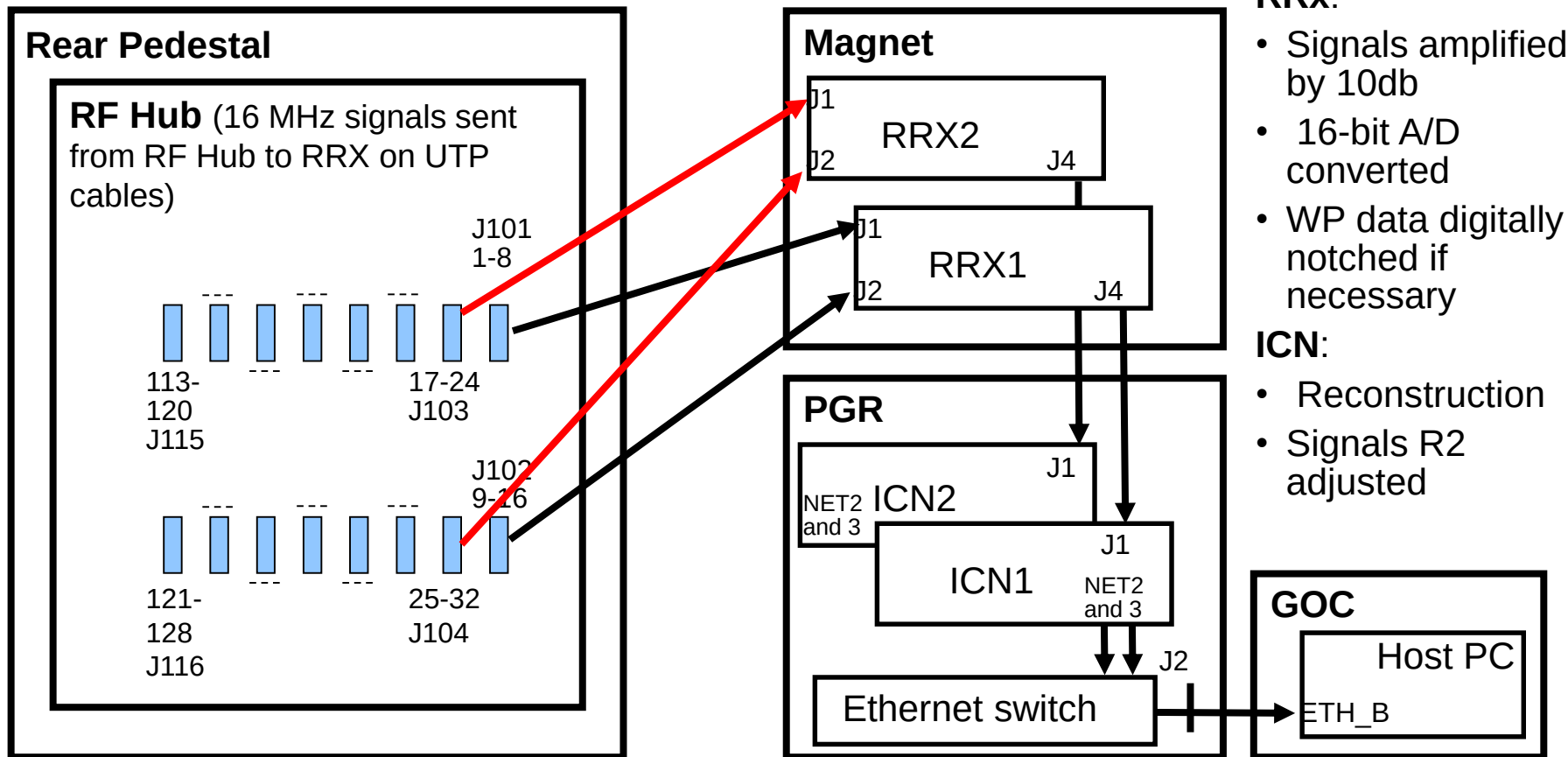


TDM (not shown) provides power to the RRXs and detects WP events from the antenna, but is *not* in the receive signal path.

Receive chain theory: 32 channels



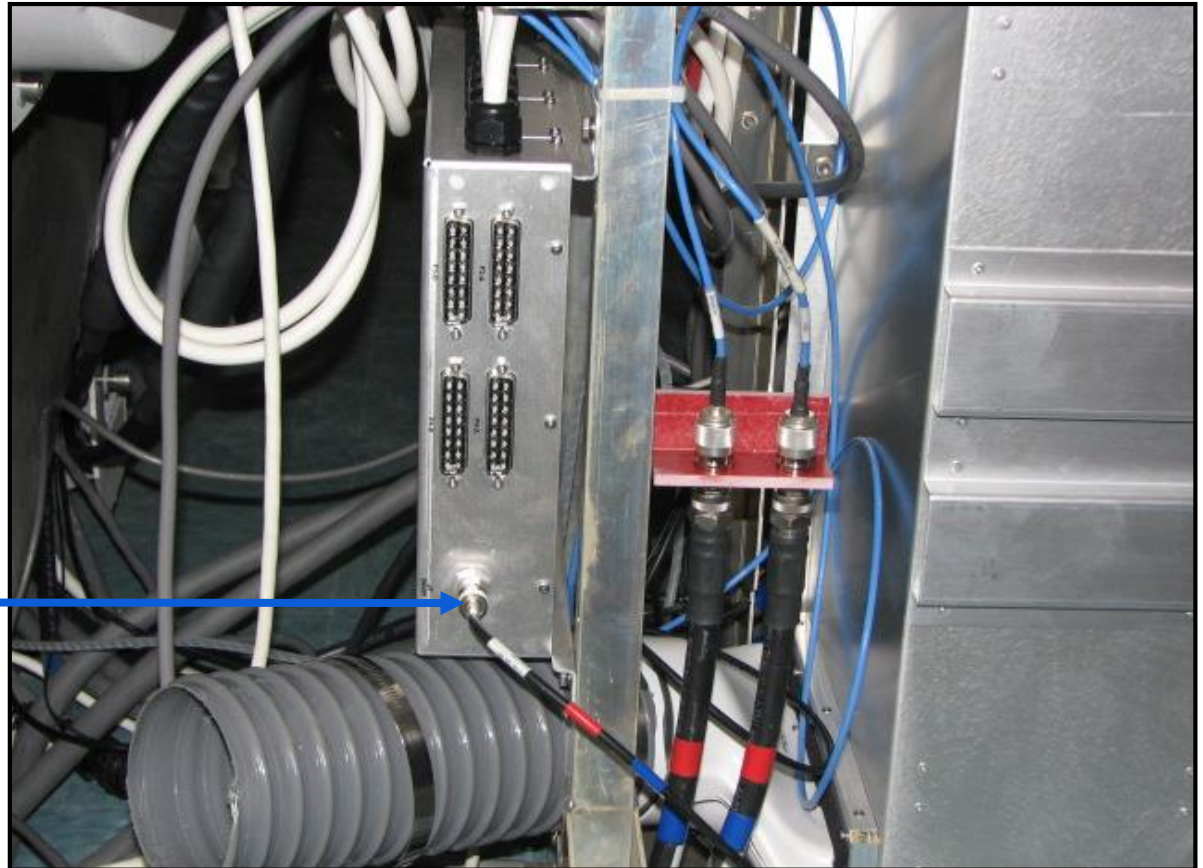
Receive Chain Theory: 32 channels



TDM (not shown) provides power to the RRXs and detects WP events from the antenna, but is *not* in the receive signal path.

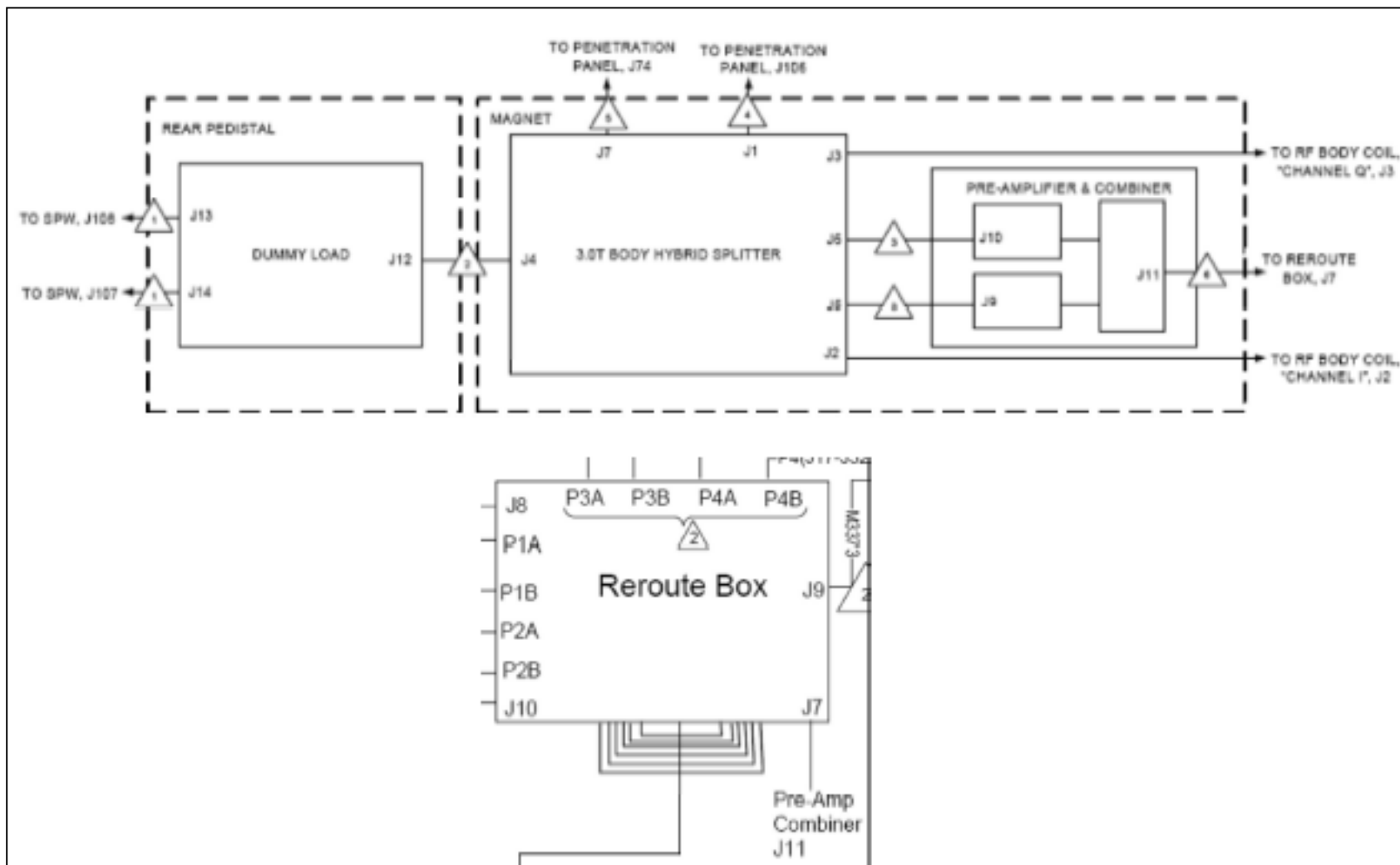
Body Receive

Body receive input



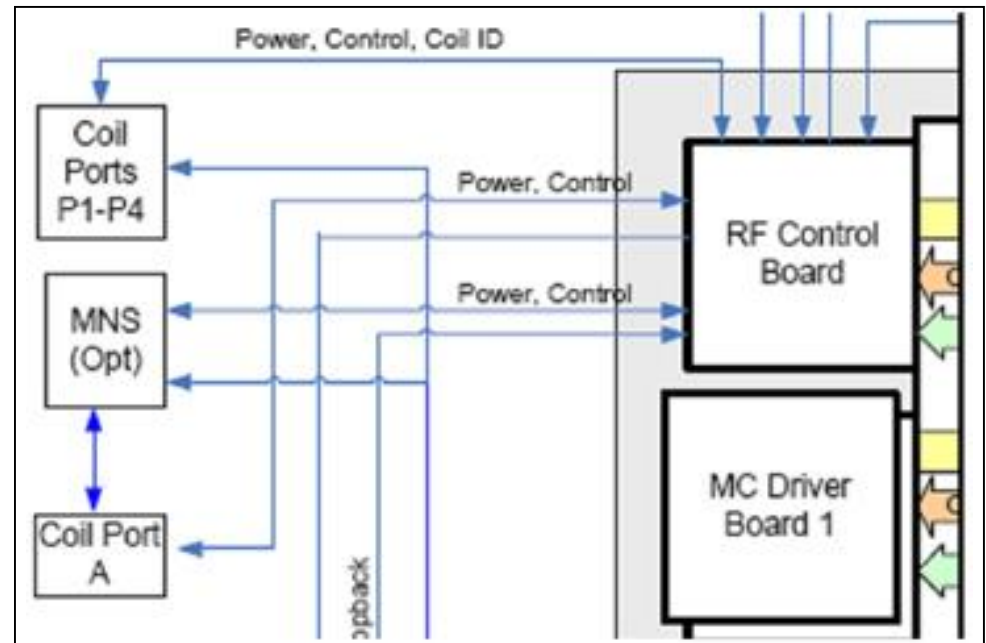
Rear view of the RF hub

Body Receive

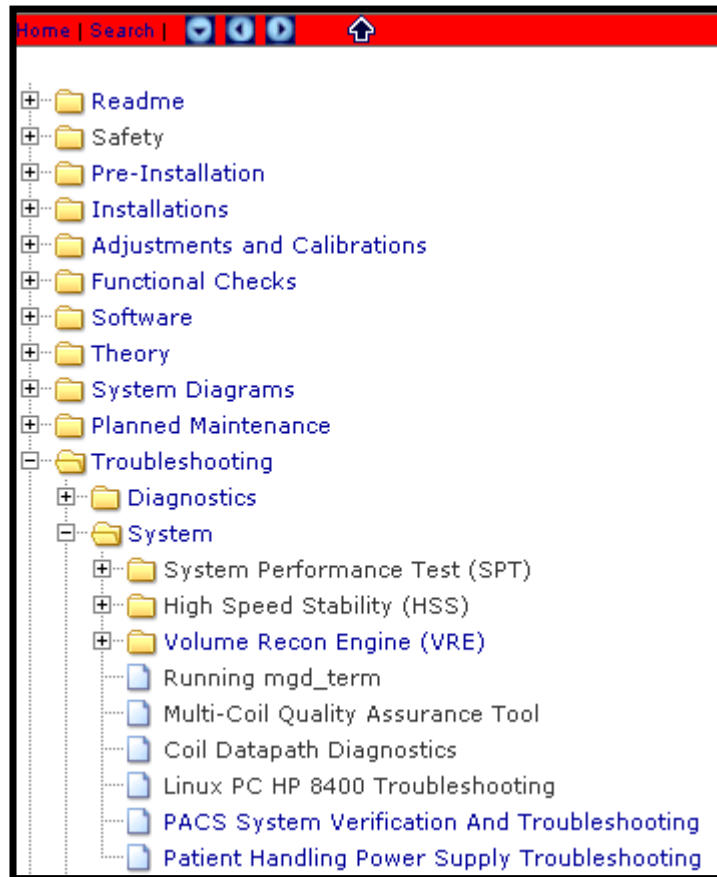


Coil ID theory

- Coils that connect into the A or P ports in the MR450 table must be Coil-ID compatible
 - P1, P2, and A ports are in the low-profile carriage assembly (LPCA) while P3 and P4 ports are inside the MR450 table
- The system will not permit the customer to scan with coils that interconnect with these connectors, but are not equipped with a coil-ID chip.

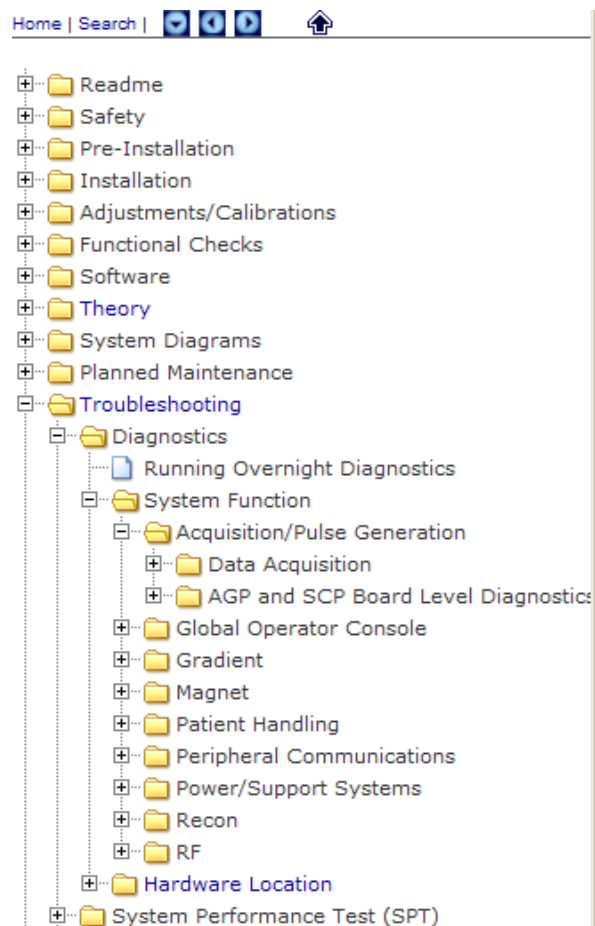


- You can access some troubleshooting information from the MR Service Documentation.

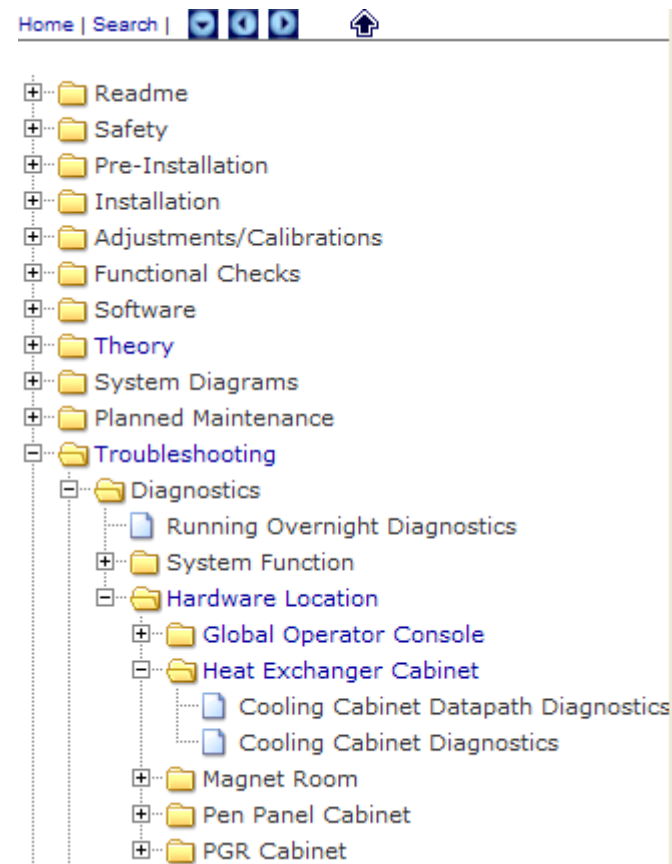


System troubleshooting procedures

- You can access diagnostics information from the MR Service Documentation redundantly under function, or under hardware location.



System Function Diagnostics



Hardware Location





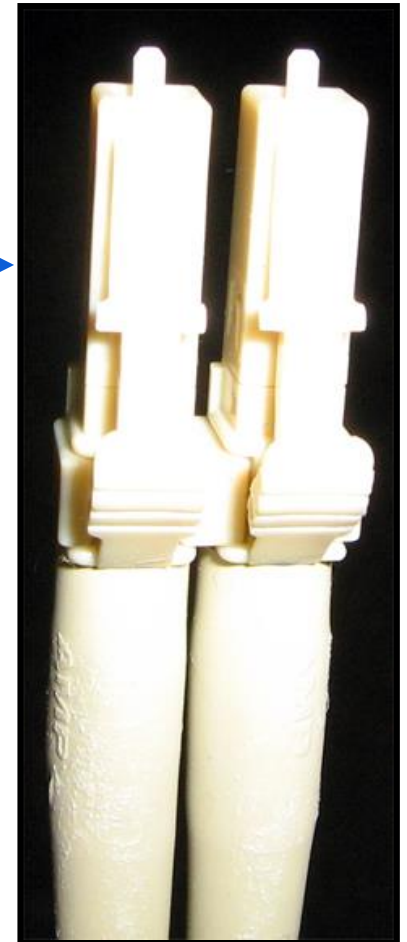
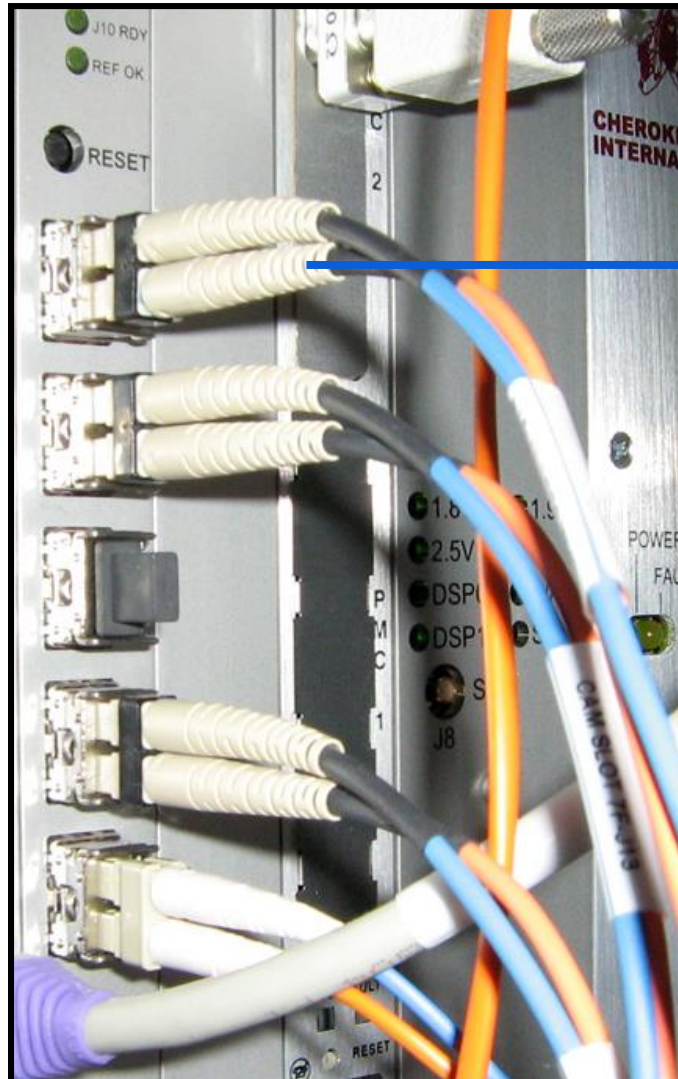
GE Optima/Discovery MR450 & MR750

Objectives

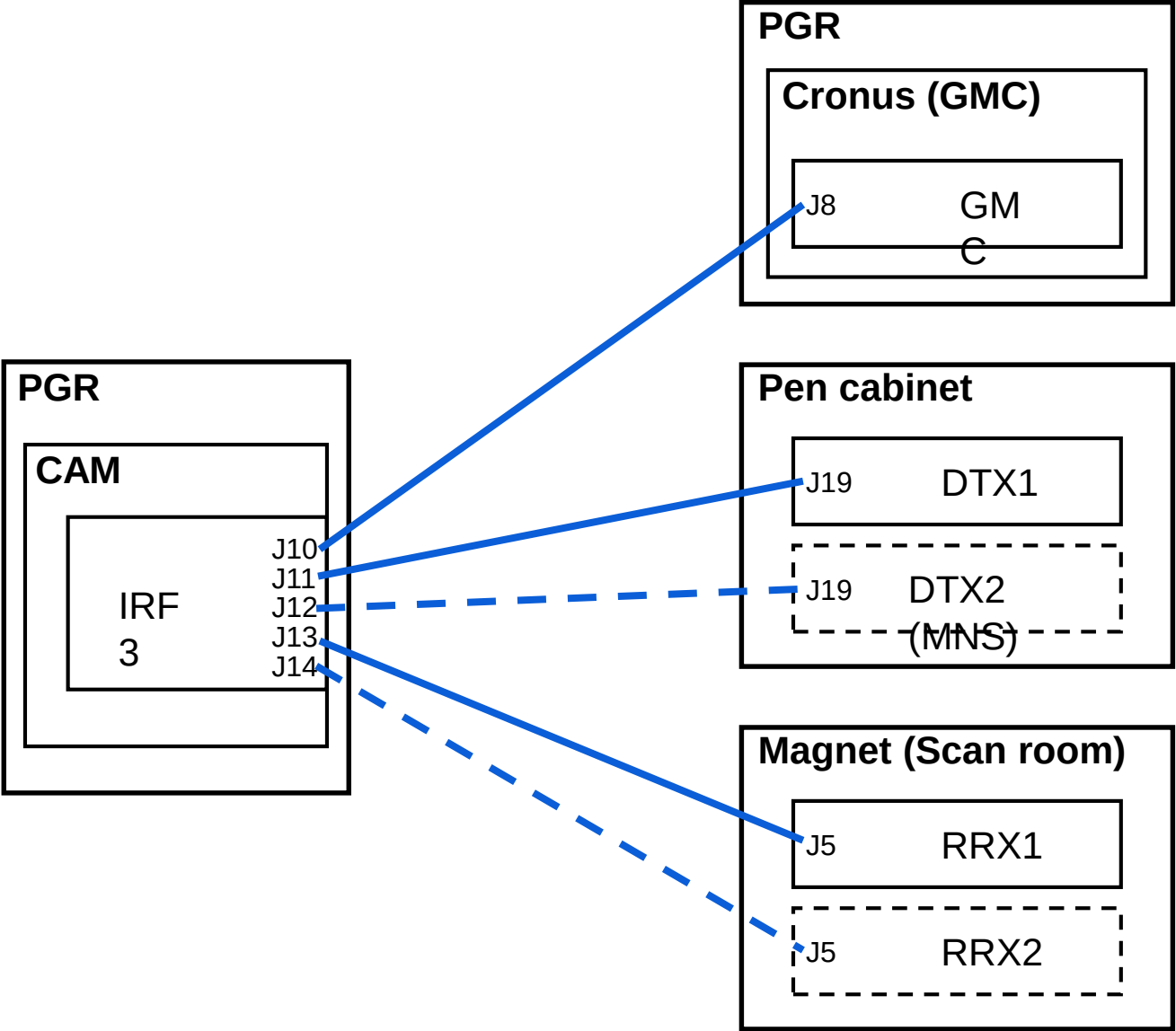
- Be able to recognize the function of the various communication links
- Use communications links for system peripheral diagnostics
- Trace signal flow using block diagrams

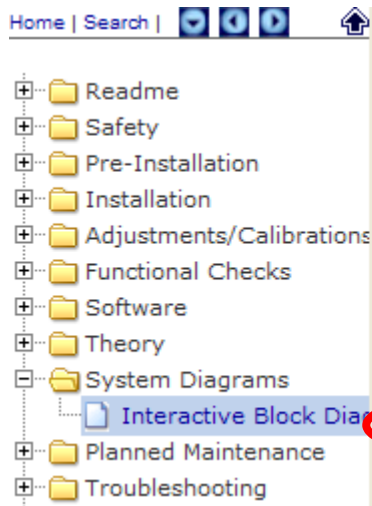
- The following are the available links for the Discovery MR:
 - DVMR Link
 - VRF Data Link
 - Infiniband Link
 - XGD Data Fiber Links
 - CAN Link
 - Clock Synchronization Fibers
 - Auxiliary Board Fiber Link
 - Ethernet Communication
 - STRB Fiber Link
 - Octavius Link
 - 1-Wire Network
 - In-Room Display Fiber Link

- PGR / CAM / IRF3



DVMR link





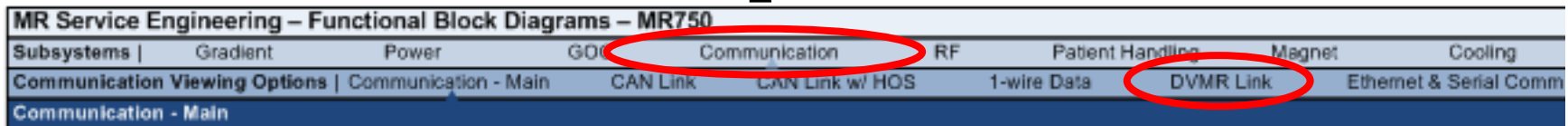
These block diagrams, along with the provided diagnostics, are intended to be the primary tools for troubleshooting the MR System from a system or subsystem level down to the specific module or board level. When using the diagnostics, the block diagrams can provide a better understanding of what circuitry the diagnostics are testing, and how to further isolate a problem if required.

Interactive Block Diagrams

[MR450w Block Diagrams](#)

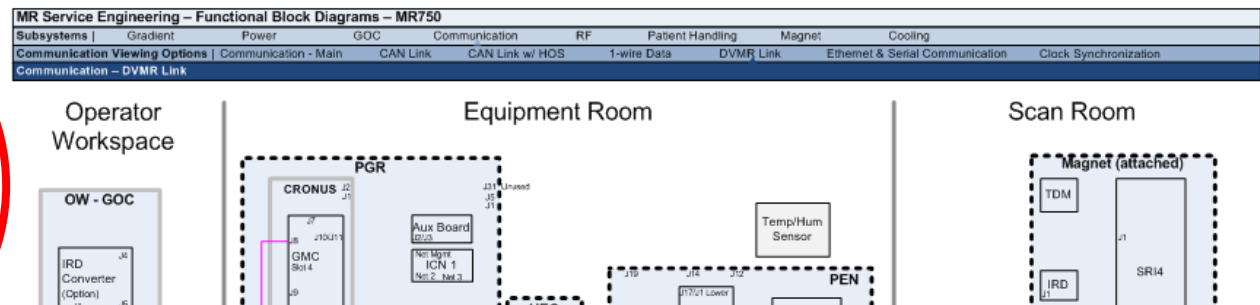
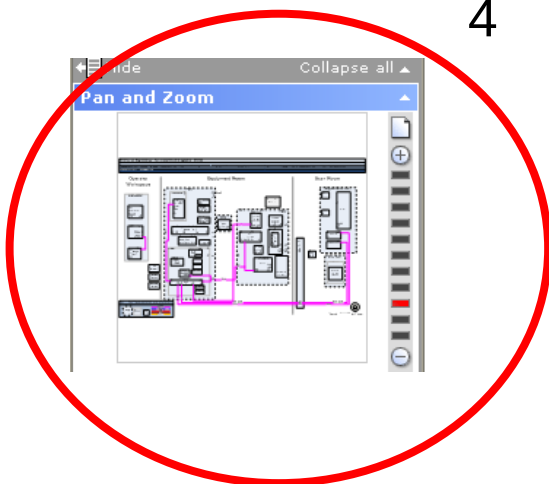
1

2



3

4





- DVMR Link traffic consists of:
 - **Status and control** signals for all devices shown
 - **Gradient waveform data**
 - **RF waveform data**



PGR

Cronus (GMC)

J8

GMC



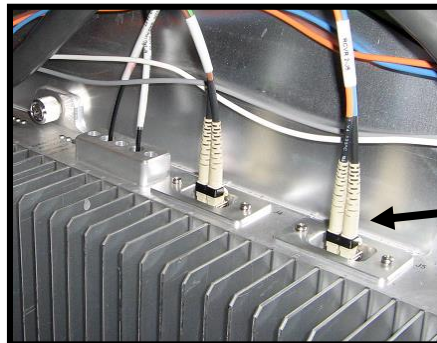
Pen cabinet

J19

DTX1

J19

DTX2 (MNS)



Magnet (Scan room)

J5

RRX1

J5

RRX2

- Other features of the DVMR Link:
 - Glass fibers
 - Continuous dual TX/RX fiber optic cable; the fibers have orange and blue jackets
 - IRF3 is like the Ethernet switch. Fiber connections at IRF3 can be made to any open port; they are not required to connect to a specific port
 - Any failure of a device on the link can cause it to hang up
 - The system identifies the port causing the hang up in the error log
 - The system identifies the port causing the hang up in the error log
 - J10 - J14 RDY LEDs on the front of the ERF3 board illuminate for active link status. The LEDs are not illuminated when the status is inactive
 - Data transfer rates are up to 2Gb/second



TX/RX

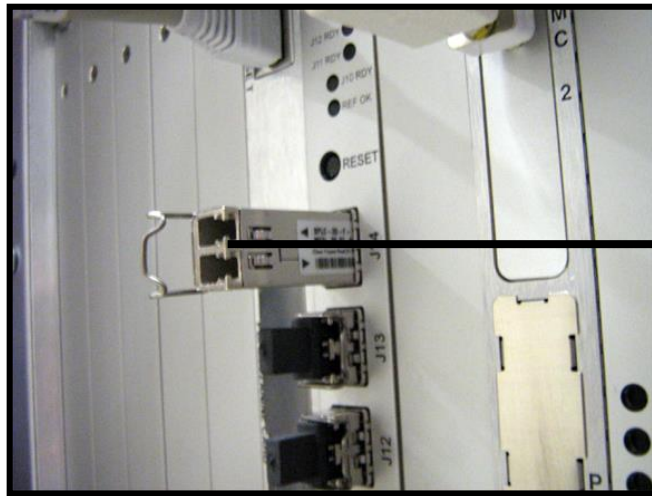
**fiber optic
cables**

Link status

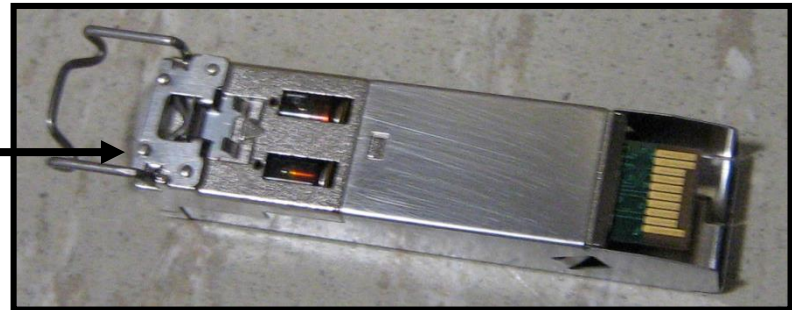


DVMR Communication Links

- All devices in DVMR link utilize removable TX/RX fiber optic modules
- You can swap TX/RX modules between devices
- 5183328-2 (newly listed FRU)



**TX/RX module partially
extended from the IRF3 board**



**Close-up of a removed TX/RX
module**

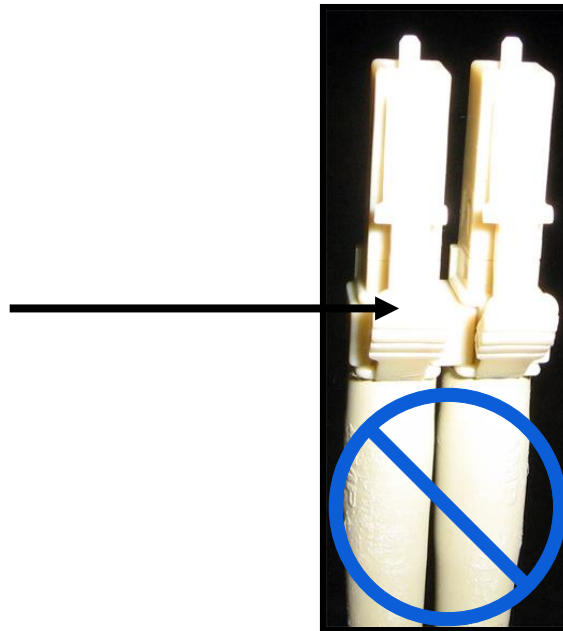
DVMR Communication Links

– While removing fiber connectors is straightforward, installing them so they lock into the TX/RX module requires technique:

- Grip the sides close to the hard plastic housing near the release latches
- Push the assembly into the TX/RX unit

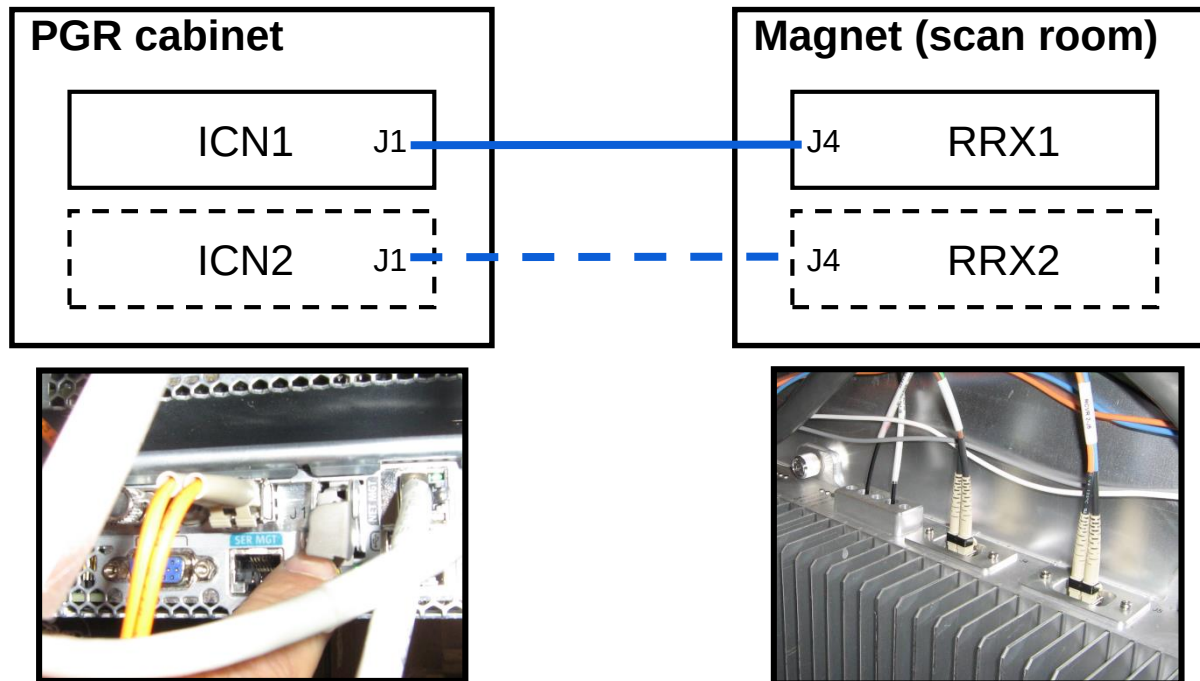
It is difficult to install fibers so that they lock into the module while holding the fibers by the boots.

Grip the sides of the connectors, not the ends

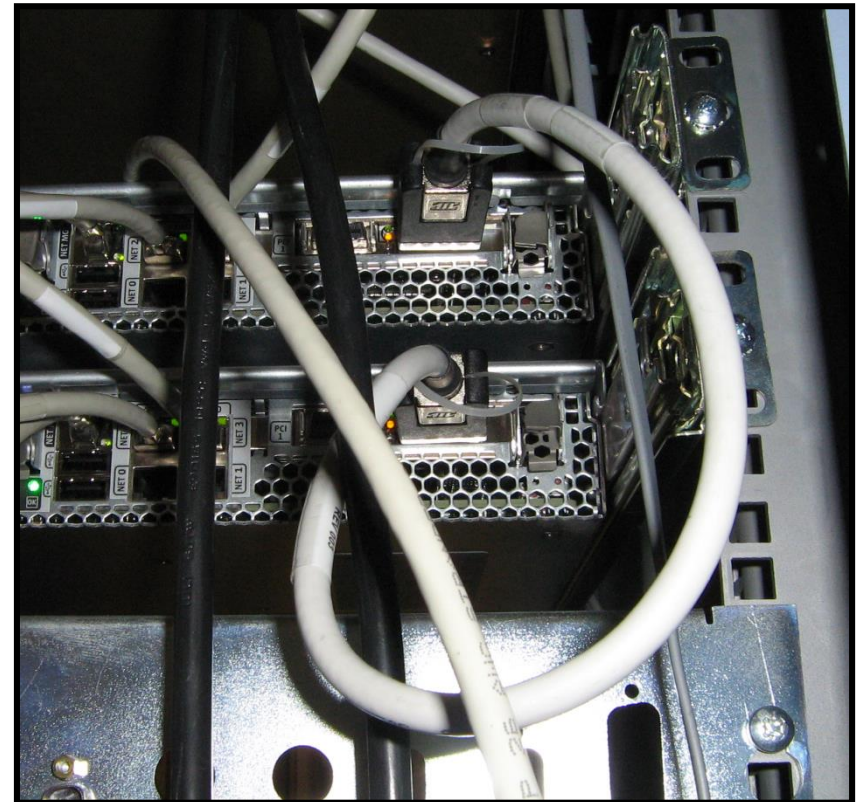
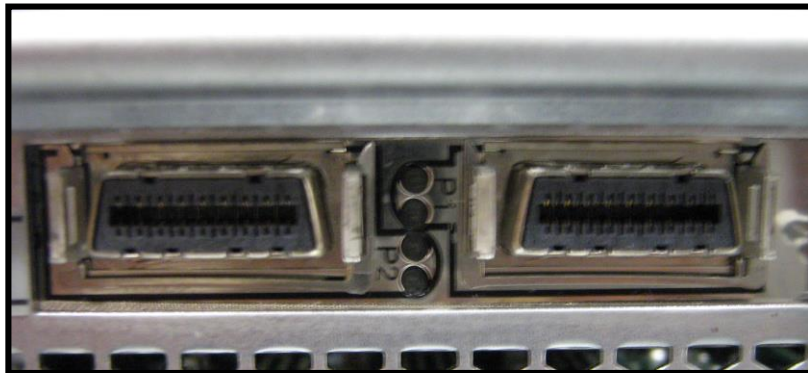
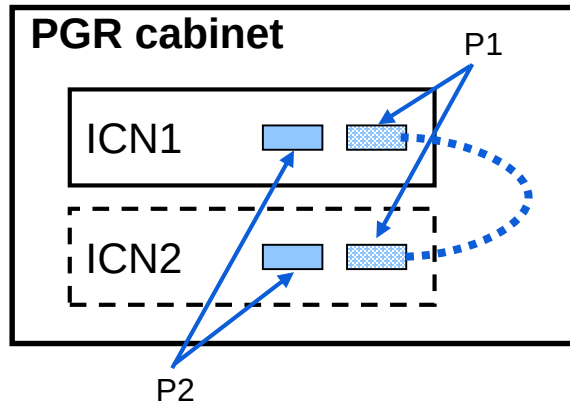


VRF Data Link

- The VRF data link:
 - is made of glass fibers
 - is a dual TX/RX fiber optic link; fibers have brown and green jackets, but otherwise look the same as DVMR Link fibers
 - is a means for transferring digitized **raw image data** from RRX to ICN
 - has fibers that can be swapped between ICNs and RRXs with no ill effects; the system can differentiate data between fibers



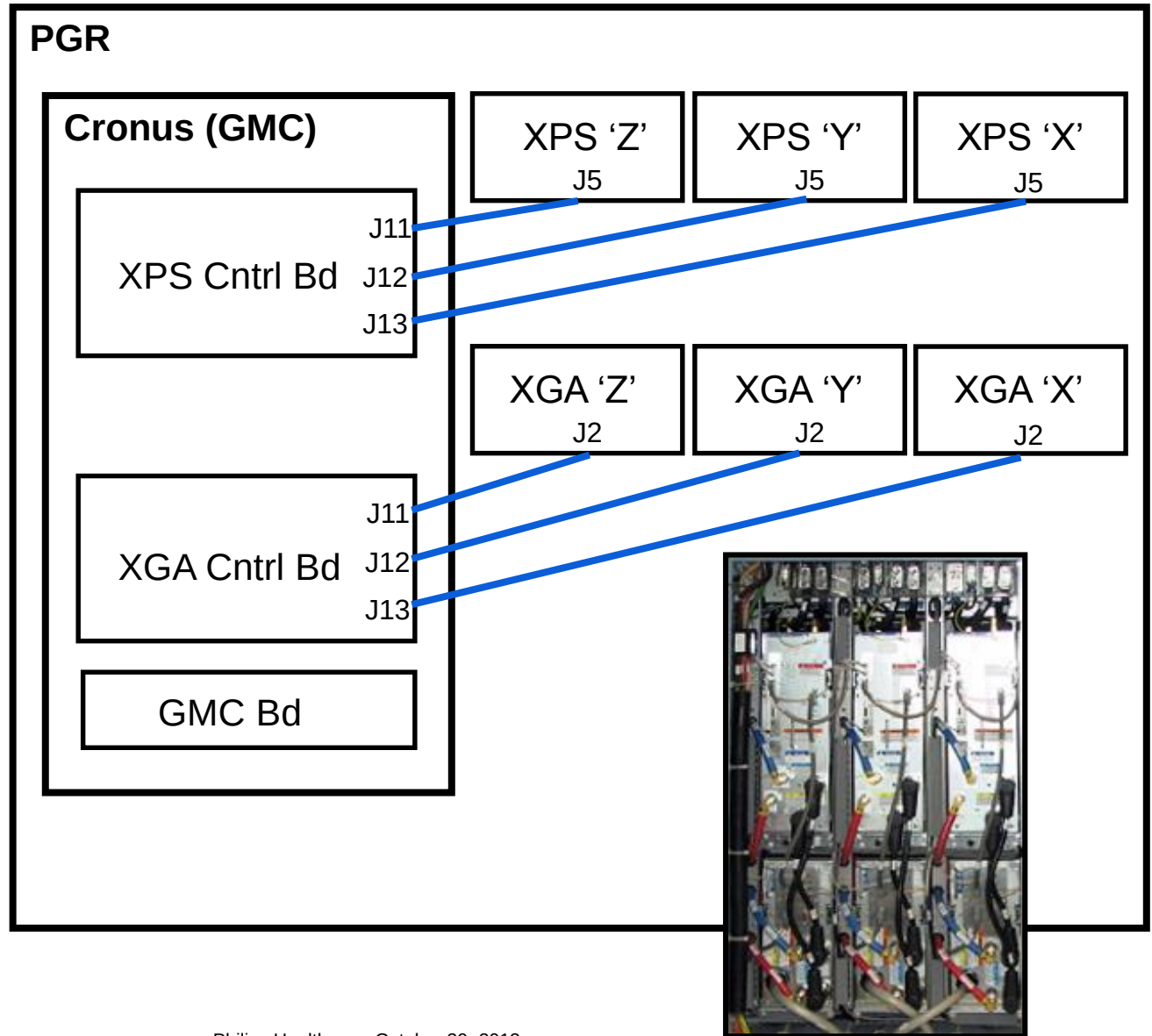
- Infiniband link:



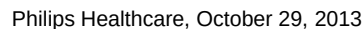
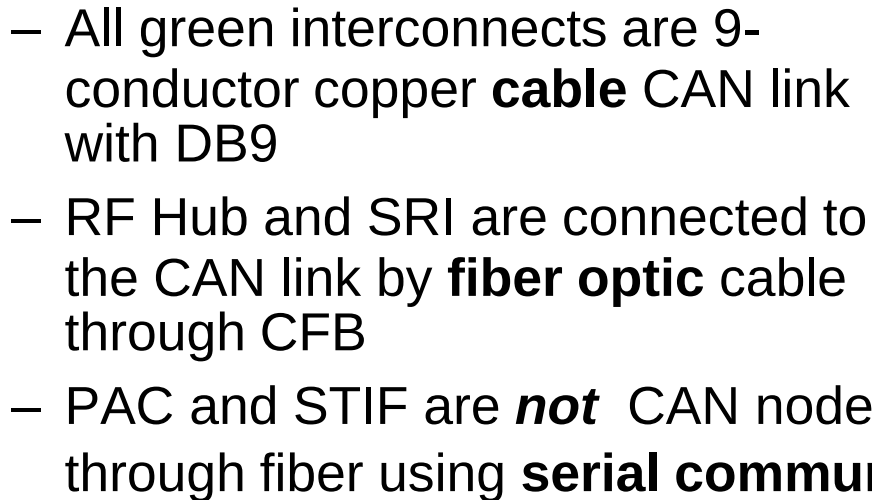
**Rear view of
ICNs**

XGD Data Fiber Links

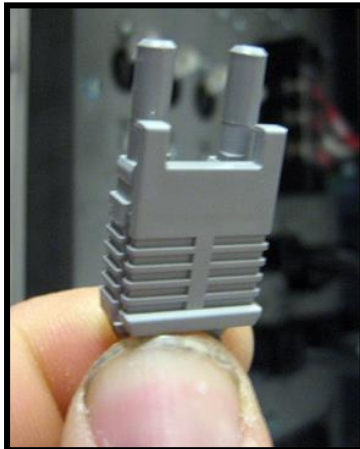
- The XGD data fiber links provide **gradient waveform data** and control information.
- Swap dual fibers to swap gradient axis data.



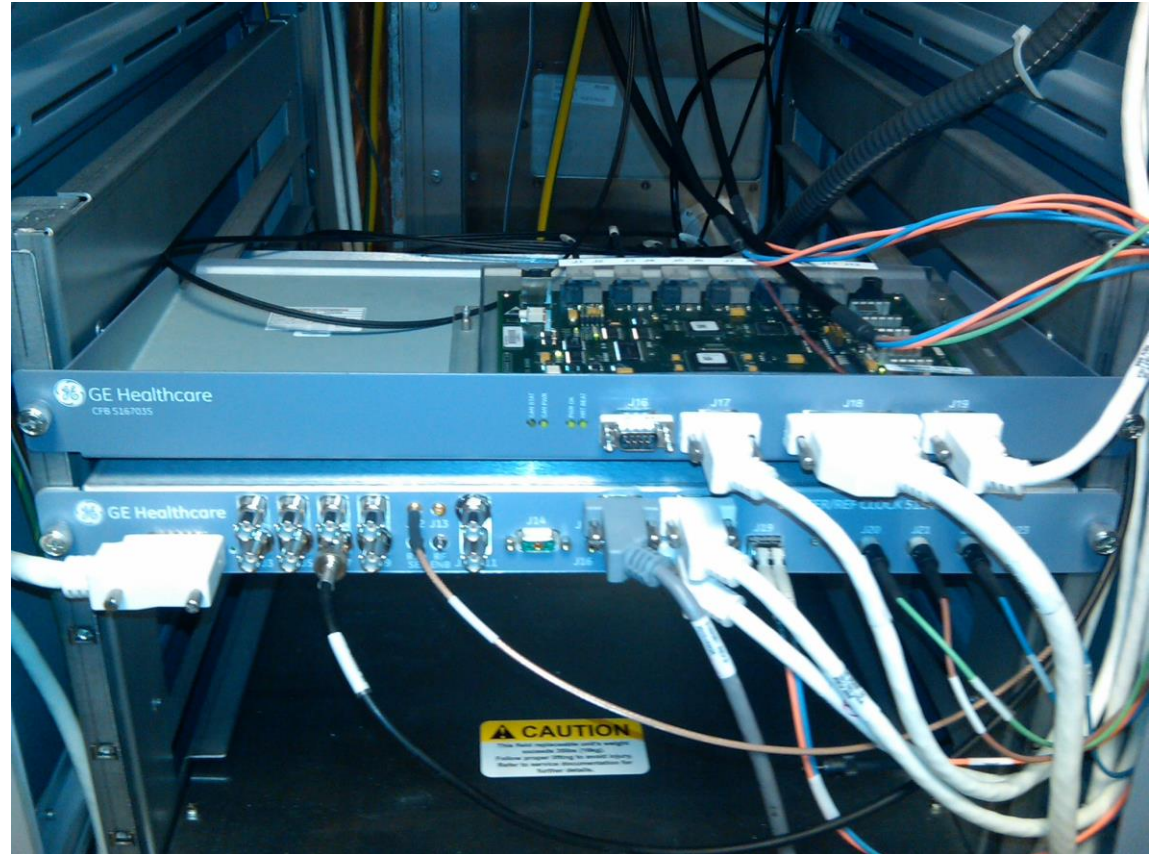
PGR



CAN devices
Node ID 2 Driver Module
Node ID 6 RF BB Amp I/F
Node ID 7 RF NB Amp I/F
Node ID 8 Power Monitor 1
Node ID 9 Power Monitor 2
Node ID 11 SRI
Node ID 20 CFB
Node ID 24 RF NB Amp
Node ID 28 RF Hub



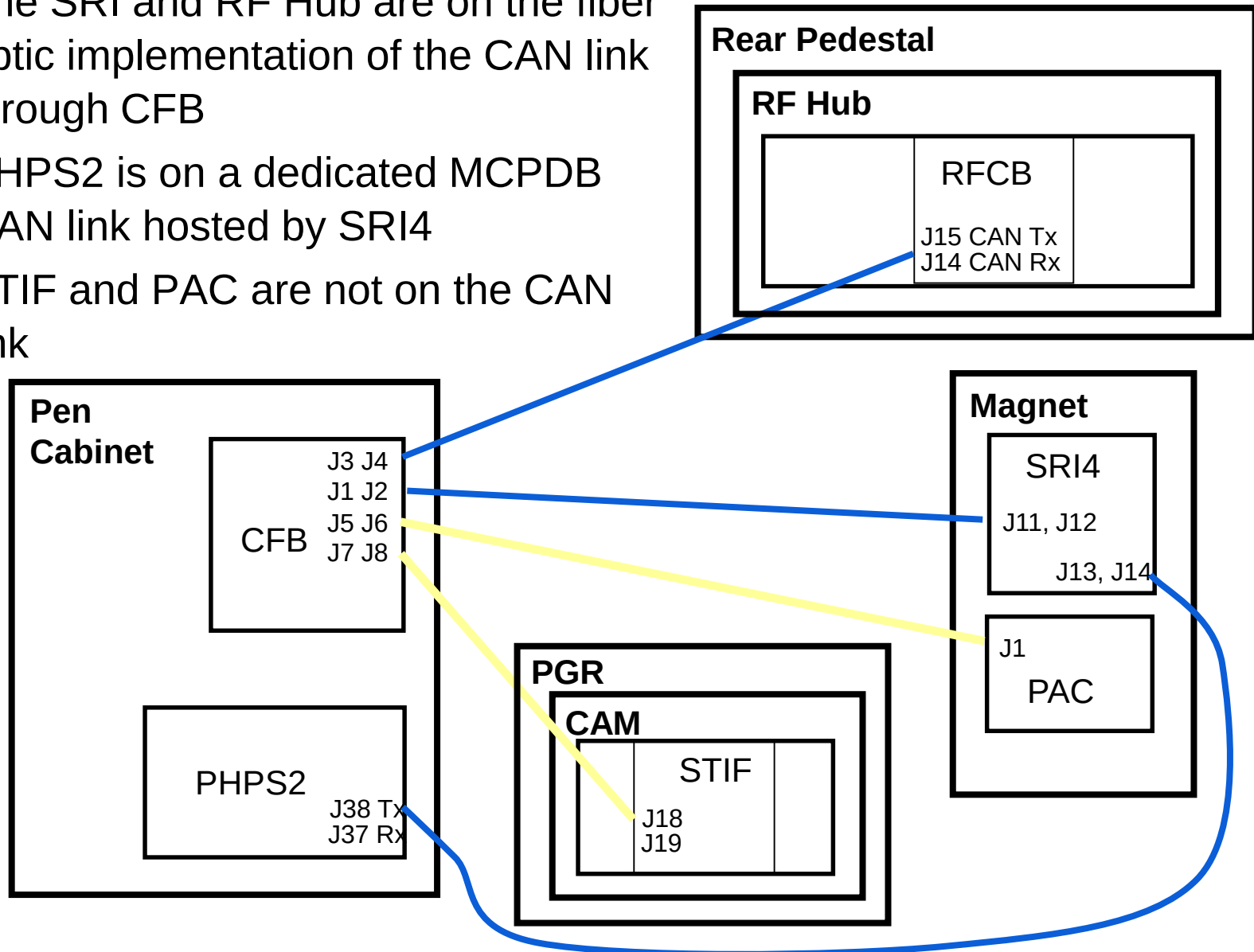
**CFB
connector**



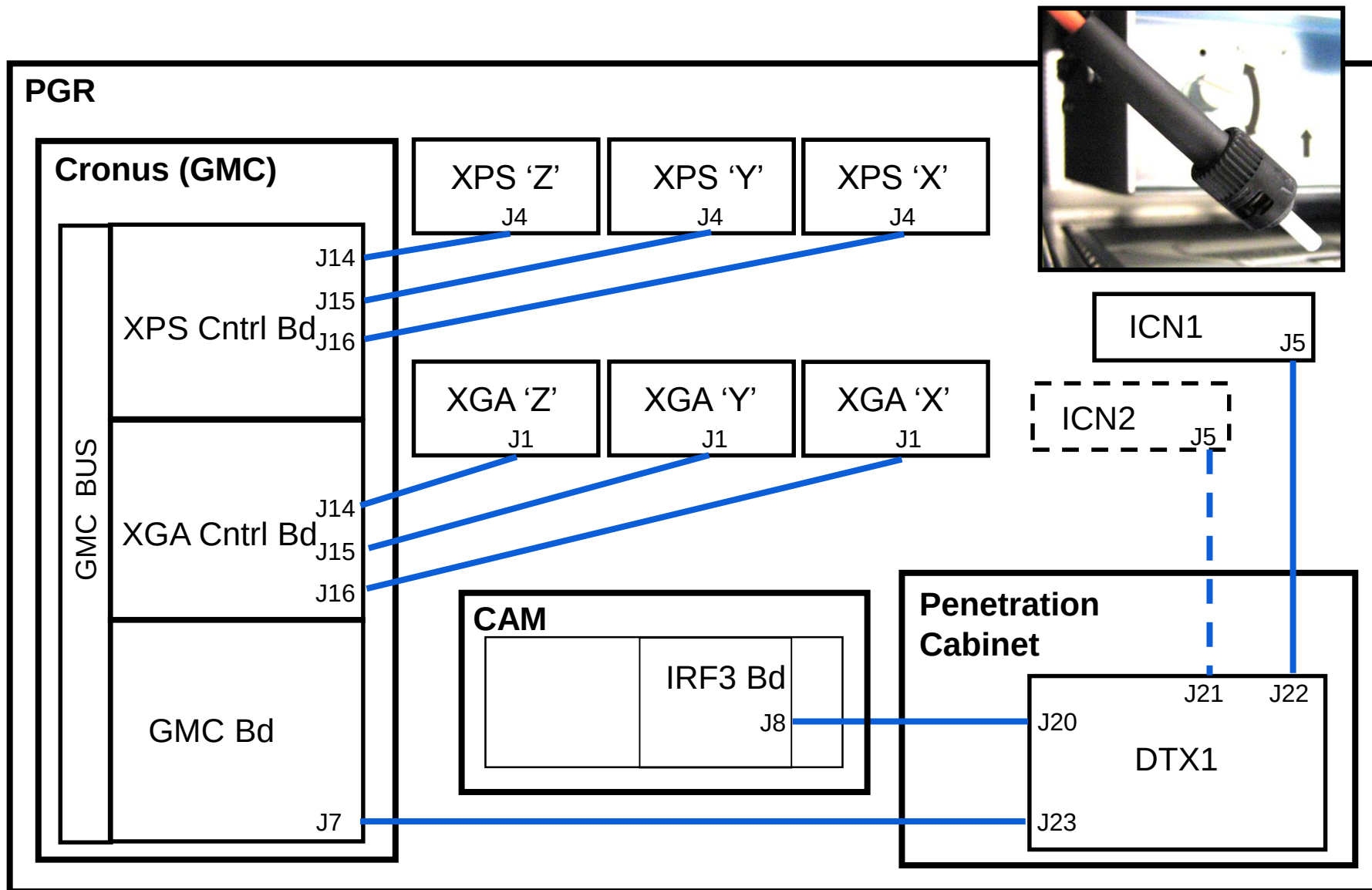
CFB board

CAN link - CFB

- The SRI and RF Hub are on the fiber optic implementation of the CAN link through CFB
- PHPS2 is on a dedicated MCPDB CAN link hosted by SRI4
- STIF and PAC are not on the CAN link



Clock Synchronization Fibers



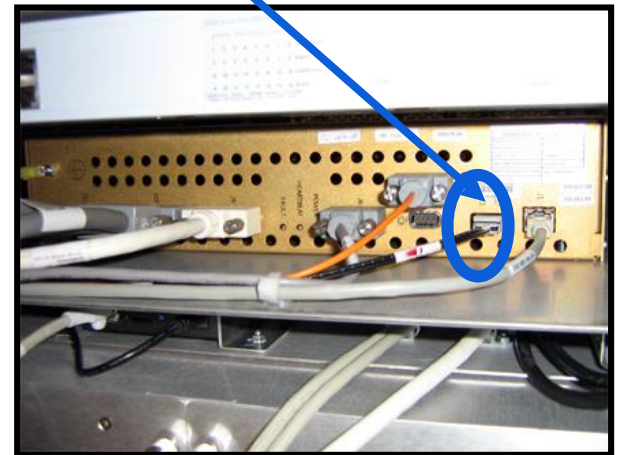
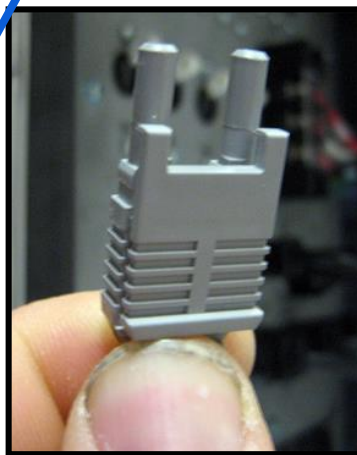
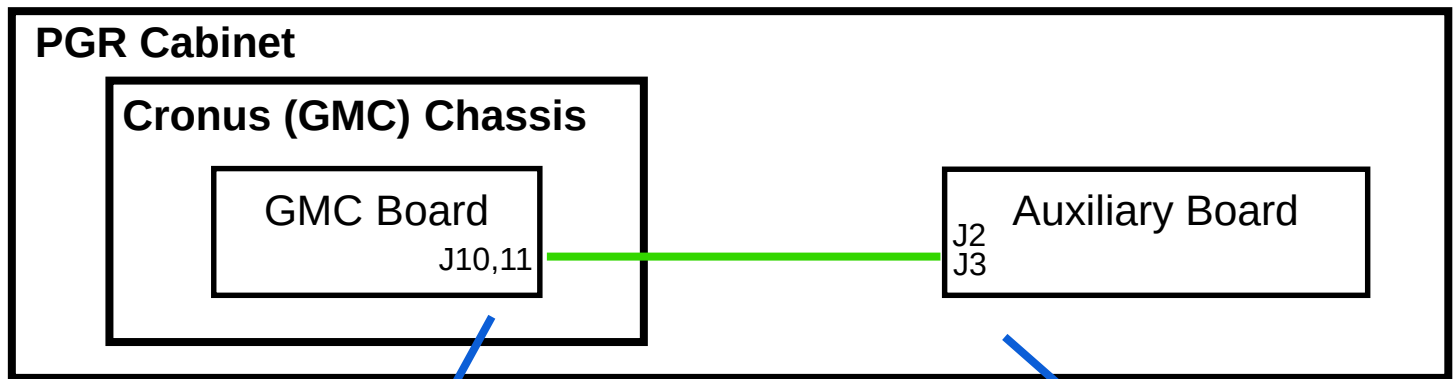
Clock Synchronization Fibers

- Single fiber; same fiber used for X, Y, and Z gradient lines between STIF and GP in all EXCITE- and HD-based products
- Clock sync fibers are not axis-specific. They can be swapped between connectors on XGA and XPS control boards



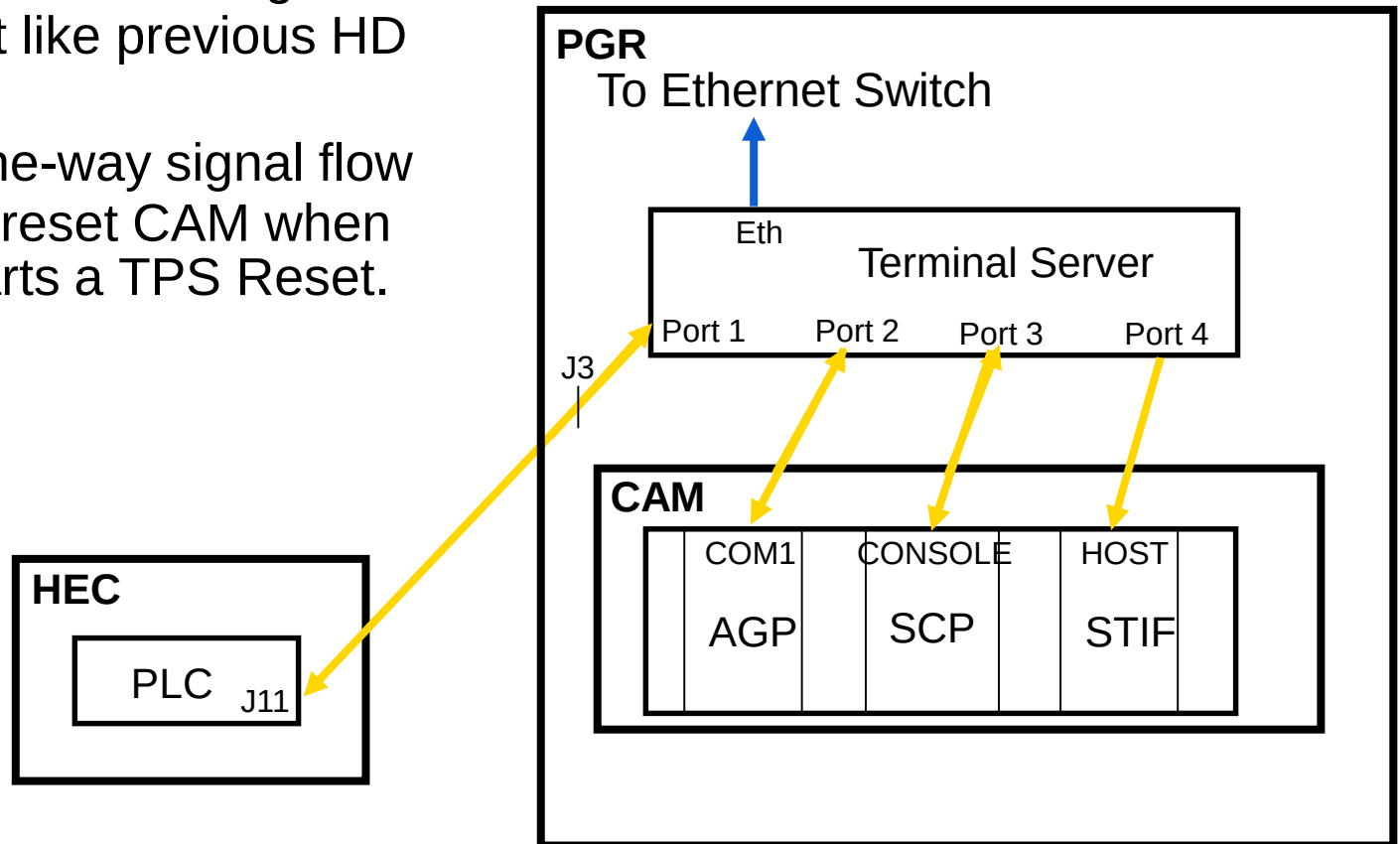
Auxiliary Board Fiber Link

- Gradient temperature, voltage, data
- Water leak sensor monitoring
- If a water leak is detected on any of the leak strips, the PDU main breaker will trip



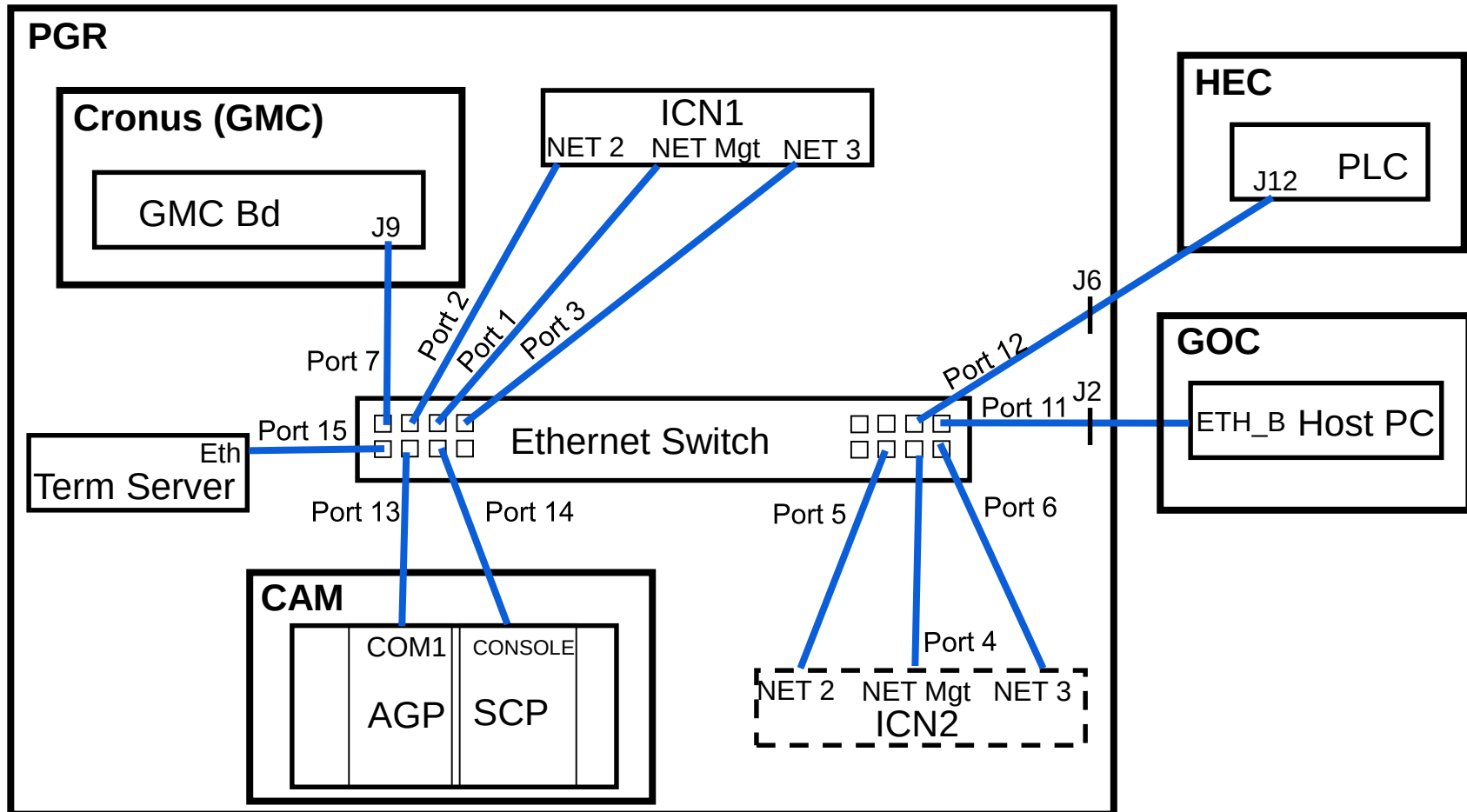
Terminal Server Communication

- Ports ARE critical, here
- Port 1 is used to automatically set the HEC PLC IP address, gateway address, and netmask during install. It is also used to manually reset PLC if it is not communicating via the Ethernet.
- Ports 2 and 3 are for diagnostic use only, just like previous HD products.
- Port 4 is a one-way signal flow from Host to reset CAM when someone starts a TPS Reset.



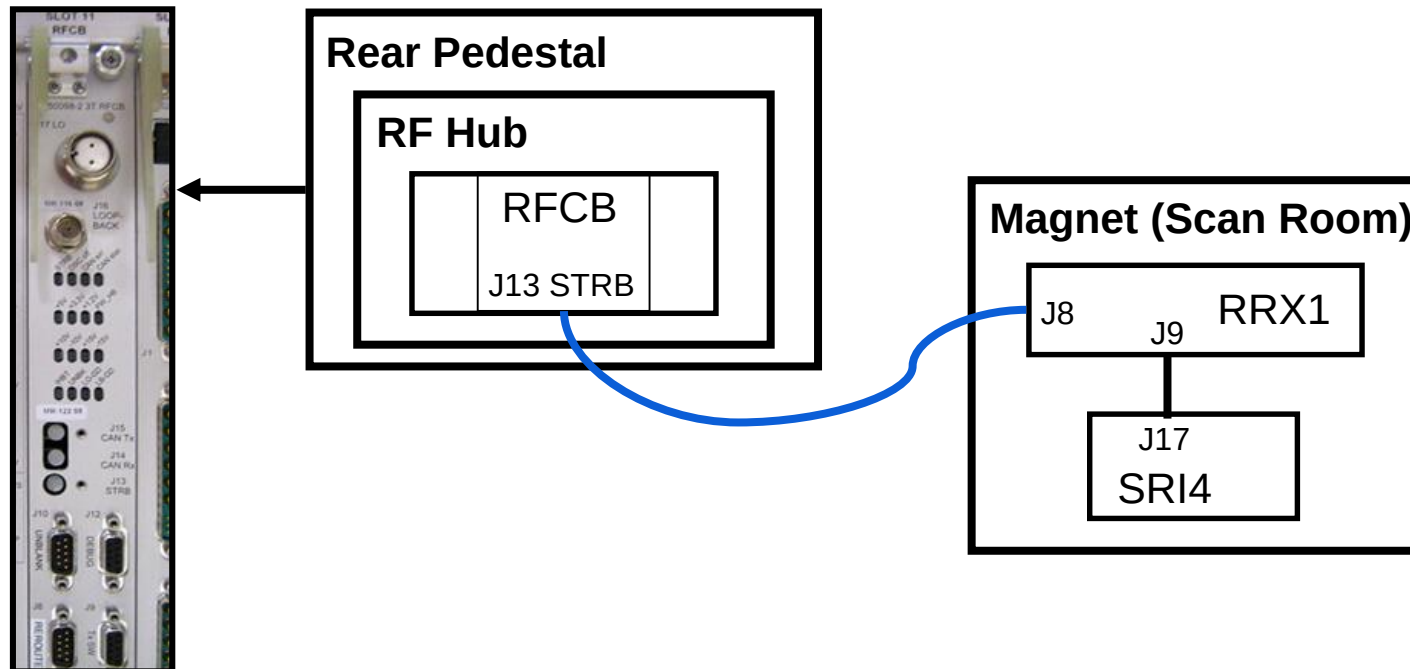
Ethernet communication

- The port on the Ethernet switch to which a device is attached is not critical
- For the purpose of drawing clarity, ports are not shown connected to the Ethernet Switch in order

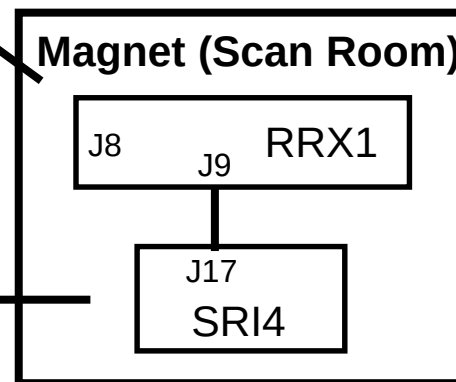
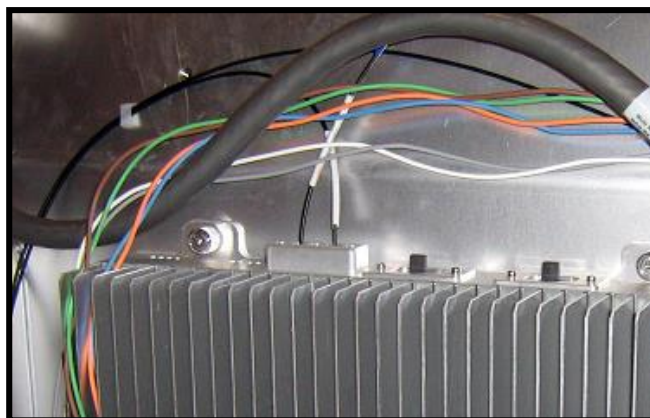


STRB Fiber Link

- The STRoBe fiber link:
 - Critical for TPS setup (RRX setup of the RF hub components; for example, R1 adjustment.)
 - *Must* work for successful diagnostics and TPS Reset completion
 - permits RRX to coordinate carriage/cradle movement with SRI during SWIFT scanning



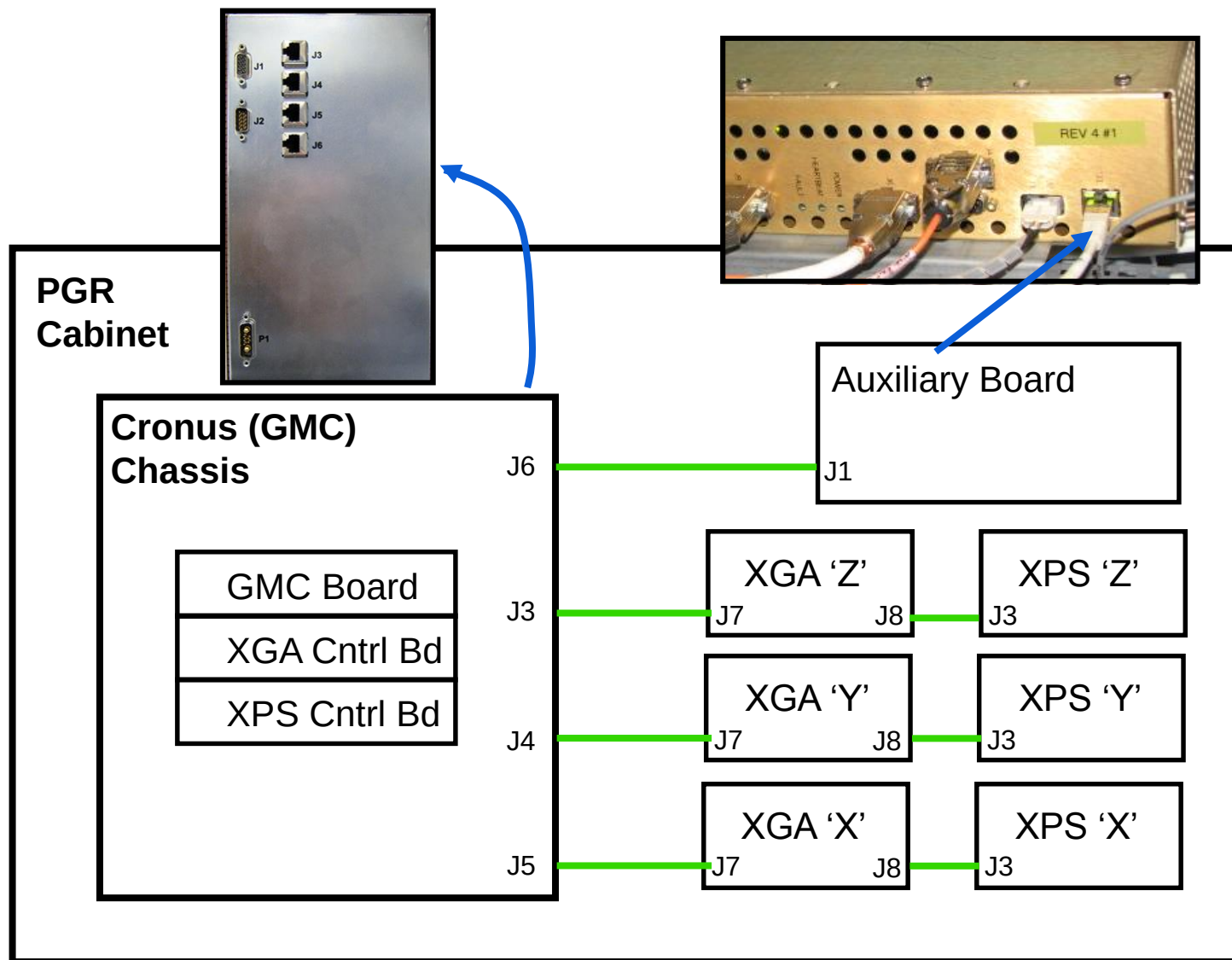
STRB fiber link



Octavius link

- The Octavius Link provides three (3) static diagnostic signals to the GMC that act as a backup in the scenario that the main fiber link J2/J3 is damaged or disconnected.
- While the system will post an error concerning any component with a faulty Octavius Link connection, it will not pause scanning.
- The Octavius link cabling connectors are RJ-45 and although the cabling is the same as that used in Ethernet, the link is not Ethernet.

Octavius Link



- Things to know:
 - The GMC Board provides Octavius Link communication bias to all components, similar to how the Driver Module provides 24V CAN Link bias to all components.
 - A Green LED (OCT POWER) illuminated means the device is receiving Octavius Link power from the GMC Board. The dark LED means that the GMC Board is not powered, has a bad connection, or has faulty cable.
 - A yellow LED (OCT STATUS) illuminated means device is communicating on a link. The dark LED means that the device has no power, has a bad cable and/or connection, or the GMC Board is not communicating.



**Octavius connection
on the front of the
XGA**

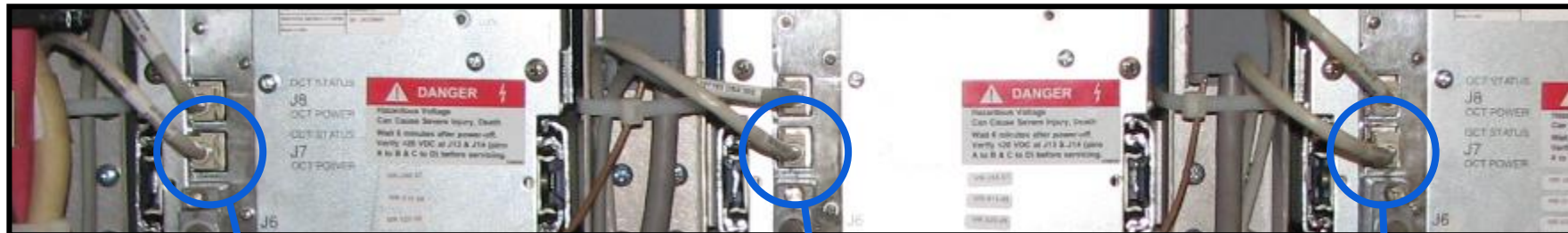
Octavius Link

XGAs

X

Y

Z



XPS

S



Auxiliary board not shown

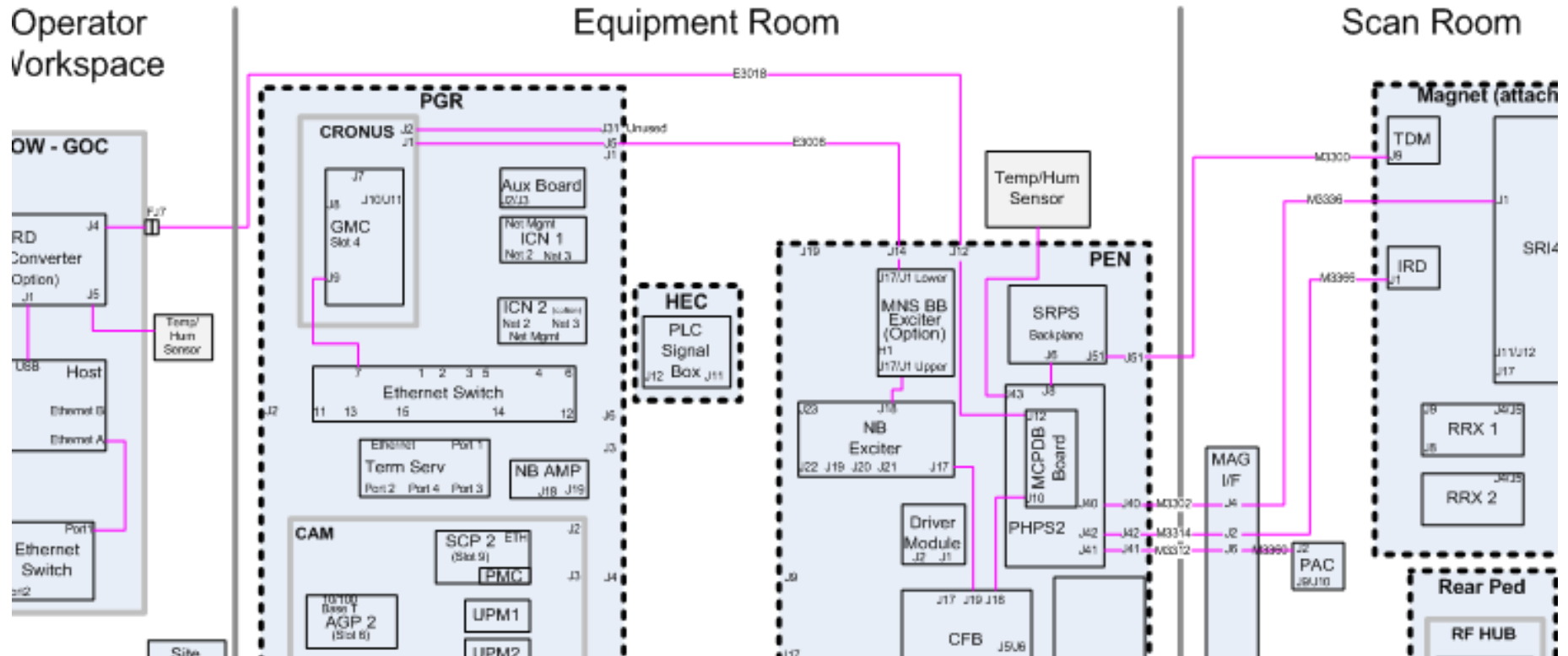
1-Wire Network

- The devices are not in series
- The devices are individually queried
- One-wire is not a single wire in this implementation
- A one-wire network provides more than just ID information
- A one-wire network provides real-time system messages;
GUI tools provide data

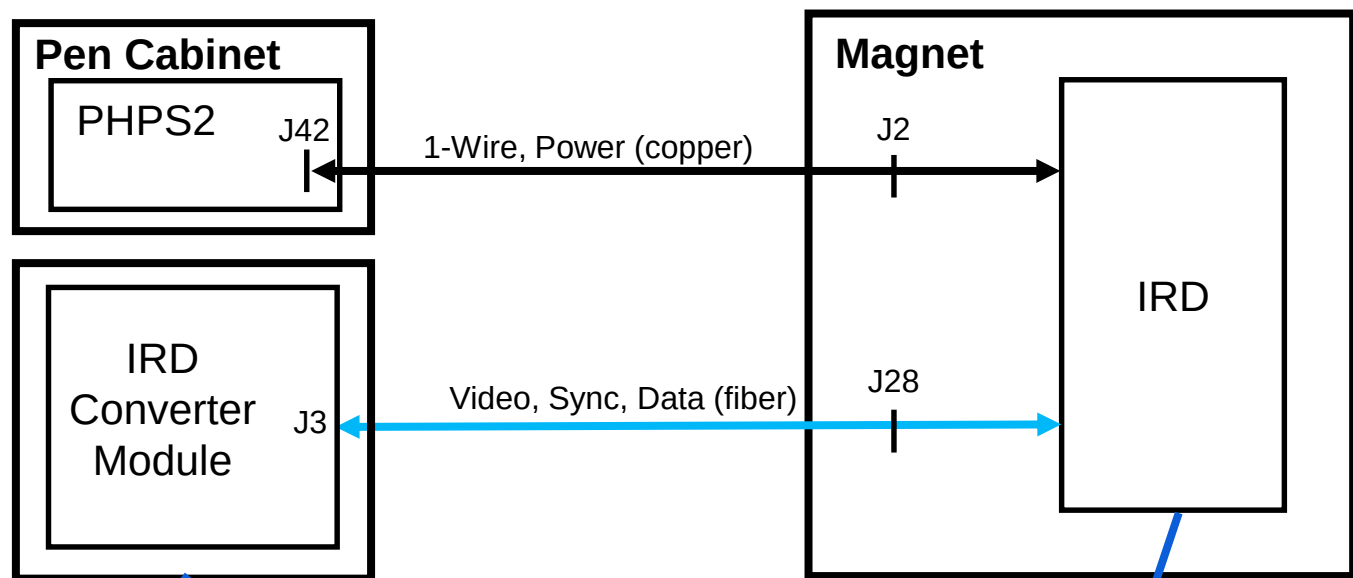
PHILIPS Service Methods – System Block diagrams –

MR Service Engineering – Functional Block Diagrams – MR750

Subsystems	Gradient	Power	GOC	Communication	RF	Post Handling	Magnet	Cooling
Communication Viewing Options	Communication - Main	CAN Link	Link w/ HOS	1-wire Data	DVMR Link	Ethernet & Serial Communication	Clock Synchronization	
Communication – CAN Link								



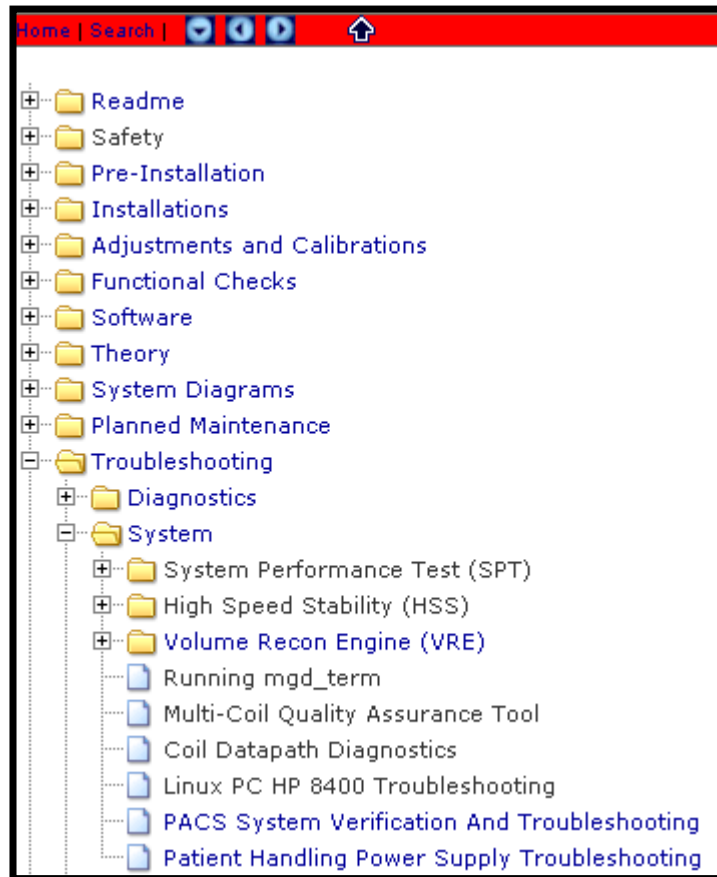
In-room Display fiber link (optional)



Summary view of communication

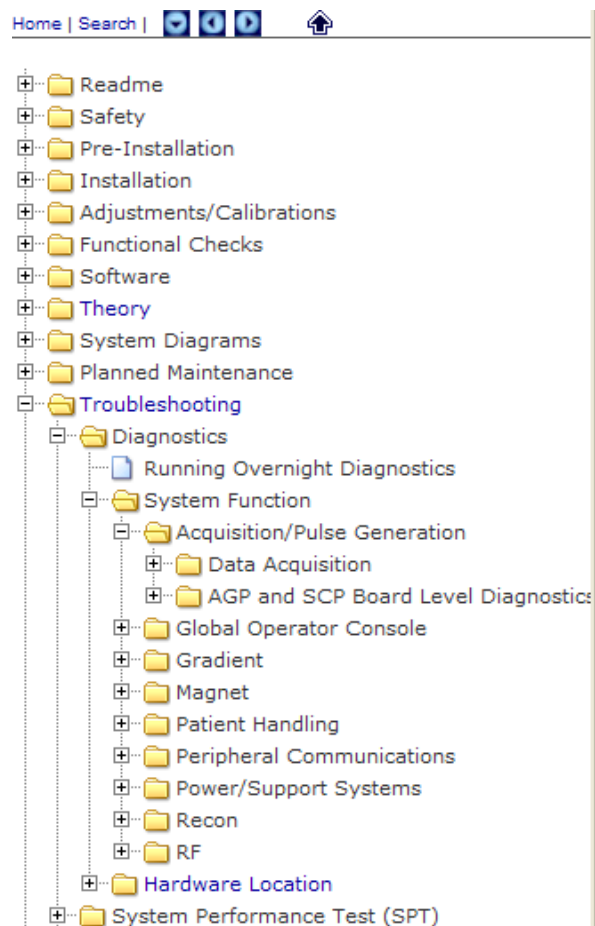
- **DVMR Link** – fast status and control
- **VRF Data Link** – raw image data
- **Infiniband Link** – permits fast sharing of image data
- **XGD Data Fiber Links** – X, Y, and Z gradient waveform data, status and control
- **CAN Link** – status and control
- **Clock Synchronization Fibers** – XGA and XPS clock signals; not unique
- **Auxiliary Board Fiber Link** – water leak, temp, voltage, prognostic data
- **Ethernet Communication** – image data, status and control
- **STRB Fiber Link** – signal routing, table motion data; managed by RRx1
- **Octavius Link** – diagnostic information, backup to fibers; not required for scanning
- **One-Wire Network** – real-time, diagnostic info; provides secondary access
- **In-Room Display Fiber Link** – video for in-room monitor

- You can access some troubleshooting information from the MR Service Documentation.

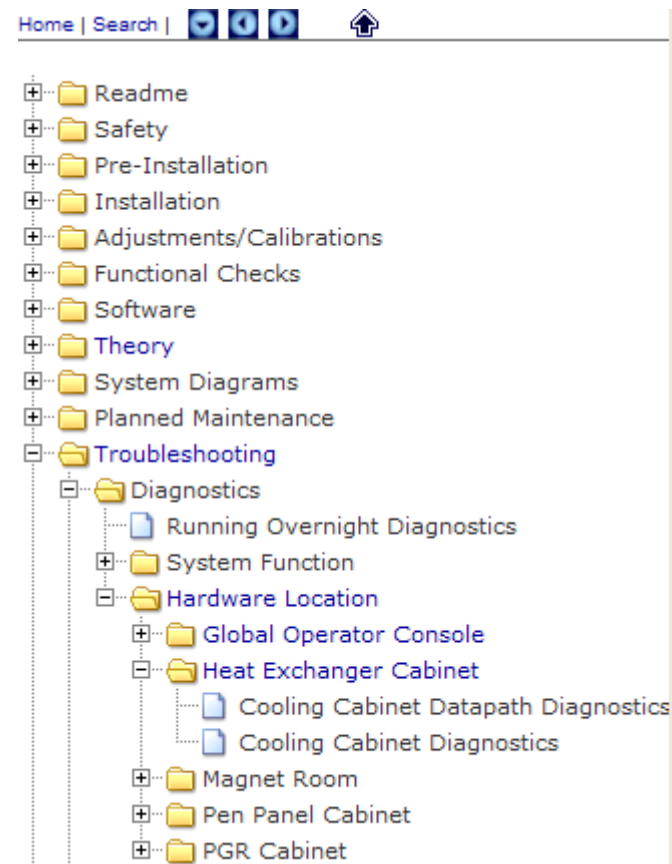


System troubleshooting procedures

- You can access diagnostics information from the MR Service Documentation redundantly under function, or under hardware location.



System Function Diagnostics



Hardware Location

